Example Problem 4–7 demonstrates the procedure of analyzing a stress element, a plane stress element in this case, and converting the stress components to three ranked principal stresses. In this case, one of the principal stresses is zero. In a design problem, we will need to further reduce the three principal stresses to an "effective stress" and make sure the yield strength of the selected material is greater than the effective stress to prevent yielding. While more is said in Chapter 5 about the concept of design stress, we now follow with the type of analysis necessary to judge the suitability of the tank design in Example Problem 4–7. Two approaches are demonstrated: the Maximum Shear Stress Theory (MSST), also called the *Tresca criterion*; and the Distortion Energy Theory, also called the *von Mises criterion*.

Using the Tresca criterion, the effective stress is

$$\sigma' = (\sigma_1 - \sigma_3)/2 = 4113/2 = 2056 > s_y \text{ (psi)}$$

Then the selected material should have a shear yield strength greater than 2013 psi. Since $s_{sy} = s_y/2$, the yield strength of the material should be greater than 4113 psi.

Using the von Mises criterion, the effective stress is

$$\sigma_{\text{eff}} = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$\sigma_{\text{eff}} = \sqrt{\frac{(4113 - 2056)^2 + (2056 - 0)^2 + (0 - 4113)^2}{2}} = 3562 \text{ psi}$$

Therefore, the yield strength of the selected material needs to be greater than 3562 psi.

Before we introduce the failure theories in Chapter 5, it is tempting to use the maximum principal stress to select the material. While considering only the maximum principle stress also results in $s_y \geq 4113$ psi, it should be noted that the solution $s_y \geq 4113$ psi is based on MSST with $\sigma_1 = 4113$ psi and $\sigma_3 = 0$. The above example also shows that when all three principal stresses are considered in the von Mises criterion, it yields a different result.

4–7 ANALYSIS OF COMPLEX LOADING CONDITIONS

The examples shown in this chapter involve relatively simple part geometries and loading conditions for which the necessary stress analysis can be performed using familiar methods of statics and strength of materials. If more complex geometries or loading conditions are involved, you may not be able to complete the required analysis to create the original stress element from which the Mohr's circle is constructed, as discussed in Section 4–4.

Consider, for example, a cast wheel for a high-performance racing car. The geometry would likely involve webs or spokes of a unique design connecting the hub to the rim to create a lightweight wheel. The loading would be a complex combination of torsion, bending, and compression generated by the cornering action of the wheel.

One method of analysis of such a load-carrying member would be accomplished by experimental stress analysis using strain gages or photoelastic techniques. The results would identify the stress levels at selected points in certain specified directions that could be used as the input to the construction of the Mohr's circle for critical points in the structure.

Another method of analysis would involve the modeling of the geometry of the wheel as a *finite-element model*. The three-dimensional model would be divided into several hundred small-volume elements. Points of support and restraint would be defined on the model, and then external loads would be applied at appropriate points. The complete data set would be input to a special type of computer analysis program called *finite-element analysis* (FEA). The output from the program lists the stresses and the deflection for each of the elements. These data can be plotted on the computer model so that the designer can visualize the stress distribution within the model. Most such programs list the principal stresses and the von Mises stress, eliminating the need to actually draw the Mohr's circle. (The application of von Mises stress is introduced in Chapter 5.) Several different finite-element analysis programs are commercially available for use on personal computers, on engineering work stations, or on mainframe computers. See Internet sites 1–5.

REFERENCE

1. Mott, Robert L. *Applied Strength of Materials*. 6th ed. Boca Raton, FL: CRC Press, 2017.

INTERNET SITES RELATED TO STRESS TRANSFORMATION

The following is a short list of the numerous companies that develop and provide finite-element analysis software for a wide variety of applications including static and dynamic structural analysis, thermal analysis, dynamic performance of mechanical systems, vibration analysis, computational fluid dynamics analysis, and other computer-aided engineering (CAE) capabilities.

1. ADINA R&D, Inc.
2. Autodesk Algor Simulation
3. ANSYS, Inc.
4. MSC Software, Inc.
5. NEi Software, Inc.

PROBLEMS

For the sets of given stresses on an element given in Table 4–2, draw a complete Mohr's circle, find the principal stresses and the maximum shear stress, and draw the principal stress element and the maximum shear stress element. Any stress components not shown are assumed to be zero.

31. Refer to Figure 3–20. For the shaft aligned with the x-axis, create a stress element on the bottom of the shaft just to the left of section B. Then draw the Mohr's circle for that element. Use $D = 0.50$ in.
32. Refer to Figure P3–48. For the shaft ABC, create a stress element on the bottom of the shaft just to the right of section B. The torque applied to the shaft at B is resisted at support C only. Draw the Mohr's circle for the stress element. Use $D = 1.50$ in.
33. Repeat Problem 32 for the shaft in Figure P3–49. Use $D = 0.50$ in.
34. Refer to Figure P3–50. For the shaft ABC, create a stress element on the bottom of the shaft just to the left of section B. The torque applied to the shaft by the crank is resisted at support B only. Draw the Mohr's circle for the stress element. Use $D = 50$ mm.
35. A short cylindrical bar having a diameter of 4.00 in is subjected to an axial compressive force of 75 000 lb and a torsional moment of 20 000 lb·in. Draw a stress element on the surface of the bar. Then draw the Mohr's circle for the element.
36. A torsion bar is used as a suspension element for a vehicle. The bar has a diameter of 20 mm. It is subjected to a torsional moment of 450 N·m and an axial tensile force of 36.0 kN. Draw a stress element on the surface of the bar, and then draw the Mohr's circle for the element.

TABLE 4–2 Given Stresses for Problems 1–30

Problem	σ_x	σ_y	τ_{xy}
1	20 ksi	0 ksi	10 ksi
2	−85 ksi	40 ksi	30 ksi
3	40 ksi	−40 ksi	−30 ksi
4	−80 ksi	40 ksi	−30 ksi
5	−120 ksi	40 ksi	−20 ksi
6	−120 ksi	40 ksi	20 ksi
7	60 ksi	−40 ksi	−35 ksi
8	120 ksi	−40 ksi	100 ksi
9	−100 MPa	0 MPa	80 MPa
10	−250 MPa	80 MPa	−110 MPa
11	50 MPa	−80 MPa	40 MPa
12	150 MPa	−80 MPa	−40 MPa
13	−150 MPa	80 MPa	−40 MPa
14	0 MPa	0 MPa	40 MPa
15	250 MPa	−80 MPa	0 MPa
16	50 MPa	−80 MPa	−30 MPa
17	400 MPa	−300 MPa	200 MPa
18	−120 MPa	180 MPa	−80 MPa
19	−30 MPa	20 MPa	40 MPa
20	220 MPa	−120 MPa	0 MPa
21	40 ksi	0 ksi	0 ksi
22	0 ksi	0 ksi	40 ksi
23	38 ksi	−25 ksi	−18 ksi
24	55 ksi	0 ksi	0 ksi
25	22 ksi	0 ksi	6.8 ksi
26	−4250 psi	3250 psi	2800 psi
27	300 MPa	100 MPa	80 MPa
28	250 MPa	150 MPa	40 MPa
29	−840 kPa	−335 kPa	−120 kPa
30	−325 kPa	−50 kPa	−60 kPa

CHAPTER
FIVE

DESIGN FOR DIFFERENT TYPES OF LOADING

THE BIG PICTURE Design for Different Types of Loading

Discussion Map

- This chapter provides additional tools you can use to design load-carrying components that are safe and reasonably efficient in their use of materials.
- You must learn how to classify the kind of loading the component is subjected to: *static*, *repeated and reversed*, *fluctuating*, *shock*, or *impact*.
- You will learn to identify the appropriate analysis techniques based on the type of load and the type of material.

Discover

Identify components of real products or structures that are subjected to static loads.

Identify components that are subjected to equal, repeated loads that reverse directions.

Identify components that experience fluctuating loads that vary with time.

Identify components that are loaded with shock or impact, such as being struck by a hammer or dropped onto a hard surface.

Using the techniques you learn in this chapter will help you to complete a wide variety of design tasks.

For the concepts considered in this chapter, the big picture encompasses a huge array of examples in which you will build on the principles of strength of materials that you reviewed in Chapters 3 and 4 and extend them from the analysis mode to the design mode. Several steps are involved, and you must learn to make rational judgments about the appropriate method to apply to complete the design.

In this chapter, you will learn how to do the following:

1. Recognize the manner of loading for a part: Is it static, repeated and reversed, fluctuating, shock, or impact?
2. Select the appropriate method to analyze the stresses produced.

3. Determine the strength property for the material that is appropriate to the kind of loading and to the kind of material: Is the material a metal or a nonmetal? Is it brittle or ductile? Should the design be based on the yield strength, ultimate tensile strength, compressive strength, endurance limit, or some other material property?
4. Specify a suitable *design factor*, often called a *factor of safety*.
5. Design a wide variety of load-carrying members to be safe under their particular expected loading patterns.

The following paragraphs show by example some of the situations to be studied in this chapter.

An ideal *static load* is one that is applied slowly and is never removed. Some loads that are applied slowly and removed and replaced very infrequently can also be considered to be static. What examples can you think of for products or their components that are subjected to static loads? Consider load-carrying members of structures, parts of furniture pieces, and brackets or support rods holding equipment in your home or in a business or factory. Try to identify specific examples, and describe them to your colleagues. Discuss how the load is applied and which parts of the load-carrying member are subjected to the higher stress levels. Some of the examples that you discovered during **The Big Picture** discussion for Chapter 3 could be used again here.

Fluctuating loads are those that vary during the normal service of the product. They typically are applied for quite a long time so the part experiences many thousands or millions of cycles of stress during its expected life. There are many examples in consumer products around your home, in your car, in commercial buildings, and in manufacturing facilities. Consider virtually anything that has moving parts. Again, try to identify specific examples, and describe them to your colleagues. How does the load fluctuate? Is it applied and then completely removed each cycle? Or is there always some level of mean or average load with an alternating load superimposed on it? Does the load swing from a positive maximum value to a negative minimum value of equal magnitude during each cycle of loading? Consider parts with rotating shafts, such as engines or agricultural, production, and construction machinery.

Consider products that have failed. You may have identified some from **The Big Picture** discussion for Chapter 3. Did they fail the first time they were used? Or did they fail after some fairly long service? Why do you think they were able to operate for some time before failure?

Can you find components that failed suddenly because the material was brittle, such as cast iron, some ceramics, or some plastics? Can you find others that failed only after some considerable deformation? Such failures are called *ductile fractures*.

What were the consequences of the failures that you have found? Was anyone hurt? Was there damage to some other valuable component or property? Or was the failure simply an inconvenience? What was the order of magnitude of cost related to the failure? The answer to some of these questions can help you make rational decisions about design factors to be used in your designs.

It is the designer's responsibility to ensure that a machine part is safe for operation under reasonably foreseeable conditions. This requires that a stress analysis be performed in which the predicted stress levels in the part are compared with the *design stress*, or that level of stress permitted under the operating conditions.

The stress analysis can be performed either analytically or experimentally, depending on the degree of complexity of the part, the knowledge about the loading conditions, and the material properties. The designer must be able to verify that the stress to which a part is subjected is safe.

The manner of computing the design stress depends on the manner of loading and on the type of material. Loading types include the following:

Static
Repeated and reversed
Fluctuating
Shock or impact
Random

Material types are many and varied. Among the metallic materials, the chief classification is between *ductile* and *brittle* materials. Other considerations include the manner of forming the material (casting, forging, rolling, machining, and so on), the type of heat treatment, the surface finish, the physical size, the environment in which it is to operate, and the geometry of the part. Different factors must be considered for plastics, composites, ceramics, wood, and others.

This chapter outlines methods of analyzing load-carrying machine parts to ensure that they are safe. Several different cases are described in which knowledge of the combinations of material types and loading patterns leads to the determination of the appropriate method of analysis. It will then be your job to apply these tools correctly and judiciously as you continue your career.

YOU ARE THE DESIGNER

Recall the task presented at the start of Chapter 4, in which you were the designer of a bracket to hold a fabric sample during a test to determine its long-term stretch characteristics. Figure 4–2 showed a proposed design.

Now you are asked to continue this design exercise by selecting a material from which to make the two bent circular bars that are welded to the rigid support. Also, you must specify a suitable diameter for the bars when a certain load is applied to the test material. ■

5–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Identify various kinds of loading commonly encountered by machine parts, including *static*, *repeated and reversed*, *fluctuating*, *shock or impact*, and *random*.
2. Define the term *stress ratio* and compute its value for the various kinds of loading.
3. Describe the stress analysis process and the concept of failure theories for mechanical design.
4. Define the *maximum shear stress theory of failure* for design with ductile materials.
5. Define the *distortion energy theory*, also called the *von Mises theory* or the *Mises-Hencky theory* for design with ductile materials.
6. Define the *maximum normal stress theory*, the *Coulomb-Mohr Theory* and the *modified Mohr theory* for design with brittle materials under static loading.
7. Define the concept of *fatigue*.
8. Define the material property of *endurance limit* and determine estimates of its magnitude for different materials.
9. Recognize the factors that affect the magnitude of endurance limit.
10. Describe the *Soderberg* and the *Goodman methods* and apply them to the design of parts subjected to fluctuating stresses.
11. Define the term *design factor*.
12. Specify a suitable value for the design factor.
13. Consider *statistical approaches*, *finite life*, *fracture mechanics*, and *damage accumulation methods* for design.

5–2 TYPES OF LOADING AND STRESS RATIO

The primary factors to consider when specifying the type of loading to which a machine part is subjected are the manner of variation of the load and the resulting variation of stress with time. Stress variations are characterized by four key values, expressed here as normal stresses:

1. Maximum stress, σ_{max}
2. Minimum stress, σ_{min}
3. Mean (average) stress, σ_m
4. Alternating stress, σ_a (*stress amplitude*)

The maximum and minimum stresses are usually computed from known information by stress analysis or finite-element methods, or they are measured using experimental stress analysis techniques. Then the mean and alternating stresses can be computed from

$$\sigma_m = (\sigma_{max} + \sigma_{min})/2 \tag{5–1}$$

$$\sigma_a = (\sigma_{max} - \sigma_{min})/2 \tag{5–2a}$$

$$\sigma_a = (\sigma_{max} - \sigma_m) \tag{5–2b}$$

The behavior of a material under varying stresses is dependent on the manner of the variation. One method used to characterize the variation is called *stress ratio*. Two types of stress ratios that are commonly used are defined as follows:

$$\begin{aligned} \text{Stress ratio } R &= \frac{\text{minimum stress}}{\text{maximum stress}} = \frac{\sigma_{min}}{\sigma_{max}} \\ \text{Stress ratio } A &= \frac{\text{alternating stress}}{\text{mean stress}} = \frac{\sigma_a}{\sigma_m} \end{aligned} \tag{5–3}$$

Stress ratio R is used in this book.

Static Stress

When a part is subjected to a load that is applied slowly, without shock, and is held at a constant value, the resulting stress in the part is called *static stress*. An example is the load on a structure due to the dead weight of the building materials. Figure 5–1 shows a diagram of stress versus time for static loading. Because $\sigma_{max} = \sigma_{min}$, the stress ratio for static stress is $R = 1.0$.

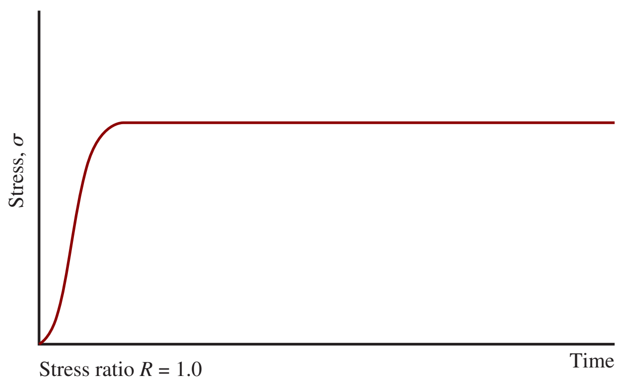


FIGURE 5–1 Static stress

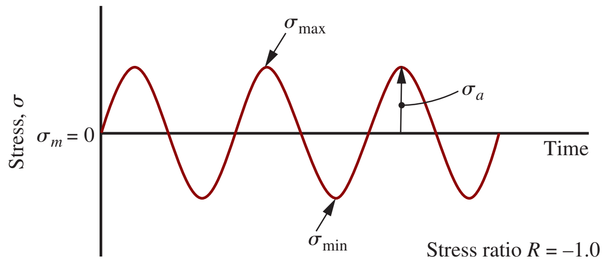


FIGURE 5–2 Repeated and reversed stress or pure oscillation

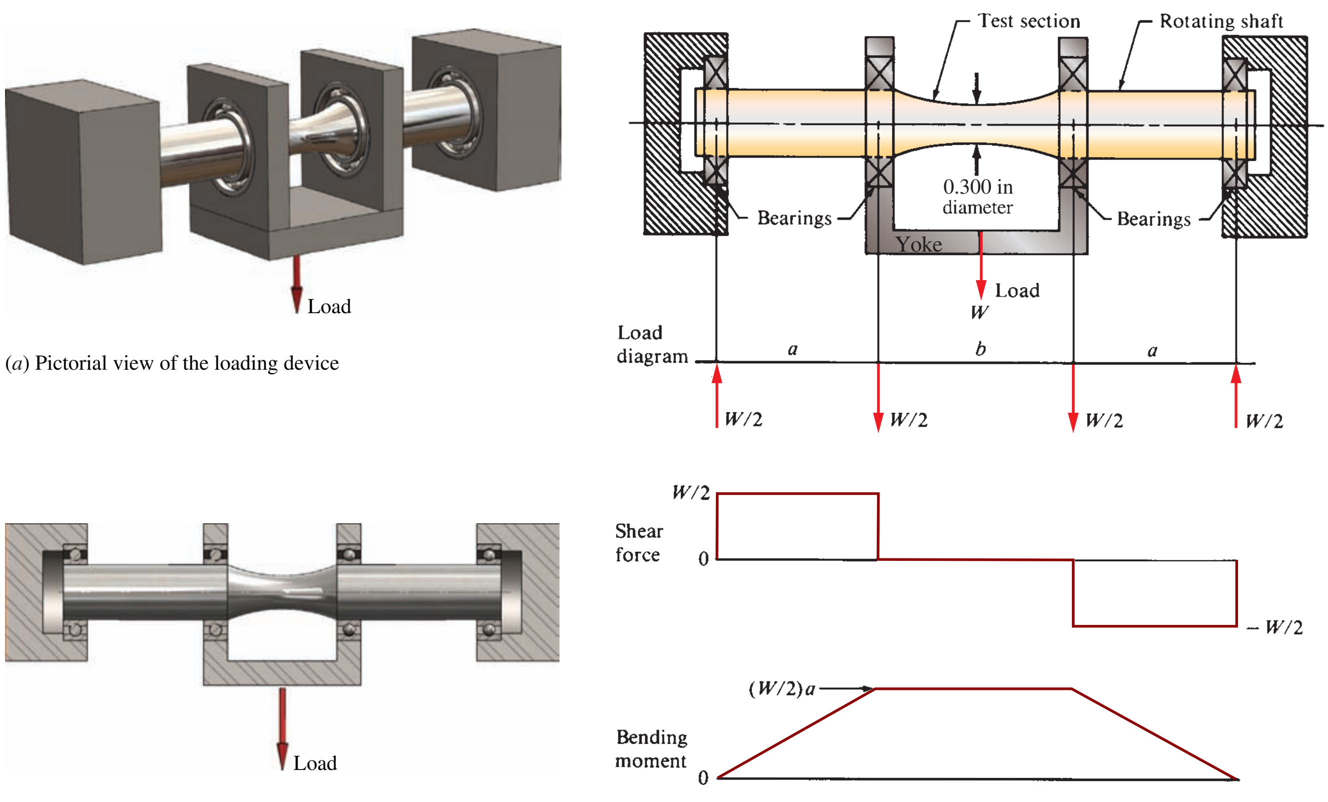
Static loading can also be assumed when a load is applied and is removed slowly and then reapplied, if the number of load applications is small, that is, under a few thousand cycles of loading.

Repeated and Reversed Stress—Pure Oscillation

A stress reversal occurs when a given element of a load-carrying member is subjected to a certain level of tensile stress followed by the *same level* of compressive stress. If this stress cycle is repeated many thousands of times, the stress is called *repeated and reversed* or *pure oscillation*. Figure 5–2 shows the diagram of stress versus time for repeated and reversed stress. Because $\sigma_{min} = -\sigma_{max}$, the stress ratio is $R = -1.0$, and the mean stress is zero.

An important example in machine design is a rotating circular shaft loaded in bending such as that shown in Figure 5–3. In the position shown, an element on the bottom of the shaft experiences tensile stress while an element on the top of the shaft sees a compressive stress of equal magnitude. As the shaft is rotated 180° from the given position, these two elements experience a complete reversal of stress. Now if the shaft continues to rotate, all parts of the shaft that are in bending see repeated, reversed stress. This is a description of the classic loading case of *repeated and reversed bending*.

This type of loading is often called *fatigue loading*, and a machine of the type shown in Figure 5–3 is often used to test the fatigue behavior of materials. The device shown is called the *R. R. Moore fatigue test device* and the material property thus measured is called *endurance limit* and this property is discussed in detail in Section 5–5. The shaft is supported by a bearing at each end while a yoke is supported on bearings. A known loading is applied to the yoke resulting in two concentrated loads being applied; one at each bearing that supports the yoke. Note from the shearing force and bending moment diagrams that this type of loading provides uniform bending moment between the yoke arms while the shearing force is zero. Thus pure bending occurs in the test section. The shaft is machined to precise dimensions with the middle portion having a very gradual taper down to a small diameter. That diameter is typically 0.300 in. With the gradual taper, the stress concentration factor is virtually 1.0. Furthermore, the shaft is polished to a fine surface finish so that machining marks do not affect the stress



(*a*) Pictorial view of the loading device

(*b*) Side view showing the shaft and four bearings

(*c*) Load, shear, and bending moment diagrams

FIGURE 5–3 R. R. Moore fatigue test device; example of reversed bending levels in the bar. The shaft is rotated by an electric motor while the system counts the number of revolutions. It also has a device to sense when the specimen breaks so that there is a known relationship between the stress level and the number of cycles to failure.

Actually, reversed bending is only a special case of fatigue loading, since any stress that varies with time can lead to fatigue failure of a part. Many materials test laboratories are using computer-controlled, repeated and reversed axial loading instead of rotating bending to acquire fatigue strength data. It is described later that there are differences between these two methods in regard to the strength values obtained. It is essential that care be exercised to determine what type of stress is used to measure the fatigue strength when using published data.

Fluctuating Stress—Pulsating Stress

When a load-carrying member is subjected to an alternating stress with a nonzero mean, the loading produces *fluctuating stress*, sometimes called *pulsating stress*. Figure 5–4 shows four diagrams of stress versus time for this type of stress. Differences among the four diagrams occur in whether the various stress levels are positive (tensile) or negative (compressive). ***Any varying stress with a nonzero mean is considered a fluctuating stress***. Figure 5–4 also shows the possible ranges of values for the stress ratio R for the given loading patterns.

A special, frequently encountered case of fluctuating stress is *repeated*, *one-direction stress*, or pure pulsating

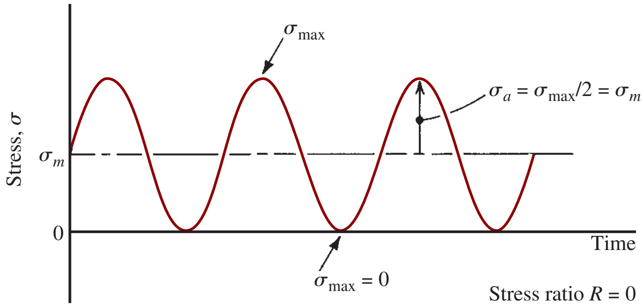


FIGURE 5–5 Repeated, one-direction stress, a special case of fluctuating stress or pure pulsating stress

stress, in which the load is applied and removed many times. As shown in Figure 5–5, the stress varies from zero to a maximum with each cycle. Then, by observation,

$$\sigma_{min} = 0$$
$$\sigma_m = \sigma_a = \sigma_{max}/2$$
$$R = \sigma_{min}/\sigma_{max} = 0$$

An example of a machine part subjected to the more general nature of fluctuating stress of the type shown in Figure 5–4(a) is shown in Figure 5–6, in which a reciprocating cam follower feeds spherical balls one at a time from a chute. The follower is held against the rotating eccentric cam by a flat spring loaded as a cantilever. Part (a) of the figure shows the entire layout of the ball feed device and part (b) shows the cross section of the flat spring. Parts (c) and (d) show two views of just the cam, follower, and flat spring. When the follower is farthest

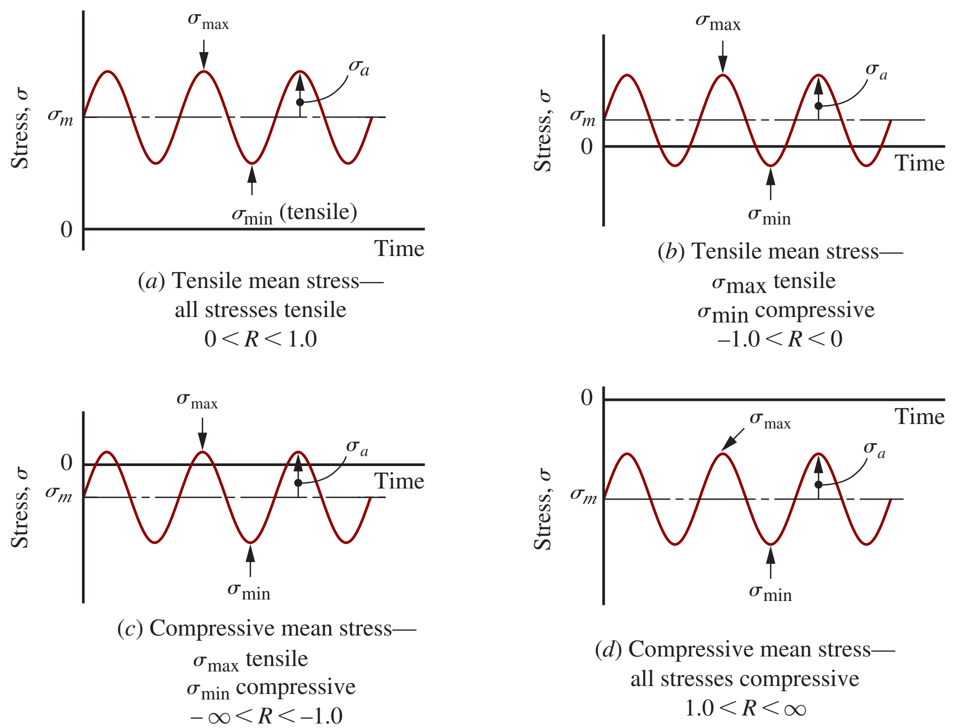
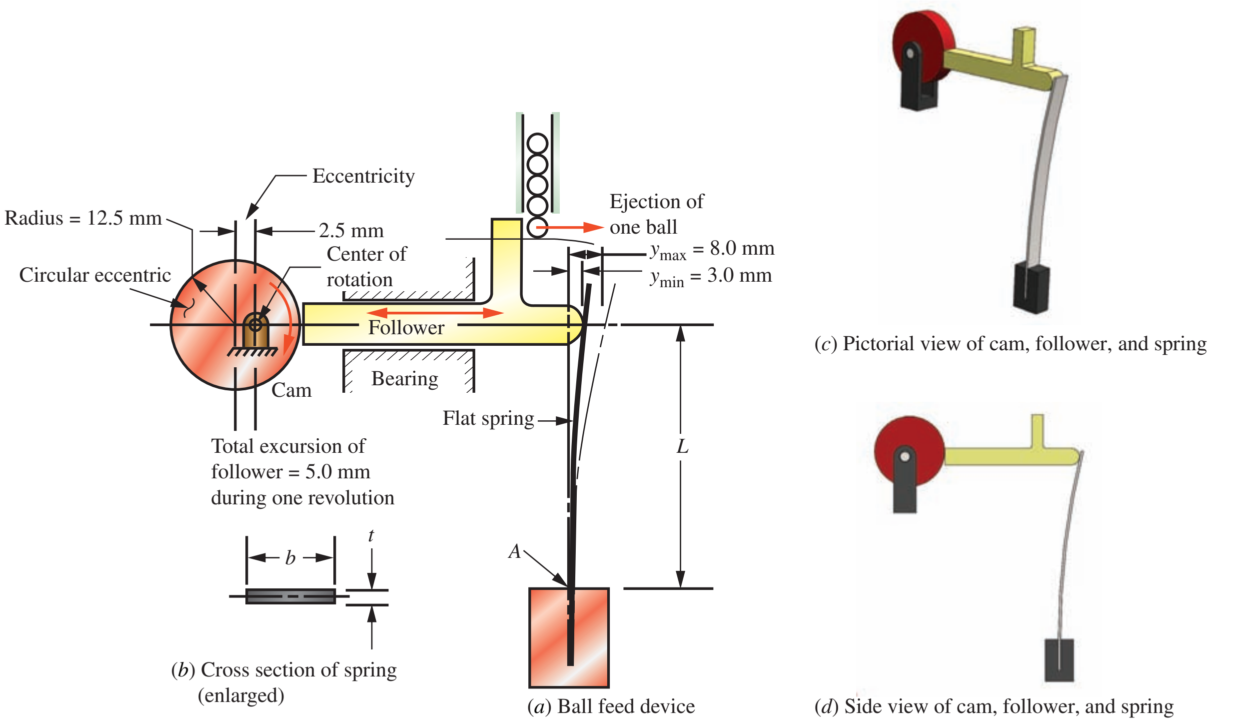


FIGURE 5–4 Fluctuating stresses—pulsating stresses



(*b*) Cross section of spring (enlarged)

(*a*) Ball feed device

(*c*) Pictorial view of cam, follower, and spring

(*d*) Side view of cam, follower, and spring

FIGURE 5–6 Example of cyclic loading in which the flat spring is subjected to fluctuating stress

to the left, the spring is deflected from its free (straight) position by an amount $y_{min} = 3.0$ mm. When the follower is farthest to the right, the spring is deflected to $y_{max} = 8.0$ mm. Then, as the cam continues to rotate, the spring sees the cyclic loading between the minimum and maximum values. Point *A* at the base of the spring on the convex side experiences the varying tensile stresses of the type shown in Figure 5–4(a). Example Problem 5–1 completes the analysis of the stress in the spring at point *A*.

Example Problem 5–1 For the flat steel spring shown in Figure 5–6, compute the maximum stress, the minimum stress, the mean stress, and the alternating stress. Also compute the stress ratio, *R*. The length *L* is 65 mm. The dimensions of the spring cross section are $t = 0.80$ mm and $b = 6.0$ mm.

Solution

Objective: Compute the maximum, minimum, mean, and alternating tensile stresses in the flat spring. Compute the stress ratio, *R*.

Given: Layout shown in Figure 5–6. The spring is steel: $L = 65$ mm.

Spring cross-sectional dimensions: $t = 0.80$ mm and $b = 6.0$ mm.

Maximum deflection of the spring at the follower $= 8.0$ mm.

Minimum deflection of the spring at the follower $= 3.0$ mm.

Analysis: Point *A* at the base of the spring experiences the maximum tensile stress. Determine the force exerted on the spring by the follower for each level of deflection using the formulas from Table A14–2, Case (a). Compute the bending moment at the base of the spring for each deflection. Then compute the stresses at point *A* using the bending stress formula, $\sigma = Mc/I$. Use Equations (5–1), to (5–3) for computing the mean, alternating stresses, and *R*.

Results: Case (a) of Table A14–2 gives the following formula for the amount of deflection of a cantilever for a given applied force:

$$y = PL^3/3EI$$

Solve for the force as a function of deflection:

$$P = 3EIy/L^3$$

Appendix 3 gives the modulus of elasticity for steel to be $E = 207$ GPa. The moment of inertia, I, for the spring cross section is found from

$$I = bt^3/12 = (6.00\text{ mm})(0.80\text{ mm})^3/12 = 0.256\text{ mm}^4$$

Then the force on the spring when the deflection y is 3.0 mm is

$$P = \frac{3(207 \times 10^9\text{ N/m}^2)(0.256\text{ mm}^4)(3.0\text{ mm})}{(65\text{ mm})^3}\frac{(1.0\text{ m}^2)}{(10^6\text{ mm}^2)} = 1.74\text{ N}$$

The bending moment at the support is

$$M = P \cdot L = (1.74\text{ N})(65\text{ mm}) = 113\text{ N}\cdot\text{mm}$$

The bending stress at point A caused by this moment is

$$\sigma = \frac{Mc}{I} = \frac{(113\text{ N}\cdot\text{mm})(0.40\text{ mm})}{0.256\text{ mm}^4} = 176\text{ N/mm}^2 = 176\text{ MPa}$$

This is the lowest stress that the spring sees in service, and therefore $\sigma_{min} = 176$ MPa.

Because the force on the spring is proportional to the deflection, the force exerted when the deflection is 8.0 mm is

$$P = (1.74\text{ N})(8.0\text{ mm})/(3.0\text{ mm}) = 4.63\text{ N}$$

The bending moment is

$$M = P \cdot L = (4.63\text{ N})(65\text{ mm}) = 301\text{ N}\cdot\text{mm}$$

The bending stress at point A is

$$\sigma = \frac{Mc}{I} = \frac{(301\text{ N}\cdot\text{mm})(0.40\text{ mm})}{0.256\text{ mm}^4} = 470\text{ N/mm}^2 = 470\text{ MPa}$$

This is the maximum stress that the spring sees, and therefore $\sigma_{max} = 470$ MPa.

Now the mean stress can be computed:

$$\sigma_m = (\sigma_{max} + \sigma_{min})/2 = (470 + 176)/2 = 323\text{ MPa}$$

Finally, the alternating stress is

$$\sigma_a = (\sigma_{max} - \sigma_{min})/2 = (470 - 176)/2 = 147\text{ MPa}$$

The stress ratio is found using Equation (5–3):

$$\text{Stress ratio } R = \frac{\text{minimum stress}}{\text{maximum stress}} = \frac{\sigma_{min}}{\sigma_{max}} = \frac{176\text{ MPa}}{470\text{ MPa}} = 0.37$$

Comments The sketch of stress versus time shown in Figure 5–4(a) illustrates the form of the fluctuating stress on the spring. In Section 5–8, you will see how to design parts subjected to this kind of stress.

Shock or Impact Loading

Loads applied suddenly and rapidly cause shock or impact. Examples include a hammer blow, a weight falling onto a structure, and the action inside a rock crusher. The design of machine members to withstand shock or impact involves an analysis of their energy-absorption capability, a topic not considered in this book. (See References 3, 6, 10, and 12.)

Random Loading

When varying loads are applied that are not regular in their amplitude, the loading is called *random*. Statistical analysis is used to characterize random loading for purposes of design and analysis. This topic is not covered in this book. See Reference 11.

5–3 FAILURE THEORIES

When a part is designed, it is of primary interest to know if the part will fail during service. As described in Section 4–3, one of the design objectives is to ensure that the material has the *strength* to sustain the *stress*. Since materials behave differently under different types of loading conditions, various failure theories have been proposed and tested. The theories presented here are accepted practices for designers to compare some defined stresses to some defined strengths. For ductile materials under static loading, a stress element is considered to have failed when yielding occurs. Thus, the failure theories are based on established yield criteria. As brittle materials do not exhibit yielding before fracture, the evaluation of failure is based on postulated fracture criteria. For cyclic loading, other failure theories for evaluating the mean and alternating stresses have been proposed. In Section 5–4, we will introduce the failure theories for static loading. Sections 5–5 and 5–6 present the concepts of fatigue strength and endurance limit that are critical in designing parts for cyclic loading. The theories for fatigue failure are presented in Section 5–7.

5–4 DESIGN FOR STATIC LOADING

Ductile Materials under Static Loading

There are two widely accepted ductile material failure theories for parts under static loading: *Maximum Shear Stress Theory* and *Distortion Energy Theory*. These theories predict failure due to yielding that produce permanent, plastic deformation.

Maximum Shear Stress Theory (MSST). The MSST of failure prediction states that a ductile material begins to yield when the maximum shear stress at a critical location in a load-carrying component exceeds that in a tensile-test specimen when yielding begins. A Mohr's circle analysis for the uniaxial tension test, discussed in Chapter 4, shows that the maximum shear stress is one-half of the applied tensile stress. At yield, then, $s_{sy} = s_y/2$. We use this approach in this book to estimate s_{sy}. Then, for design, the stress element at the location of interest is safe when:

$$\tau_{max} \leq \frac{s_{sy}}{N} = \frac{s_y}{2N} \tag{5–4}$$

where N is the design factor (to be discussed in Section 5–9). As shown in Section 4–4, the maximum shear stress is the Tresca stress:

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} \leq \frac{s_{sy}}{N} \tag{5–5}$$

Thus, because $s_{sy} = s_y/2$,

$$\sigma_1 - \sigma_3 \leq \frac{s_y}{N} \tag{5–6}$$

where σ_1 and σ_3 are the maximum and minimum principal stresses. Both Equations (5–5) and (5–6) evaluate the diameter of the Mohr's circle against the yield strength of the material. When considering all three principal stresses, two additional cases should also be noted. For the condition that all of the principal stresses are in tension, $\sigma_1 > \sigma_2 > \sigma_3 > 0$, the stress element is safe when

$$\sigma_1 \leq \frac{s_y}{N} \tag{5–7}$$

For the condition that all of the principal stresses are in compression, $0 > \sigma_1 > \sigma_2 > \sigma_3$, the stress element is safe when

$$\sigma_3 \geq -\frac{s_y}{N} \tag{5–8}$$

This concept can be illustrated by the *yield locus* shown in Figure 5–7. The numerical scales on the graph are normalized to the yield strength, $\sigma_1/s_y = 1.0$. The materials are even materials for which the values of yield strength

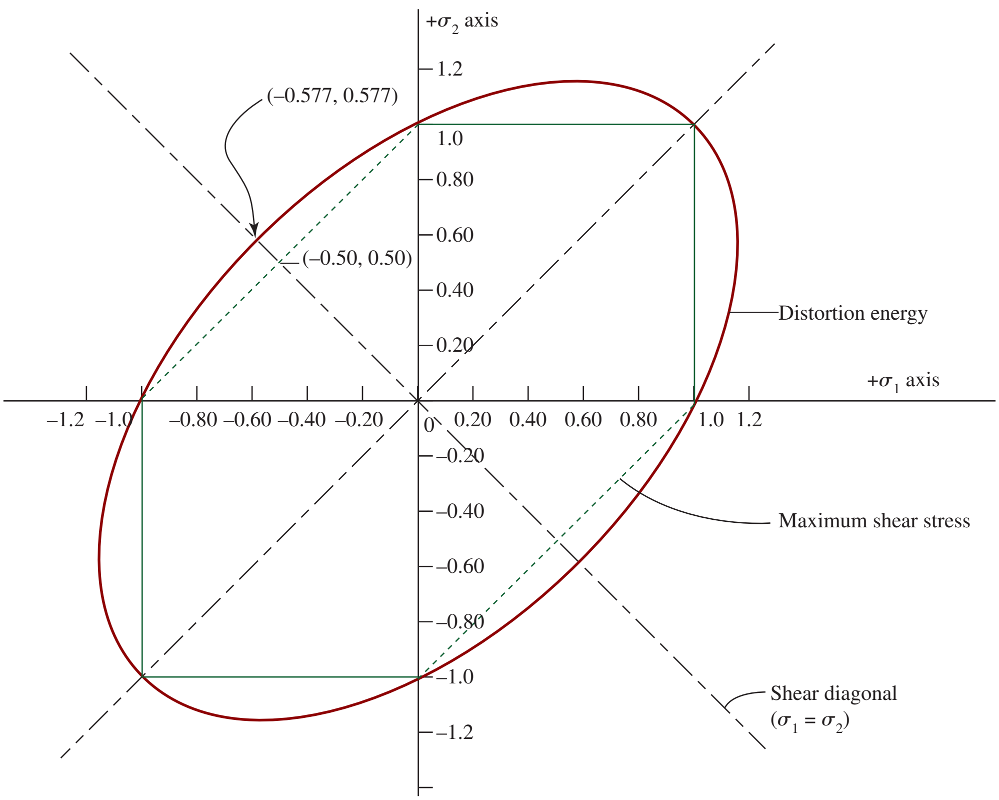


FIGURE 5–7 Maximum shear stress and distortion energy failure criteria

in tension and compression are the same, a characteristic of ductile materials. Plotting the principal stresses, the stress states that lie within the hexagon are predicted to be safe, while those outside would predict failure.

The maximum shear stress method of failure prediction has been shown by experimentation to be somewhat conservative. It is relatively easy to use and is often chosen by designers. For more precise analysis, the distortion energy method is preferred.

Distortion Energy Theory (DET). The distortion energy Theory has been shown to be the best predictor of failure for ductile materials under static loads. It is applied using the von Mises stress, σ_e, which was described in Section 4–4.

The von Mises stress can be presented in different forms, including the general form for 3D stress element:

$$\sigma_e = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}} \tag{5–9}$$

A stress element is considered safe when

$$\sigma_e \leq \frac{s_y}{N} \tag{5–10}$$

A number of variations based on 2D stress state can be derived from Equation (5–9). For example, the von Mises stress of the stress element $\sigma_1 > \sigma_2 > 0$ and $\sigma_3 = 0$ is expressed as

$$\sigma_e = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2} \tag{5–11}$$

It is also helpful to visualize the distortion energy failure prediction theory by plotting the von Mises yield locus on a graph, as shown in Figure 5–7. The yield locus is an ellipse centered at the origin and passing through the yield strength on each axis, in both the tensile and compressive regions. As the concept demonstrated in the MSST, the stress states that lie within the ellipse are predicted to be safe, while those outside would predict failure.

The comparison of the MSST and DET is shown in Figure 5–7. With data showing that the distortion energy theory is the best predictor, it can be seen that the MSST is generally conservative and that it coincides with the distortion energy ellipse at six points. In other regions, it is as much as 16% lower. Note the 45° diagonal line through the second and fourth quadrants, called the *pure shear diagonal*. It was demonstrated in Section 4–6 that for pure shear the stress state is

$$\sigma_1 = \tau_{xy}, \qquad \sigma_2 = 0, \qquad \sigma_3 = -\tau_{xy}$$

Substituting into Equation (5–9):

$$\sigma_e = \sqrt{\frac{(\tau_{xy} - 0)^2 + (0 + \tau_{xy})^2 + (-\tau_{xy} - \tau_{xy})^2}{2}} = \sqrt{3}\tau_{xy} \leq \frac{s_y}{N} \tag{5–12}$$

and thus

$$\tau_{xy} \leq \frac{s_y}{\sqrt{3}} = 0.577 s_y \tag{5–13}$$

for $N = 1$. This predicts yielding when the shear stress is $0.577s_y$. The MSST predicts failure at $0.50s_y$, thus quantifying the conservatism of the MSST.

Brittle Materials under Static Loading

Brittle materials do not yield. Thus, their failure prediction is based on fracture criteria. Brittle materials are typically "non-even" meaning that their compressive strength is higher than the tensile strength. This characteristic is reflected in the three failure theories that are presented here.

Maximum Normal Stress Theory (MNST). The MNST states that a material will fracture when the maximum normal stress (either tension or compression) exceeds the ultimate strength of the material as obtained from a standard tensile or compressive test. Its use is limited, namely for brittle materials under pure uniaxial static tension or compression. When applying this theory, any stress concentration factor, K_t, at the region of interest should be applied to the computed stress because brittle materials do not yield and therefore cannot redistribute the increased stress. Given the principal stresses, $\sigma_1 > \sigma_2 > \sigma_3$, the following equations apply the maximum normal stress theory for safe design.

$$K_t\sigma_1 \leq \frac{s_{ut}}{N} \tag{5–14}$$

$$K_t\sigma_3 \geq \frac{s_{uc}}{N} \tag{5–15}$$

where s_{ut} and s_{uc} are the uniaxial ultimate strength in tension and compression, respectively. It should be noted that the value of s_{uc} is negative. Figure 5–8 shows the failure line defined by MNST. The stress states within the square area are predicted to be safe.

Coulomb-Mohr Theory (CMT). As discussed in the previous section, the MNST is useful for failure prediction of brittle materials subjected to uniaxial tension and uniaxial compression. For a biaxial stress element, a failure criterion known as the Coulomb-Mohr Theory can better predict failure by fracture. The following equation expresses the basis for the CMT:

$$\frac{\sigma_1}{s_{ut}} + \frac{\sigma_3}{s_{uc}} = \frac{1}{N} \tag{5–16}$$

The fracture prediction based on the CMT is an uneven hexagon as shown in Figure 5–8. Equation (5–16) represents the line connecting the points (s_{ut}, 0) and (0, s_{uc}). It can be observed that the CMT is an extension of MSST with consideration of the non-even material

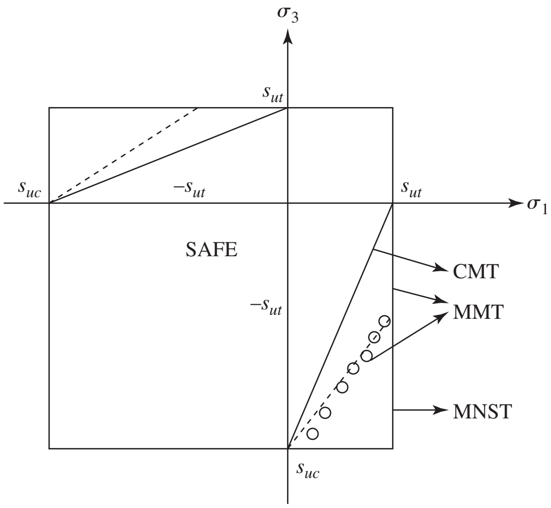


FIGURE 5–8 Failure predictions of brittle materials under static loading

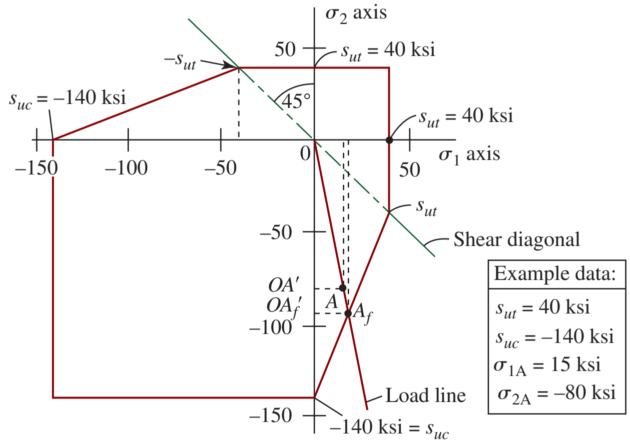


FIGURE 5–9 Modified Mohr diagram with example data and a load line plotted

property of brittle materials. Similar to MSST, when considering all three principal stresses, two additional cases should also be evaluated. When all of the principal stresses are in tension, $\sigma_1 > \sigma_2 > \sigma_3 > 0$, the stress element is safe when

$$\sigma_1 \leq \frac{s_{ut}}{N} \tag{5–17}$$

When all of the principal stresses are in compression, $0 > \sigma_1 > \sigma_2 > \sigma_3$, the stress element is safe when

$$\sigma_3 \geq \frac{s_{uc}}{N} \tag{5–18}$$

Note that the value of s_{uc} is negative.

Modified Mohr Theory (MMT). The Modified Mohr Theory is a semi-empirical data fitting method that best predicts fracture of brittle material under static loading, especially for the stress states in the fourth quadrant as depicted in Figure 5–8 where the circles illustrate examples of experimental results. Failure is predicted when the stress state is outside of the region defined by connecting the points (s_{ut}, s_{ut}), $(-s_{ut}, s_{ut})$, $(s_{uc}, 0)$, (s_{uc}, s_{uc}), $(0, s_{uc})$, $(s_{ut}, -s_{ut})$, and (s_{ut}, s_{ut}).

For design, because of the many different shapes and dimensions of safe-stress zones, it is suggested that a rough plot be made of the pertinent part of the modified Mohr diagram from actual material strength data as shown in Figure 5–9.Then the actual values of σ_1 and σ_2 can be plotted to ensure that they lie within the safe zone of the diagram as shown in Figure 5–9. A *load line* can be an aid in determining the design factor, N, using the modified Mohr diagram. The assumption is made that stresses increase proportionally as loads increase. Apply the following steps for an example stress state, A, for which $\sigma_{1A} = 15$ ksi and $\sigma_{2A} = -80$ ksi. The material is Grade 40A gray cast iron having $s_{ut} = 40$ ksi and $s_{uc} = -140$ ksi.

1. Draw the modified Mohr diagram as shown in Figure 5–9.
2. Plot point A (15, −80).
3. Draw the load line from the origin through point A until it intersects the failure line on the diagram at the point labeled A_f.
4. Determine the distances $OA = 81.4$ ksi and $OA_f = 96.0$ ksi by scaling the diagram.
5. Compute the design factor from $N = OA_f/OA = 96.0/81.4 = 1.18$.

Summary of Static Loading Failure Theories

It is shown in this section that the selection of static loading failure theories mainly depends on the material (ductile or brittle). The information needed for failure prediction includes the principal stresses and the yield strength or ultimate tensile/compressive strength as shown in the various equations. A summary of the presented failure theories is shown in Table 5–1.

5–5 ENDURANCE LIMIT AND MECHANISMS OF FATIGUE FAILURE

Whenever a machine element is subjected to cyclical loading characterized by the patterns like those shown in Figures 5–2, 5–4, and 5–5, the loading is generally called *fatigue loading,* common to machinery components. Yield strength and ultimate tensile strength of a material are not adequate to represent the ability of materials to resist fatigue loading. This section presents the concept of *endurance limit*, sometimes called *fatigue limit,* that must be used in such cases. Components may fail at stress levels lower than ultimate or yield strengths after experiencing applied stresses for several cycles. Fatigue failures are often classified as either *low-cycle*

TABLE 5–1 Summary of static loading failure theories

	Static Loading Failure Theories	Equations
Ductile Materials	Maximum Shear Stress Theory	(5–6), (5–7), (5–8)
	Distortion Energy Theory	(5–9), (5–10)
Brittle Materials	Maximum Normal Stress Theory	(5–14), (5–15)
	Coulomb-Mohr Theory	(5–16), (5–17), (5–18)
	Modified Mohr Theory	Graphic method

fatigue (LCF) or *high-cycle fatigue* (HCF) because the mechanism of failure is different for each. While no specific dividing line can be defined, designers often use up to 1000 cycles (10^3) for LCF and higher numbers of cycles—up to infinite life—as HCF.

Fatigue failures start at small surface cracks, internal imperfections, or even at grain boundaries in the material in areas subjected to tensile stress. With repeated applications of stress, the cracks grow and progress to larger areas of the cross section. Eventually, the component fails, often suddenly and catastrophically. Such failures frequently occur within areas of stress concentration such as keyways or grooves in shafts, steps in the size of a cross section, notches, or other geometric discontinuities as discussed in Section 3–22. Even surface roughness from machining or accidental nicks and scratches can serve as points of crack initiation. Therefore, designers must consider the possibility of fatigue failure when sizing critical sections of components. Manufacturers must also understand this phenomenon and produce parts with good finishes that are free from damage. End users of critical components must also handle them with care.

In low-cycle fatigue, local stresses experience high strain levels, approaching or exceeding the strain at yield of the material. Such events may be due to accidental overloading, or infrequently encountered situations during fabrication of a component, installation into an assembly, shock during transportation or handling, evasive maneuvers, takeoffs or landings of aircraft, launch of a ship or spacecraft, initial testing, seismic shock during an earthquake, or operating for prolonged periods near the limits of the capability of a system. The high strain may cause microscopic cracks that progress to ultimate failure. Prediction of the life of a component under such conditions falls under the analysis procedure called *fracture mechanics* that requires extensive knowledge of the geometry of the crack and the ability to characterize how a specific material behaves in the highly localized region of high strain and stress around the crack. A life prediction method called *strain-life* is used. References 1, 2, 7, and 9 and Internet sites 1–8 contain extensive detail about these topics. This book will not cover the topic of fracture mechanics encountered in low-cycle fatigue and will concentrate on designing to prevent high-cycle fatigue.

The endurance limit of a material under high-cycle fatigue loading is determined from tests that apply cyclic patterns of stress for long periods of time, and data are obtained for the number of cycles to failure for a given stress level. As expected, higher stress levels produce failure at fewer numbers of cycles and lower stresses permit higher numbers of cycles—up to a point. For many common materials used in machinery, a stress level is reached where a virtually unlimited number of cycles of stress can be applied without fatigue failure. This stress level is called the *endurance limit* or *fatigue limit* of the material. In this book, we use the symbol s_n for this property.

Data for endurance limit are reported on charts such as that shown in Figure 5–10, called a *stress-life diagram*. The vertical axis is the *stress amplitude*, σ_a, as defined by Equation (5–2) and shown in Figure 5–2, and it is assumed that the same amplitude occurs for each cycle for many thousands of cycles. The horizontal axis is the number of cycles to failure, N. Both axes are logarithmic scales, resulting in the data plotting as straight lines. The data are average values of endurance limit through the scattered data points taken from numerous tests at each stress level. The transition from the sloped line to the horizontal line at the fatigue limit for any given material typically occurs at approximately one million cycles (10^6), and the curves shown are idealized, showing the break to be sharp. The following equation represents the sloped portion of the curve:

$$s_a = s_n(N)^b \tag{5–19}$$

where

s_a = stress amplitude level for a given number of cycles to failure

N = number of cycles to failure at a given stress level

s_n = fatigue limit or endurance limit of the material

b = exponent related to the slope of the curve

Data for these properties for many materials are presented in References 1, 2, 7, and 9, and Internet sites 1–8 describe software programs that contain sizable databases of such data. Table 5–2 shows the data for the five

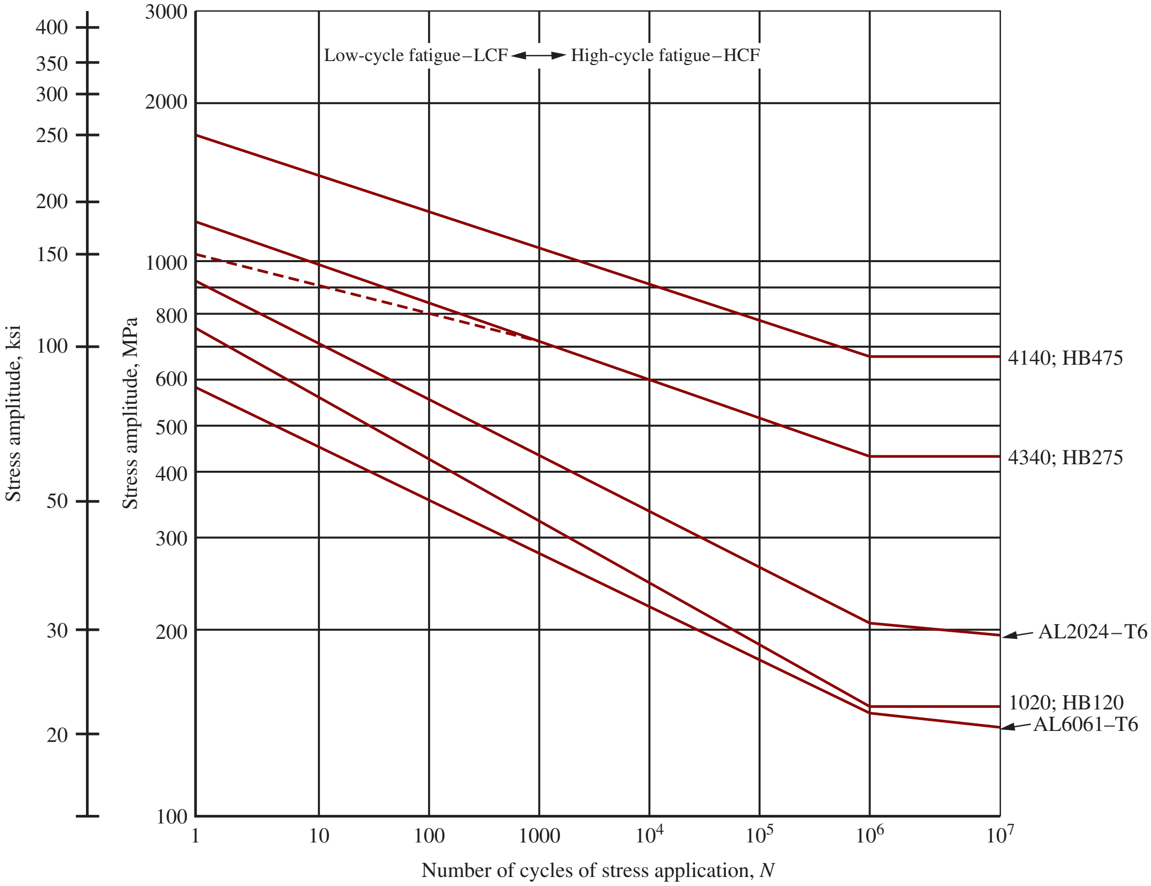


FIGURE 5–10 Representative endurance limits and endurance strengths at lower numbers of cycles

selected materials shown in Figure 5–10 for one plain carbon steel, two alloy steels, and two aluminum alloys, taken from Internet site 1.

The last column for *curve intercept*, s'_f, represents the stress value where the curve intersects the vertical axis. This value has no further use as discussed later for low-cycle fatigue.

Note the difference between the curves for the three steels and those for the two aluminum alloys. The steels exhibit a true fatigue limit resulting in the horizontal line to the right of 10^6 cycles, and should never fail by fatigue at higher numbers of cycles of loading. The curves for aluminum continue to drop after 10^6 cycles, although at a much reduced slope. Therefore, data for endurance limits of aluminum, many other nonferrous metals, and some very-high-strength ferrous metals are quoted as a value of s_n at a stated number of cycles, typically 10^6 or 10^7. For higher numbers of cycles, additional data should be sought.

The rotating bending test, as shown in Figure 5–3, has been used for many years to acquire endurance limit data and much of the reported data are based on this

TABLE 5–2 Materials and Fatigue Property Data for Curves Shown in Figure 5–10

Material	Condition	Ultimate Strength, s_u		Fatigue Limit, s_n		Curve Slope, b	Curve Intercept, s'_f	
		MPa	ksi	MPa	ksi		MPa	ksi
Carbon and alloy steels								
1020	HB120	393	57	142	21	−0.121	754	109
4340	HB275	1048	152	430	62	−0.075	1211	176
4140	HB475	2033	295	663	96	−0.070	1745	253
Aluminum								
6061	T6	310	45	138	20	−0.102	565	82
2024	T6	476	69	205	30	−0.110	938	136

test. The specimen has a small diameter (typically 0.30 in or 7.62 mm) and is highly polished to eliminate any effect of surface texture. The shape and manner of loading produces pure bending with zero shearing stress and no stress concentrations in the central section. The magnitude of the load can be varied to produce a desired stress level and the shaft is rotated until it breaks. The total number of revolutions to failure is recorded. The test produces the classic repeated and reversed stress shown in Figure 5–2 having a mean stress of zero, stress amplitude of s_a, and stress ratio, $R = -1.0$.

In recent times, other methods have gained favor, particularly programmable servo-controlled axial tension testing devices. Test specimens can be loaded in many different patterns, simulating any of the conditions shown in Figures 5–2, 5–4, 5–5, and others. When the load is reversed and repeated, seemingly similar to the rotating bending test, the stress cycle shown in Figure 5–2 is produced, resulting in a mean stress of zero and a stress ratio, $R = -1.0$. However, an important difference in the behavior of the material occurs because the stress distribution created in the specimens is ideally uniform across the entire section. Note that for rotating bending, only the outermost part of the cylindrical specimen sees the maximum stress and the stress decreases linearly to zero at the center of the bar. Fatigue failures are more likely to initiate in regions of high tensile stress. Because in the axial load test all of the material is subjected to the highest stress, reported endurance limit data are typically lower than those for the rotating bending test by approximately 20%. This situation is discussed more in Section 5–6.

The servo-controlled testing devices are also used to evaluate the effect of loading that produces a fluctuating or pulsating cyclical stress pattern with a nonzero mean stress such as those shown in Figures 5–4 and 5–5. The *mean stress effect* is discussed later.

References 4 and 11 include many tables of data for fatigue strengths of materials along with additional detail on the nature of fatigue failures. Data for the endurance limit should be used wherever it is available, either from test results or from reliable published data. However, such data are not always available. Reference 5 suggests the following approximation for the basic rotating bending endurance limit for wrought steel having $s_u \leq 1500$ MPa (220 ksi).

$$\text{Endurance limit} = s_n = 0.50\,(\text{ultimate tensile strength})$$

$$s_n = 0.50 s_u \tag{5–20}$$

Low-cycle Fatigue

That part of Figure 5–10 to the left of the line for 1000 cycles is the low-cycle fatigue region and the discussion above does not apply. In fact, the parts of the lines from $N = 1000$ down to $N = 1$ are not used at all, except for providing a convenient way of drawing the sloped part of the curves—a straight line from the curve intercept to the fatigue limit. Failure at a single cycle ($N = 1$), of course, occurs at the ultimate tensile strength for the material, s_u. Some designers add a supplementary line from $N = 1000$ to $N = 1$, as shown in dashed form for 4340 steel. However, it is recommended that the strain-life technique mentioned earlier be used in this region.

5–6 ESTIMATED ACTUAL ENDURANCE LIMIT, s'_n

If the actual material characteristics or operating conditions for a machine part are different from those for which the basic endurance limit was determined, the fatigue strength must be reduced from the reported value. Some of the factors that decrease the endurance limit are discussed in this section. The discussion relates only to the endurance limit for materials subjected to reversed and repeated bending stress. Cases involving endurance limit in shear are discussed separately in Section 5–11.

We begin by presenting a procedure for estimating the *actual endurance limit*, s'_n, for the material for the part being designed. It involves applying several factors to the basic endurance limit for the material. Additional elaboration on the factors follows.

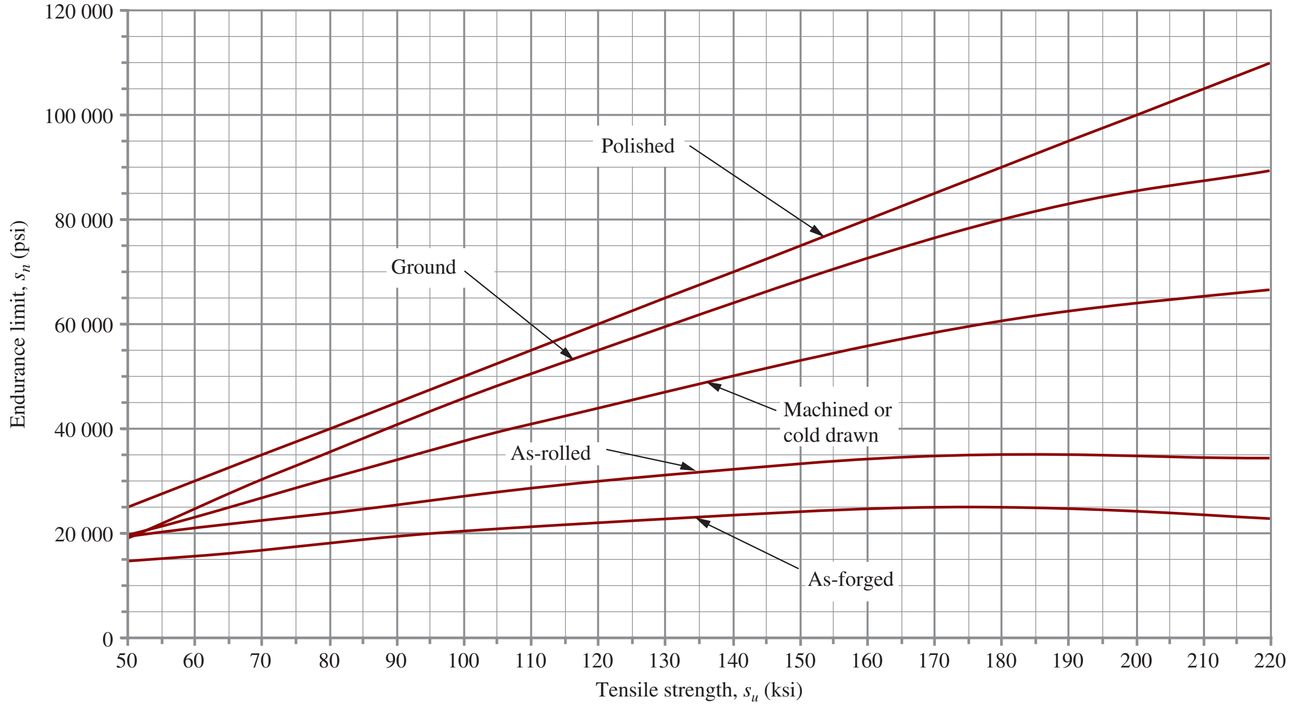
PROCEDURE FOR ESTIMATING ACTUAL ENDURANCE LIMIT, s'_n

1. Specify the material for the part and determine its ultimate tensile strength, s_u, considering its condition, as it will be used in service.
2. Specify the manufacturing process used to produce the part with special attention to the condition of the surface in the most highly stressed area.
3. Use Figure 5–11 to estimate the modified endurance limit, s_n.
4. Apply a material factor, C_m, from the following list.

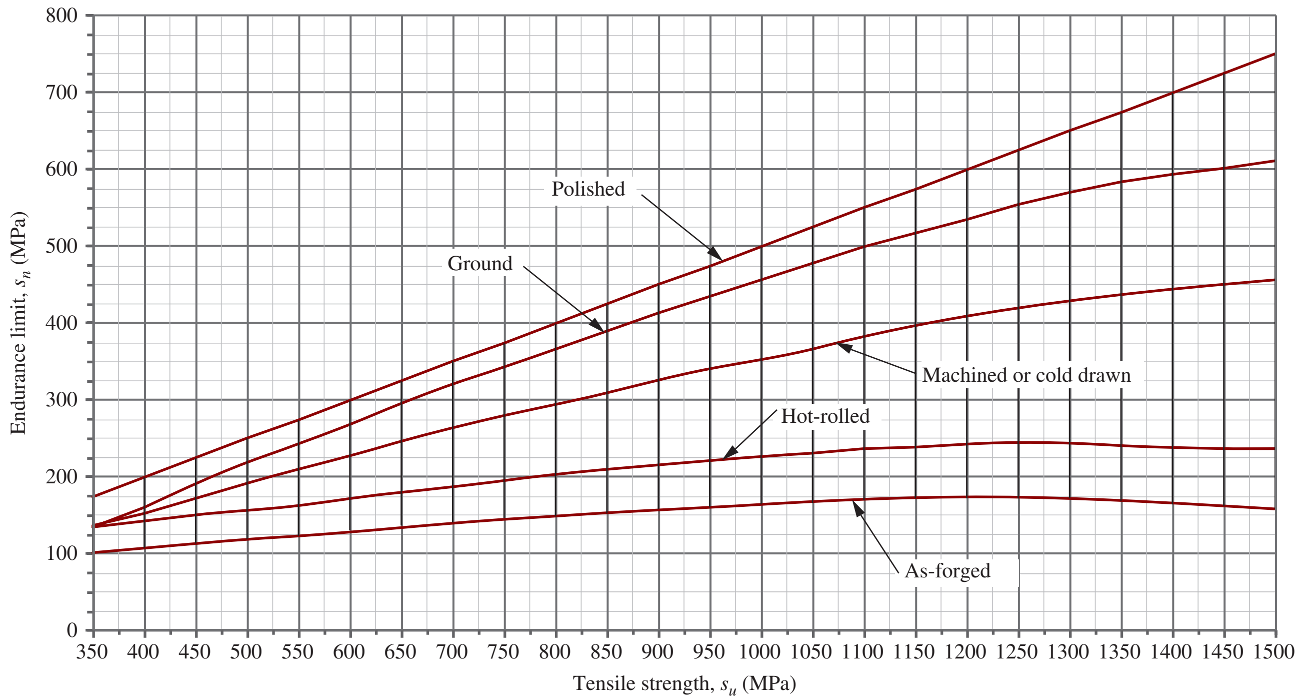
Wrought steel:	$C_m = 1.00$
Cast steel:	$C_m = 0.80$
Powdered steel:	$C_m = 0.76$
Malleable cast iron:	$C_m = 0.80$
Gray cast iron:	$C_m = 0.70$
Ductile cast iron:	$C_m = 0.66$

5. Apply a type-of-stress factor: $C_{st} = 1.0$ for bending stress; $C_{st} = 0.80$ for axial tension.
6. Apply a reliability factor, C_R, from Table 5–3.
7. Apply a size factor, C_s, using Figure 5–12 and Table 5–4 as guides.
8. Compute the estimated actual endurance limit, s'_n, from

$$s'_n = s_n(C_m)(C_{st})(C_R)(C_s) \tag{5–21}$$



(*a*) U.S. customary units



(*b*) SI metric units

FIGURE 5–11 Endurance limit modified for surface condition versus tensile strength for wrought steel

These are the only factors that will be used consistently in this book. If data for other factors can be determined from additional research, they should be multiplied as additional terms in Equation (5–21). In most cases, we suggest accounting for other factors for which reasonable data cannot be found by adjusting the value of the design factor as discussed in Section 5–9.

Stress concentrations caused by sudden changes in geometry are, indeed, likely places for fatigue failures to occur. Care should be taken in the design and manufacture of cyclically loaded parts to keep stress concentration factors to a low value. We will apply stress concentration factors to the computed stress rather than to the endurance limit. See Section 5–11.

While 12 factors affecting endurance limit are discussed in the following section, note that the procedure just given includes only the first five. They are *surface finish*, *material factor*, *type-of-stress factor*, *reliability factor*, and *size factor*. The others are mentioned to alert you to the variety of conditions you should investigate as you complete a design. However, generalized data are difficult to acquire for all factors. Special testing or additional literature searching should be done when conditions exist for which no data are provided in this book. The end-of-chapter references contain a huge amount of such information (see References 2, 4, 7, 9, 11, and 13–16).

Surface Finish

Any deviation from a polished surface reduces endurance limit because the rougher surface provides sites where locally increased stresses or irregularities in the material structure promote the initiation of microscopic cracks that can progress to fatigue failures. Manufacturing processes, corrosion, and careless handling produce detrimental surface roughening.

Figure 5–11, adapted from data in Reference 8, shows estimates for the endurance limit s_n compared with the ultimate tensile strength of wrought steels for several practical surface conditions. The data first estimate the endurance limit for the polished specimen to be 0.50 times the ultimate strength and then apply a factor related to the surface condition. U.S. Customary units are used in Figure 5–11(a) while SI units are shown in Figure 5–11(b). Project vertically from the s_u axis to the appropriate curve and then horizontally to the endurance limit axis.

The data from Figure 5–11 should not be extrapolated for s_u > 220 ksi or 1500 MPa without specific testing as empirical data reported in Reference 4 are inconsistent at higher strength levels.

Note that the curve labeled *Polished* is actually the straight line, $s_n = 0.50s_u$, implying a factor of 1.0 because endurance limit test specimens are polished.

Ground surfaces are fairly smooth and reduce the endurance limit by a factor of approximately 0.90 for $s_u < 160$ ksi (1100 MPa), decreasing to about 0.80 for $s_u = 220$ ksi (1500 MPa). Machining or cold drawing produce a somewhat rougher surface because of tooling marks resulting in a reduction factor in the range of 0.80 to 0.60 over the range of strengths shown. The outer part of a hot-rolled steel has a roughened oxidized scale that produces a reduction factor from 0.72 to 0.30 if a part is used in the as-rolled condition. For a forged part, not subsequently machined, the factor ranges from 0.57 to 0.20.

From these data, it should be obvious that you must give special attention to surface finish for critical surfaces exposed to fatigue loading in order to benefit from the steel's basic strength. Also, critical surfaces of fatigue-loaded parts must be protected from nicks, scratches, and corrosion because they drastically reduce fatigue strength.

Material Factors, C_m

Metal alloys having similar chemical composition can be wrought, cast, or made by powder metallurgy to produce the final form. Wrought materials are usually rolled or drawn, and they typically have higher endurance limit than cast materials. The grain structure of many cast materials or powder metals and the likelihood of internal flaws and inclusions tend to reduce their endurance limit. Reference 5 provides data from which the *material factors* listed in step 4 of the procedure outlined previously are taken.

Type-of-Stress Factor, C_{st}

As discussed in Section 5–6, most endurance limit data are obtained from tests using a rotating cylindrical bar subjected to repeated and reversed bending in which the outer part experiences the highest stress. Stress levels decrease linearly to zero at the center of the bar. Because fatigue cracks usually initiate in regions of high tensile stress, a relatively small proportion of the material experiences such stresses. Contrast this with the case of a bar subjected to direct axial tensile stress for which *all* of the material experiences the maximum stress. There is a higher statistical probability that local flaws anywhere in the bar may start fatigue cracks. The result is that the endurance limit of a material subjected to repeated and reversed axial stress is approximately 80% of that from repeated and reversed bending. In this book, we assume that basic endurance limit data are obtained from rotating bending tests and recommend the factors $C_{st} = 1.0$ for bending stress and $C_{st} = 0.80$ for axial loading.

Reliability Factor, C_R

The data for endurance limit for steel shown in Figure 5–11 represent average values derived from many tests of specimens having the appropriate ultimate strength and surface conditions. Naturally, there is variation among

TABLE 5–3 Approximate Reliability Factors, C_R

Desired reliability	C_R
0.50	1.0
0.90	0.90
0.99	0.81
0.999	0.75

the data points; that is, half are higher and half are lower than the reported values on the given curve. The curve, then, represents a reliability of 50%, indicating that half of the parts would fail. Obviously, it is advisable to design for a higher reliability, say, 90%, 99%, or 99.9%. A factor can be used to estimate a lower endurance limit that can be used for design to produce the higher reliability values. Ideally, a statistical analysis of actual data for the material to be used in the design should be obtained. By making certain assumptions about the form of the distribution of strength data, Reference 8 reports the values in Table 5–3 as approximate reliability factors, C_R.

Size Factor, C_S—Circular Sections in Rotating Bending

Recall that the basic endurance limit data were taken for a specimen with a circular cross section that has a diameter of 0.30 in (7.62 mm) and that it was subjected to repeated and reversed bending while rotating. Therefore, each part of the surface is subjected to the maximum tensile bending stress with each revolution. Furthermore, the most likely place for fatigue failure to initiate is in the zone of maximum tensile stress within a small distance of the outer surface.

Data from References 5 and 8 show that as the diameter of a rotating circular bending specimen increases, the endurance limit decreases because the stress gradient (change in stress as a function of radius) places a greater proportion of the material in the highly stressed region. Figure 5–12 and Table 5–4 show the size factor to be used in this book, adapted from Reference 8. These data can be used for either solid or hollow circular sections.

Size Factor, C_S—Other Conditions

We need different approaches to determining the size factor when a part with a circular section is subjected to repeated and reversed bending but it is *not rotating*, or if the part has a noncircular cross section. Here we show a procedure adapted from Reference 8 that focuses on the volume of the part that experiences 95% or more of the maximum stress. It is in this volume that fatigue failure is most likely to be initiated. Furthermore, in order to relate the physical size of such sections to the

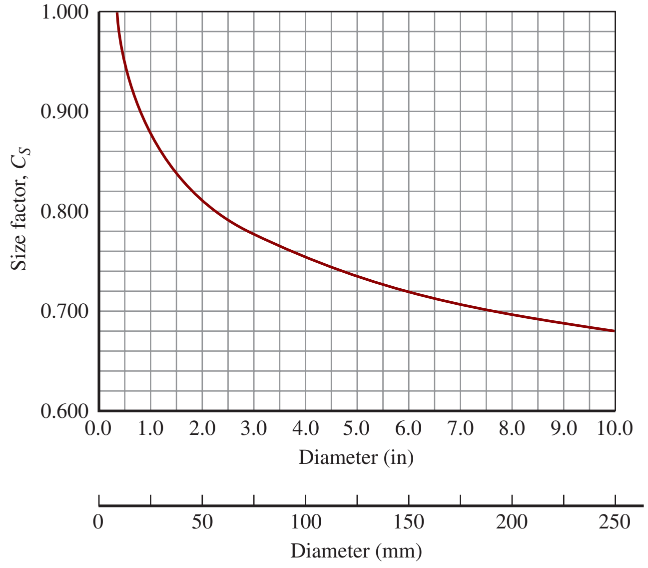


FIGURE 5–12 Size factor, C_s

size factor data in Figure 5–12, we develop an equivalent diameter, D_e.

When the parts in question have a uniform geometry over the length of interest, the volume is the product of the length and the cross-sectional area. We can compare different shapes by considering a unit length for each and looking only at the areas. As a base, let's begin by determining an expression for that part of a circular section subjected to 95% or more of the maximum rotating bending stress, calling this area, A_{95}. Because the stress is directly proportional to the radius, we need the area of the thin ring between the outside surface with the full diameter D and a circle whose diameter is $0.95D$, as shown in Figure 5–13(a). Then,

$$A_{95} = (\pi/4)[D^2 - (0.95D)^2] = 0.0766D^2 \quad \textbf{(5–22)}$$

You should demonstrate that this same equation applies to a hollow circular section as shown in Figure 5–13(b). This verifies that the data for size factor shown in Figure 5–12

TABLE 5–4 Size Factors

Size range	For D in inches
U.S. customary units	
$D \le 0.30$	$C_s = 1.0$
$0.30 < D \le 2.0$	$C_s = (D/0.3)^{-0.11}$
$2.0 < D < 10.0$	$C_s = 0.859 - 0.021\,25D$
SI units	
Size range	**For D in mm**
$D \le 7.62$	$C_s = 1.0$
$7.62 < D \le 50$	$C_s = (D/7.62)^{-0.11}$
$50 < D < 250$	$C_s = 0.859 - 0.000\,837D$

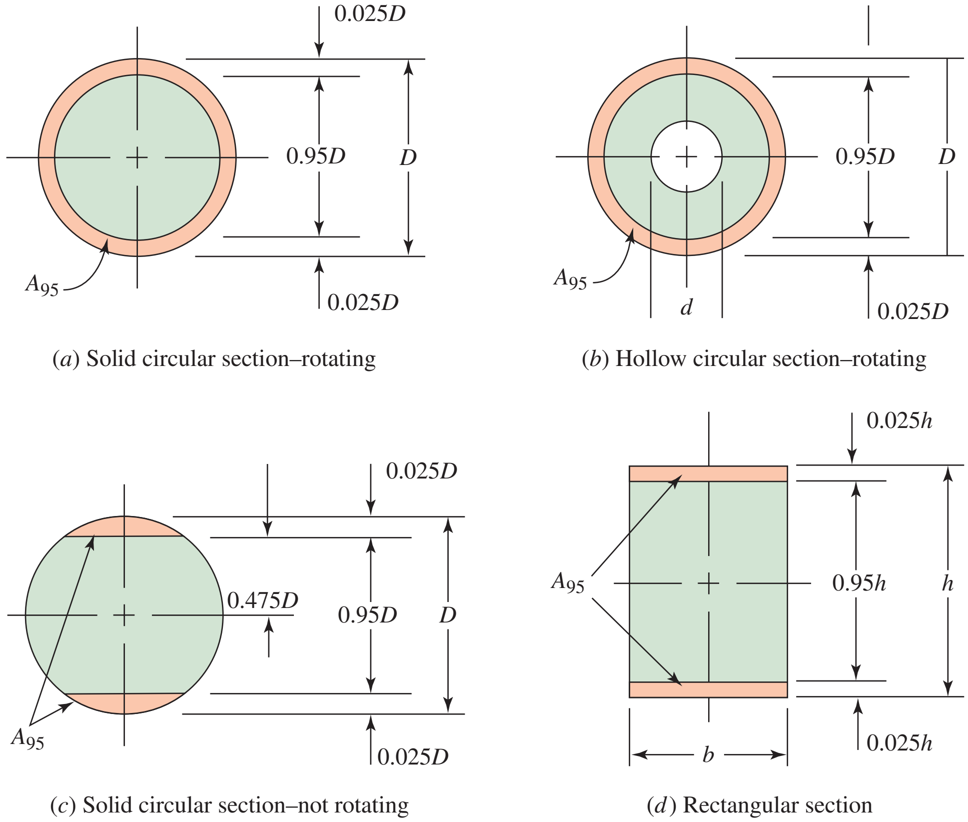


FIGURE 5–13 Geometry of sections for computing A_{95} area

and Table 5–4 apply directly to either the solid or hollow circular sections when they experience rotating bending.

Nonrotating Circular Section in Repeated and Reversed Flexure. Now consider a solid circular section that does not rotate but that is flexed back and forth in repeated and reversed bending. Only the top and bottom segments beyond a radius of $0.475D$ experience 95% or higher of the maximum bending stress, as shown in Figure 5–13(c). By using properties of a segment of a circle, it can be shown that

$$A_{95} = 0.0105D^2 \tag{5–23}$$

Now we can determine the *equivalent diameter*, D_e, for this area by equating Equations (5–22) and (5–23) while designating the diameter in Equation (5–21), as D_e and then solving for D_e.

$$0.0766D_e^2 = 0.0105D^2$$
$$D_e = 0.370D \tag{5–24}$$

This same equation applies to a hollow circular section. The diameter D_e can be used in Figure 5–12 or in Table 5–4 to find the size factor.

Rectangular Section in Repeated and Reversed Flexure. The A_{95} area is shown in Figure 5–13(d) as the two strips having a thickness of $0.025h$ at the top and the bottom of the section. Therefore,

$$A_{95} = 0.05hb$$

Equating this to A_{95} for a circular section gives,

$$0.0766D_e^2 = 0.05hb$$
$$D_e = 0.808\sqrt{hb} \tag{5–25}$$

This diameter can be used in Figure 5–12 or in Table 5–4 to find the size factor.

Other shapes can be analyzed in a similar manner.

Any Shape in Repeated Direct Axial Tensile Stress. This special case is based on the concept that the greatest likelihood of initiating a fatigue failure is in zones of highest tensile stress. For bending and torsion, the highest tensile stress occurs in the outermost parts of the cross section and that is the basis for the data in Figure 5–12. However, for axial tensile stress, all parts of the cross section experience the same level of tensile stress and, therefore, all parts are equally susceptible to initiation of a fatigue crack.

For this case, use $C_s = 1.0$ regardless of the size of the member.

Other Factors

The following factors are not included quantitatively in problem solutions in this book because of the difficulty of finding generalized data. However, you should consider each one as you engage in future designs and seek additional data as appropriate.

Flaws. Internal flaws of the material, especially likely in cast parts, are places in which fatigue cracks initiate. Critical parts can be inspected by x-ray techniques for internal flaws. If they are not inspected, a higher-than-average design factor should be specified for cast parts, and a lower endurance limit should be used.

Temperature. Most materials have a lower endurance limit at high temperatures. The reported values are typically for room temperatures. Operation above 500°F (260°C) will reduce the endurance limit of most steels. See Reference 8.

Nonuniform Material Properties. Many materials have different strength properties in different directions because of the manner in which the material was processed. Rolled sheet or bar products are typically stronger in the direction of rolling than they are in the transverse direction. Fatigue tests are likely to have been run on test bars oriented in the stronger direction. Stressing of such materials in the transverse direction may result in lower endurance limit.

Nonuniform properties are also likely to exist in the vicinity of welds because of incomplete weld penetration, slag inclusions, and variations in the geometry of the part at the weld. Also, welding of heat-treated materials may alter the strength of the material because of local annealing near the weld. Some welding processes may result in the production of residual tensile stresses that decrease the effective endurance limit of the material. Annealing or normalizing after welding is often used to relieve these stresses, but the effect of such treatments on the strength of the base material must be considered.

Residual Stresses. Fatigue failures typically initiate at locations of relatively high tensile stress. Any manufacturing process that tends to produce residual tensile stress will decrease the endurance limit of the component. Welding has already been mentioned as a process that may produce residual tensile stress. Grinding and machining, especially with high material removal rates, also cause undesirable residual tensile stresses. Critical areas of cyclically loaded components should be machined or ground in a gentle fashion.

Processes that produce residual *compressive* stresses can prove to be beneficial. Shot blasting and peening are two such methods. *Shot blasting* is performed by directing a high-velocity stream of hardened balls or pellets at the surface to be treated. *Peening* uses a series of hammer blows on the surface. Crankshafts, springs, gears, and other cyclically loaded machine parts can benefit from these methods.

Corrosion and Environmental Factors. Endurance limit data are typically measured with the specimen in air. Operating conditions that expose a component to water, salt solutions, or other corrosive environments can significantly reduce the effective endurance limit. Corrosion may cause harmful local surface roughness and may also alter the internal grain structure and chemistry of the material. Steels exposed to hydrogen are especially affected adversely.

Nitriding. Nitriding is a surface-hardening process for alloy steels in which the material is heated to 950°F (514°C) in a nitrogen atmosphere, typically ammonia gas, followed by slow cooling. Improvement of endurance limit of 50% or more can be achieved with nitriding.

Effect of Stress Ratio on Endurance Limit. Figure 5–14 shows the general variation of endurance-strength data for a given material when the stress ratio R varies from -1.0 to $+1.0$, covering the range of cases including the following:

- Repeated, reversed stress (Figure 5–2); $R = -1.0$
- Partially reversed fluctuating stress with a tensile mean stress [Figure 5–4(b)]; $-1.0 < R < 0$
- Repeated, one-direction tensile stress (Figure 5–5); $R = 0$
- Fluctuating tensile stress [Figure 5–4(a)]; $0 < R < 1.0$
- Static stress (Figure 5–1); $R = 1$

Note that Figure 5–14 is only an example, and it should not be used to determine actual data points. If such data are desired for a particular material, specific data for that material must be found either experimentally or in published literature.

The most damaging kind of stress among those listed is the repeated, reversed stress with $R = -1$. (See Reference 4.) Recall that the rotating shaft in bending as shown in Figure 5–2 is an example of a load-carrying member subjected to a stress ratio, $R = -1$.

Fluctuating stresses with a compressive mean stress, as shown in Figures 5–4(c) and (d), do not significantly affect the endurance limit of the material because fatigue failures tend to originate in the regions of tensile stress.

Note that the curves of Figure 5–14 show estimates of the endurance limit, s_n, as a function of the ultimate tensile strength for steel. These data apply to ideal polished specimens and do not include any of the other factors discussed in this section. For example, the curve for $R = -1.0$ (reversed bending) shows that the endurance limit for steel is approximately 0.5 times the ultimate strength ($0.50 \times s_u$) for large numbers of cycles of loading (approximately 10^5 or higher). This is a good general estimate for steels. The chart also shows that types of loads producing R greater than -1.0 but less than 1.0 have less of an effect on the endurance limit. This illustrates that using data from the reversed bending test is the most conservative.

We will not use Figure 5–14 directly for problems in this book because our procedure for

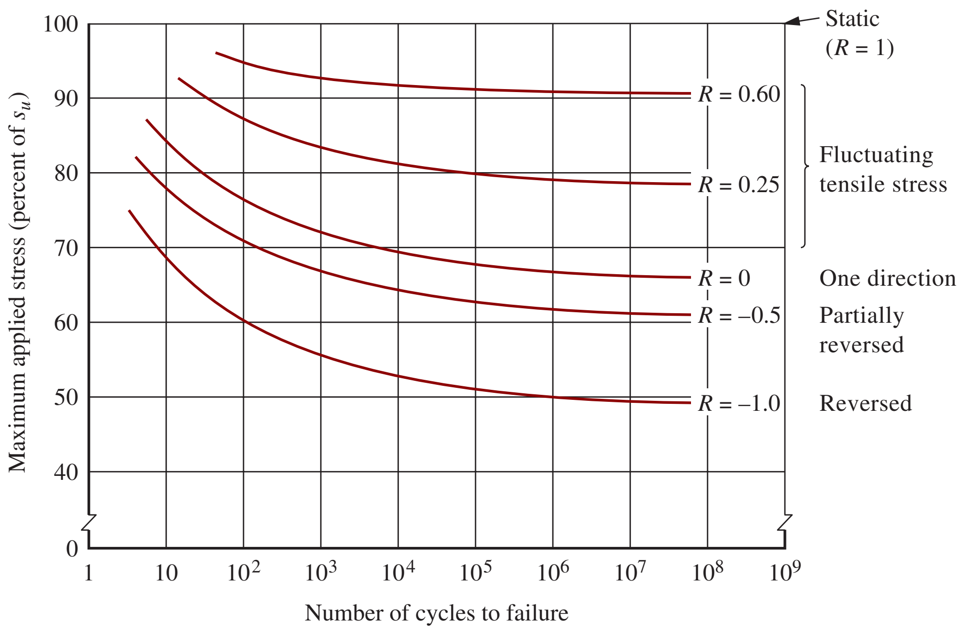


FIGURE 5–14 Effect of stress ratio, R, on endurance limit of a material

estimating the actual endurance limit starts with the use of Figure 5–11, which presents data from reversed bending tests. Therefore, the effect of stress ratio is already included. Section 5–7 includes methods of analysis for loading cases in which the fluctuating stress produces a stress ratio different from $R = -1.0$.

Example Problems for Estimating Actual Endurance Limit

This section shows two examples that demonstrate the application of the *Procedure for Estimating Actual Endurance Limit, s_n'*.

Example Problem 5–2 Estimate the actual endurance limit of SAE 1050 cold-drawn steel when used in a circular shaft subjected to rotating bending only. The shaft will be machined to a diameter of approximately 1.75 in.

Solution

Objective: Compute the estimated actual endurance limit of the shaft material.

Given: SAE 1050 cold-drawn steel, machined.
Size of section: $D = 1.75$ in.
Type of stress: Reversed, repeated bending.

Analysis: Use the Procedure for Estimating Actual Endurance Limit, s_n'.

Step 1: The ultimate tensile strength: $s_u = 100$ ksi from Appendix 3.

Step 2: Diameter is machined.

Step 3: From Figure 5–11, $s_n = 38$ ksi.

Step 4: Material factor for wrought steel: $C_m = 1.0$.

Step 5: Type-of-stress factor for reversed bending: $C_{st} = 1.0$.

Step 6: Specify a desired reliability of 0.99. Then $C_R = 0.81$ (Design decision).

Step 7: Size factor for circular section with $D = 1.75$ in.
From Figure 5–12, $C_s = 0.83$.

Step 8: Use Equation (5–21) to compute the estimated actual endurance limit.

$$s_n' = s_n(C_m)(C_{st})(C_R)(C_s) = 38\text{ ksi}(1.0)(1.0)(0.81)(0.83) = 25.5\text{ ksi}$$

Comments: This is the level of stress that would be expected to produce fatigue failure in a rotating shaft due to the action of reversed bending. It accounts for the basic endurance limit of the wrought SAE 1050 cold-drawn material, the effect of the machined surface, the size of the section, and the desired reliability.

Example Problem 5–3 Estimate the actual endurance limit of cast steel having an ultimate strength of 120 ksi when used in a bar subjected to a reversed, repeated, bending load. The bar will be machined to a rectangular cross section, 1.50 in wide × 2.00 in high.

Solution Objective Compute the estimated actual endurance limit of the bar material.

Given Cast steel, machined: $s_u = 120$ ksi.
Size of section: $b = 1.50$ in., $h = 2.00$ in rectangular
Type of stress: Repeated, reversed bending.

Analysis Use the Procedure for Estimating Actual Endurance Limit, s_n'.

Step 1: The ultimate tensile strength is given to be $s_u = 120$ ksi.

Step 2: Surfaces are machined.

Step 3: From Figure 5–11, $s_n = 44$ ksi.

Step 4: Material factor for cast steel: $C_m = 0.80$.

Step 5: Type-of-stress factor for bending: $C_{st} = 1.00$.

Step 6: Specify a desired reliability of 0.99. Then $C_R = 0.81$ (Design decision).

Step 7: Size factor for rectangular section: First use Equation (5–25) to determine the equivalent diameter,

$$D_e = 0.808\sqrt{hb} = 0.808\sqrt{(2.00\text{ in})(1.50\text{ in})} = 1.40\text{ in}$$

Then from Figure 5–12, $C_s = 0.85$.

Step 8: Use Equation (5–21), to compute the estimated actual endurance limit.

$$s_n' = s_n(C_m)(C_{st})(C_R)(C_s) = 44\text{ ksi}(0.80)(1.00)(0.81)(0.85) = 24.2\text{ ksi}$$

5–7 DESIGN FOR CYCLIC LOADING

Ductile Materials under Cyclic Loading

Similar to the static loading failure theories, prediction of fatigue failure involves comparison of the stresses to the strengths. The stresses required are the mean stress and alternating stress introduced in Section 5–2. The strengths considered are endurance limit and either yield strength or ultimate strength of the material. The concept of fatigue failure theories can be illustrated by introducing the *Yield Line* in the σ_m-σ_a coordinate system as shown in Figure 5–15 where the abscissa is σ_m and

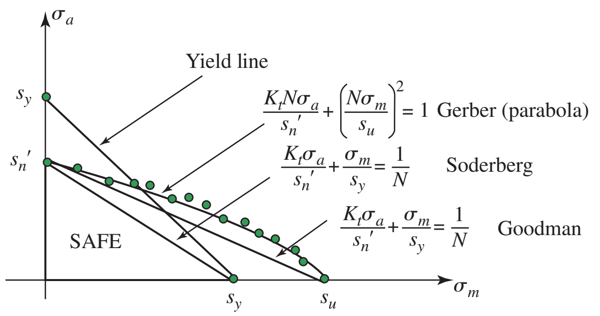


FIGURE 5–15 Fatigue failure theories for cyclic loading

the ordinate is σ_a. The yield strength of the material is identified on both axes. By connecting these two points, the Yield Line is defined and it is represented by the following equation:

$$\frac{\sigma_a}{s_y} + \frac{\sigma_m}{s_y} = 1 \qquad (5\text{–}26)$$

A failure theory can be postulated that, when plotting the mean and alternating stresses of a stress element, the element is safe if the stress state is below the Yield Line in the triangle region. Examining the stress state at $(s_y, 0)$ where the mean stress is at the yield strength and the alternating stress is zero (static loading), the postulated theory seems valid as in static loading the element is considered safe before the stress reaches the yield strength. However, for the stress state at $(0, s_y)$, the element is subjected to repeated and reversed cyclic loading with alternating stress reaching the yield strength. As it is clear that failure would have occurred when the alternating stress exceeds the endurance limit of the material, modification of yield line is needed to predict fatigue failure. Three empirical fatigue failure criteria are presented below and they are shown in Figure 5–15.

Soderberg Criterion. The Soderberg criterion gives a conservative fatigue failure prediction. As shown in

Figure 5–15, the Soderberg line connects s_y on the σ_m axis to s'_n on the σ_a axis. Its equation is

$$\frac{K_t\sigma_a}{s'_n} + \frac{\sigma_m}{s_y} = \frac{1}{N} \tag{5–27}$$

where K_t is the stress concentration factor and N is the factor of safety. For $N = 1$, the safe zone is the triangular area below the Soderberg line. It is recommended that any stress concentration factor be applied to the alternating stress component. However, as it is not necessary to apply a stress concentration factor to the mean stress because the material is ductile and experimental evidence shows that the presence of a stress concentration does not affect the contribution of mean stress to fatigue failure.

Goodman Criterion. The Goodman criterion is perhaps the most-often used analysis for machine components subjected to cyclic loads. The criterion can be expressed by connecting the ultimate strength s_u on the σ_m axis to s'_n on the σ_a axis. Its equation is

$$\frac{K_t\sigma_a}{s'_n} + \frac{\sigma_m}{s_u} = \frac{1}{N} \tag{5–28}$$

For failure prediction, combinations of mean and alternating stresses that lie under the Goodman line are considered to be safe. It can be observed that the main difference between the Goodman and the Soderberg criterion is the stress states near the lower right-hand side of the safe zone where the alternating stress is low and the mean stress is beyond the yield strength of the material. Therefore, the possibility of yielding must be considered as described next.

Checking for Early Cycle Yielding. The Goodman line appears to allow a pure mean stress above the yield strength of the material. While it is possible for ductile material to acquire additional strength to prevent further yielding (due to work hardening), early cycle yielding is not commonly acceptable. As such, an evaluation for early cycle yielding is needed. It is desirable to limit any possible stress to below the yield strength, and any stress concentration should be considered for both the alternating and the mean stress to ensure that even infrequently applied high stresses will not cause damaging strains that could precipitate fatigue cracks. Applying a design factor to the yield line results in a design equation to protect against yield as,

$$\frac{K_t\sigma_a}{s_y} + \frac{K_t\sigma_m}{s_y} = \frac{1}{N} \tag{5–29}$$

Gerber Criterion. The Gerber Criterion is a parabolic function that starts at s_u on the σ_m axis and ends at s'_n on the σ_a axis:

$$\frac{K_t N\sigma_a}{s'_n} + \left(\frac{N\sigma_m}{s_u}\right)^2 = 1 \tag{5–30}$$

As can be observed from Figure 5–15, the Gerber line passes generally through the array of experimental data points (examples of experimental results shown as circles in the figure) for failures under specific combinations of mean and alternating stresses. The starting and end points of the Gerber criterion is the same as that of the Goodman criterion, and it can be seen that the Goodman line is slightly conservative. While the Gerber criterion provides more accurate failure predictions, the parabolic function is somewhat complicated to implement. See References 5 and 8 for more details on the Gerber Criterion.

Smith Diagram for Showing the Effect of Mean Stress on Fatigue

Some stress analysts, for example many in Germany, use the *Smith Diagram* to map a region of acceptable combinations of mean and alternating stresses. Additional materials property data are required to apply this method as compared with the Goodman approach defined earlier and used in this book. Figure 5–16 shows one approach to drawing the Smith Diagram for bending stress. Data required are as follows:

1. Fatigue limit strength for completely reversed and repeated bending stress ($R = -1$), called s_n in this book. Some call this loading pattern *pure oscillation*, $s_{n(-1)}$, as shown in Figure 5–2. The mean stress is zero [$\sigma_m = 0$] and the stress oscillates between $+\sigma_{max}$ and $-\sigma_{min}$ having the same absolute value. The alternating stress, $\sigma_a = \sigma_{max}$.
2. Fatigue limit strength for repeated, one-direction bending stress ($R = 0$), sometimes called *pure pulsating stress*, $s_{n(0)}$, and shown in Figure 5–5. The stress oscillates between the value of σ_{max} and $\sigma_{min} = 0$. The alternating stress and the mean stress are equal: $\sigma_a = \sigma_m = \sigma_{max}/2$.
3. Ultimate strength, s_u.

The steps used to draw the diagram in Figure 5–16 are as follows:

1. Draw the vertical axis for alternating stress, σ_a, and the horizontal axis for mean stress, σ_m.
2. Draw a dashed line at 45° from the origin. This represents the line of increasing mean stress.
3. On the vertical axis, plot two points: Point I at $+\sigma_{max} = s_{n(-1)}$ and Point II at $-\sigma_{min} = -s_{n(-1)}$ for the pure oscillation case.
4. Plot data for the pure pulsating case: Point III at $s_{n(0)}/2$, 0 and Point IV at $s_{n(0)}/2$, $s_{n(0)}$.
5. Draw a line from Point I through Point IV toward its intersection with the 45° line.
6. Draw another line from Point II through Point III toward its intersection with the 45° line.

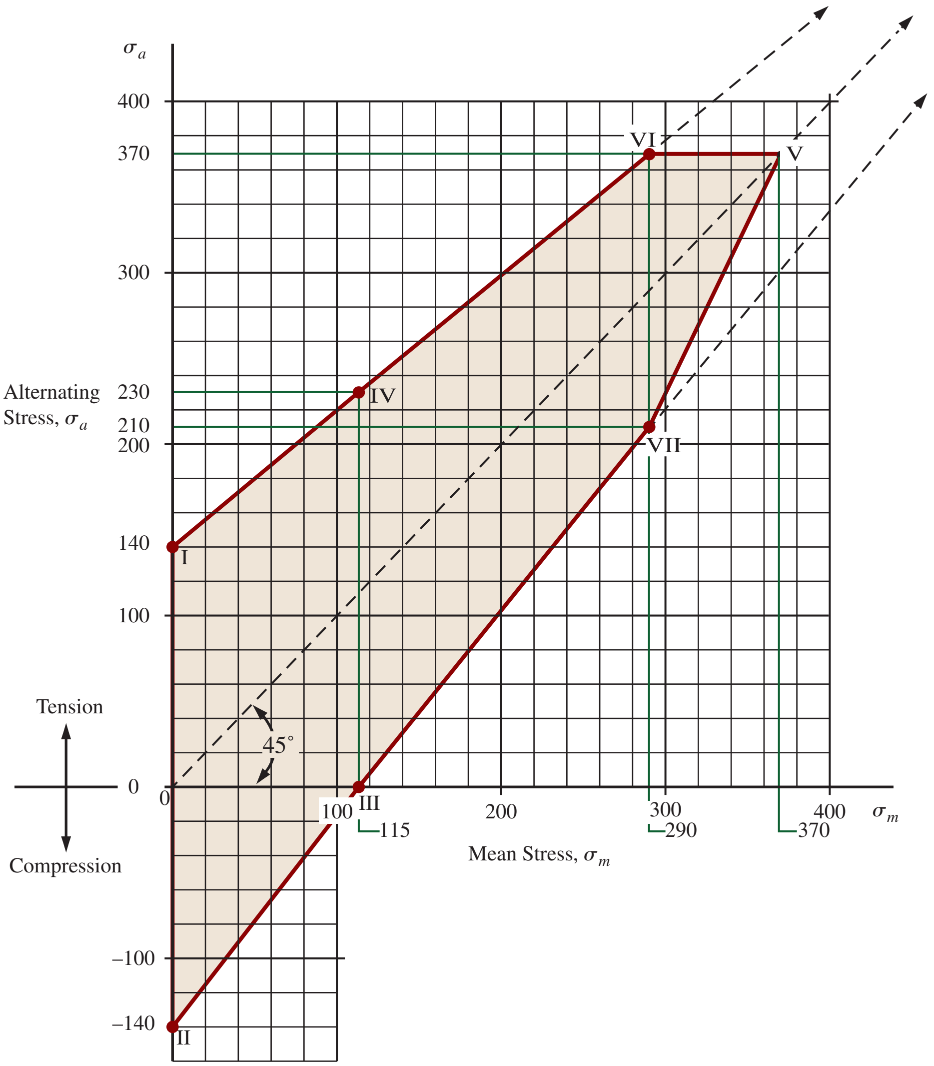


FIGURE 5–16 Example of a Smith diagram for fatigue due to fluctuating bending stress

7. Draw a horizontal line from the value of the ultimate strength on the vertical axis to its intersection with the 45° line, calling that intersection Point V.
8. Call the intersection of the line from Step 7 and the line from Step 5 Point VI.
9. Drop a vertical line from Point VI to its intersection with the line from Step 6, calling that intersection Point VII.
10. Draw a line from Point V to Point VII.
11. The resulting polygon from Points I → VI → V → *VII → II → I* encloses an area of theoretically safe combinations of mean and alternating stresses.
12. The upper part of the polygon from Points I → VI → V is a plot of the maximum stress that can be applied to a component.
13. The lower part of the polygon from Points V → VII → II is a plot of the minimum stress corresponding to any maximum stress directly vertically above.
14. For any vertical line drawn through the upper and lower parts of the polygon:
    a. The top intersection is $+\sigma_{max}$.
    b. The intersection with the dashed line is σ_m.
    c. The lower intersection is σ_{min}.
15. Actual allowable applied stresses, of course, would be lower than these points according to the desired design factor, *N*.

The example data used for drawing Figure 5–16 are as follows:

Ultimate strength = s_u = 370 MPa
Fatigue strength for pure oscillation, $s_{n(-1)}$ = 140 MPa
Fatigue strength for pure pulsation, $s_{n(0)}$ = 230 MPa and $s_{n(0)}/2$ = 115 MPa

Point coordinates:

	Horizontal coordinate	Vertical coordinate	
I	0	140 MPa	[Step 3]
II	0	−140 MPa	[Step 3]
III	115 MPa	0	[Step 4]
IV	115 MPa	230 MPa	[Step 4]
V	370 MPa	370 MPa	[Step 7]
VI	290 MPa (read from diagram)	370 MPa	[Step 8]
VII	290 MPa	210 MPa (read from diagram)	[Step 9]

This method is an approximation of data from fatigue tests from many combinations of mean and alternating stresses. A modification of this method is to limit any applied stress to the yield strength of the material instead of the ultimate strength. Other charts of similar nature are often drawn for torsional shear stresses and for alternating direct tension–compression.

Brittle Materials under Cyclic Loading

In the past, other than cast iron in compression, brittle materials were not commonly used for cyclic loading. Recent work has increased the acceptance of composite and ceramic materials for fatigue applications. For brittle materials, stress is significantly increased at locations of stress concentration and crack tips. It is recommended to apply a stress concentration factor to both mean and alternating stresses, similar to the approach of checking early cycle yielding, Equation (5–29), when using the fatigue failure theory.

Summary of Cyclic Loading Failure Theories

In the three cyclic loading failure theories introduced, the mean and alternating stresses are presented as σ_m and σ_a, indicating one-dimensional loading. The failure criteria can be extended to multi-dimensional loading by replacing σ_m and σ_a with σ'_m and σ'_a:

Soderberg criterion:

$$\frac{K_t\sigma'_a}{s'_n} + \frac{\sigma'_m}{s_y} = \frac{1}{N} \tag{5–31}$$

Goodman criterion:

$$\frac{K_t\sigma'_a}{s'_n} + \frac{\sigma'_m}{s_u} = \frac{1}{N} \tag{5–32}$$

Gerber criterion:

$$\frac{K_tN\sigma'_a}{s'_n} + \left(\frac{N\sigma'_m}{s_u}\right)^2 = 1 \tag{5–33}$$

TABLE 5–5 Summary of Fatigue Failure Theories

Fatigue Failure Criteria	Equations
Soderberg	(5–31)
Goodman	(5–32), (5–34)
Gerber	(5–33), (5–34)
Smith Diagram	Graphic method

Check for early cycle yielding:

$$\frac{K_t\sigma'_a}{s_y} + \frac{K_t\sigma'_m}{s_y} = \frac{1}{N} \tag{5–34}$$

where σ'_m is the mean stress and σ'_a is the alternating stress based on Tresca (MSST) or von Mises (DET) criterion. The calculation of σ'_m and σ'_a is accomplished by drawing two Mohr's circles, one for the mean stresses and one for the alternating stresses. From these circles, determine the three principal stresses. Then compute the "effective stresses" based on MSST or DET for both the mean and the alternating components. Since one-dimensional loading is a special case of three-dimensional loading, Equations (5–27) to (5–30) can be replaced by Equations (5–31) to (5–34). A summary of the presented fatigue failure theories is given in Table 5–5.

5–8 RECOMMENDED DESIGN AND PROCESSING FOR FATIGUE LOADING

Throughout this chapter, we have described factors that influence the fatigue life of components subjected to cyclical loads. This section presents some summary recommendations for designers when specifying size, shape, and processing techniques for a given component. See Reference 15 for additional methods and supporting data.

1. Where critical stress levels are encountered, prepare the surface with low roughness and specify a processing method that will produce well rounded valleys between peaks. Examples are to use a turning tool for cylindrical parts with a broad tip radius or a ball end mill with a large diameter for milling operations.
2. Design parts having inherent abrupt changes in geometry with low values of stress concentrations. An important example is to provide large radii at sites of change in diameter or width of a component. Consult Section 3–22 and Appendix 18.
3. Design the part so that the predominant direction of processing is parallel to the major axis of the part.
4. Increase the fatigue strength of critical areas of a component by using processing methods that leave

compressive residual stresses. Examples include shot peening, cold working by rolling or burnishing.

5. Avoid processing methods that produce tensile residual stresses that decrease the fatigue strength of the material. Examples are (a) aggressive grinding operations that rapidly remove large amounts of material, (b) some heat-treating operations that use rapid quenching from high temperatures, and (c) some welding processes near the heat-affected zone. If such processes are used, supplementary processing is recommended to relieve the tensile residual stress.
6. If processing leaves surface irregularities such as burning, cracks, decarburized materials, or rough welds, take additional steps to remove those irregularities by light finishing cuts, electropolishing, or lapping.
7. Control the entire path of manufacturing processes to ensure that (a) design specifications are met; (b) material properties are within the values used in design analyses; and (c) parts are handled carefully to prohibit accidental surface damage from nicks, scratches, or corrosion.

5–9 DESIGN FACTORS

The term *design factor, N*, is a measure of the relative safety of a load-carrying component. In most cases, the strength of the material from which the component is to be made is divided by the design factor to determine a *design stress*, σ_d, sometimes called the *allowable stress*. Then the actual stress to which the component is subjected should be less than the design stress. For some kinds of loading, it is more convenient to set up a relationship from which the design factor, N, can be computed from the actual applied stresses and the strength of the material. Still in other cases, particularly for the case of the buckling of columns, as discussed in Chapter 6, the design factor is applied to the *load* on the column rather than the strength of the material.

Sections 5–4 and 5–7 present methods for computing the design stress or design factor for several different kinds of loading and materials.

The designer must determine what a reasonable value for the design factor should be in any given situation. Often the value of the design factor or the design stress is governed by codes established by standards-setting organizations such as the American Society of Mechanical Engineers, the American Gear Manufacturers Association, the U.S. Department of Defense, the Aluminum Association, or the American Institute of Steel Construction. For structures, local or state building codes often prescribe design factors or design stresses. Some companies have adopted their own policies specifying design factors based on past experience with similar conditions.

In the absence of codes or standards, the designer must use judgment to specify the desired design factor. Part of the design philosophy, discussed in Section 5–10, identifies issues such as the nature of the application, environment, nature of the loads on the component to be designed, stress analysis, material properties, and the degree of confidence in data used in the design processes. All of these considerations affect the decision about what value for the design factor is appropriate. This book will use the following guidelines.

Ductile Materials

1. **N = 1.25 to 2.0.** Design of structures under static loads for which there is a high level of confidence in all design data.
2. **N = 2.0 to 2.5.** Design of machine elements under dynamic loading with average confidence in all design data. (Typically used in problem solutions in this book.)
3. **N = 2.5 to 4.0.** Design of static structures or machine elements under dynamic loading with uncertainty about loads, material properties, stress analysis, or the environment.
4. **N = 4.0 or higher.** Design of static structures or machine elements under dynamic loading with uncertainty about some combination of loads, material properties, stress analysis, or the environment. The desire to provide extra safety to critical components may also justify these values.

Brittle Materials

5. **N = 3.0 to 4.0.** Design of structures under static loads for which there is a high level of confidence in all design data.
6. **N = 4.0 to 8.0.** Design of static structures or machine elements under dynamic loading with uncertainty about loads, material properties, stress analysis, or the environment.

Sections 5–11 and 5–12 provide guidance on the introduction of the design factor into the design process with particular attention to the selection of the strength basis for the design and the computation of the design stress. In general, design for static loading involves applying the design factor to the yield strength or ultimate strength of the material. Dynamic loading requires the application of the design factor to the endurance limit using the methods described in Section 5–6 to estimate the actual endurance limit for the conditions under which the component is operating.

5–10 DESIGN PHILOSOPHY

It is the designer's responsibility to ensure that a machine part is safe for operation under reasonably foreseeable conditions. You should evaluate carefully the application in which the component is to be used, the environment in which it will operate, the nature of applied loads, the types of stresses to which the component will be exposed, the type of material to be used, and the degree of confidence you have in your knowledge about the application. Some general considerations are as follows:

1. ***Application.*** Is the component to be produced in large or small quantities? What manufacturing techniques will be used to make the component? What are the consequences of failure in terms of danger to people and economic cost? How cost-sensitive is the design? Are small physical size or low weight important? With what other parts or devices will the component interface? For what life is the component being designed? Will the component be inspected and serviced periodically? How much time and expense for the design effort can be justified?
2. ***Environment.*** To what temperature range will the component be exposed? Will the component be exposed to electrical voltage or current? What is the potential for corrosion? Will the component be inside a housing? Will guarding protect access to the component? Is low noise important? What is the vibration environment?
3. ***Loads.*** Identify the nature of loads applied to the component being designed in as much detail as practical. Consider all modes of operation, including startup, shut down, normal operation, and foreseeable overloads. The loads should be characterized as *static, repeated and reversed, fluctuating, shock*, or *impact* as discussed in Section 5–2. Key magnitudes of loads are the *maximum, minimum*, and *mean*. Variations of loads over time should be documented as completely as practical. Will high mean loads be applied for extended periods of time, particularly at high temperatures, for which creep must be considered? This information will influence the details of the design process.
4. ***Types of Stresses.*** Considering the nature of the loads and the manner of supporting the component, what kinds of stresses will be created: direct tension, direct compression, direct shear, bending, or torsional shear? Will two or more kinds of stresses be applied simultaneously? Are stresses developed in one direction (*uniaxially*), two directions (*biaxially*), or three directions (*triaxially*)? Is buckling likely to occur?
5. ***Material.*** Consider the required material properties of yield strength, ultimate tensile strength, ultimate compressive strength, endurance limit, stiffness, ductility, toughness, creep resistance, corrosion resistance, and others in relation to the application, loads, stresses, and the environment. Will the component be made from a ferrous metal such as plain carbon, alloy, stainless, or structural steel, or cast iron? Or will a nonferrous metal such as aluminum, brass, bronze, titanium, magnesium, or zinc be used? Is the material brittle (percent elongation $< 5\%$) or ductile (percent elongation $> 5\%$)? Ductile materials are highly preferred for components subjected to fatigue, shock, or impact loads. Will plastics be used? Is the application suitable for a composite material? Should you consider other nonmetals such as ceramics or wood? Are thermal or electrical properties of the material important?
6. ***Confidence.*** How reliable are the data for loads, material properties, and stress calculations? Are controls for manufacturing processes adequate to ensure that the component will be produced as designed with regard to dimensional accuracy, surface finish, and final as-made material properties? Will subsequent handling, use, or environmental exposure create damage that can affect the safety or life of the component? These considerations will affect your decision for the design factor, N, to be discussed in the next section.

All design approaches must define the relationship between the applied stresses on a component and the strength of the material from which it is to be made, considering the conditions of service. The strength basis for design can be yield strength in tension, compression, or shear; ultimate strength in tension, compression, or shear; endurance limit; or some combination of these. The goal of the design process is to achieve a suitable *design factor*, N (sometimes called a *factor of safety*) that ensures the component is safe. That is, the strength of the material must be greater than the applied stresses. Design factors are discussed in the next section.

The sequence of design analysis will be different depending on what has already been specified and what is left to be determined. For example,

1. ***Geometry of the component and the loading are known:*** We apply the desired design factor, N, to the actual expected stress to determine the required strength of the material. Then a suitable material can be specified.
2. ***Loading is known and the material for the component has been specified:*** We compute a *design stress* by applying the desired design factor, N, to the appropriate strength of the material. This is the maximum allowable stress to which any part of the component can be exposed. We can then complete the stress analysis to determine what shape and size of the component will ensure that stresses are safe.
3. ***Loading is known, and the material and the complete geometry of the component have been specified:*** We compute both the expected maximum applied stress and the design stress. By comparing these stresses, we can determine the resulting design factor, N, for the proposed design and judge its acceptability. A redesign may be called for if the design factor is either too low (unsafe) or too high (over designed).

Practical Considerations. While ensuring that a component is safe, the designer is expected to also make the design practical to produce, considering several factors.

- Each design decision should be tested against the cost of achieving it.
- Material availability must be checked.
- Manufacturing considerations may affect final specifications for overall geometry, dimensions, tolerances, or surface finish.
- In general, components should be as small as practical unless operating conditions call for larger size or weight.
- After computing the minimum acceptable dimension for a feature of a component, standard or preferred sizes should be specified using normal company practice or tables of preferred sizes such as those listed in Appendix 2.
- Before a design is committed to production, tolerances on all dimensions and acceptable surface finishes must be specified so the manufacturing engineer and the production technician can specify suitable manufacturing processes.
- Surface finishes should only be as smooth as required for the function of a particular area of a component, considering appearance, effects on fatigue strength, and whether or not the area mates with another component. Producing smoother surfaces increases cost dramatically. See Chapter 13.
- Tolerances should be as large as possible while maintaining acceptable performance of the component. The cost to produce smaller tolerances rises dramatically. See Chapter 13.
- The final dimensions and tolerances for some features may be affected by the need to mate with other components. Proper clearances and fits must be defined, as discussed in Chapter 13. Another example is the mounting of a commercially available bearing on a shaft for which the bearing manufacturer specifies the nominal size and tolerances for the bearing seat on the shaft. Chapter 16 gives guidelines for clearances between the moving and stationary parts where either boundary or hydrodynamic lubrication is used.
- Will any feature of the component be subsequently painted or plated, affecting the final dimensions?

Deformations. Machine elements can also fail because of excessive deformation or vibration. From your study of strength of materials, you should be able to compute deformations due to axial tensile or compressive loads, bending, torsion, or changes in temperature. Some of the basic concepts are reviewed in Chapter 3. For more complex shapes or loading patterns, computer-based analysis techniques such as finite-element analysis or beam analysis software are important aids.

Criteria for failure due to deformation are often highly dependent on the machine's use. Will excessive deformation cause two or more members to touch when they should not? Will the desired precision of the machine be compromised? Will the part look or feel too flexible (flimsy)? Will parts vibrate excessively or resonate at the frequencies experienced during operation? Will rotating shafts exhibit a critical speed during operation, resulting in wild oscillations of parts carried by the shaft?

This chapter will not pursue the quantitative analysis of deformation, leaving that to be your responsibility as the design of a machine evolves. Later chapters do address some critical cases such as the interference fit between two mating parts (Chapter 13), the position of the teeth of one gear relative to its mating gear (Chapter 9), the radial clearance between a journal bearing and the shaft rotating within it (Chapter 16), and the deformation of springs (Chapter 18). Also, Section 5–11, as a part of the general design procedure, suggests some guidelines for limiting deflections.

5–11 GENERAL DESIGN PROCEDURE

The earlier parts of this chapter have provided guidance related to the many factors involved in design of machine elements that must be safe when carrying the applied loads. This section brings these factors together so that you can complete the design. The general design procedure described here is meant to give you a feel for the process. It is not practical to provide a completely general procedure. You will have to adapt it to the specific situations that you encounter.

The procedure is set up assuming that the following factors are known or can be specified or estimated:

- General design requirements: Objectives and limitations on size, shape, weight, desired precision, and so forth
- Nature of the loads to be carried
- Types of stresses produced by the loads
- Type of material from which the element is to be made
- General description of the manufacturing process to be used, particularly with regard to the surface finish that will be produced
- Desired reliability

GENERAL DESIGN PROCEDURE

1. Specify the objectives and limitations, if any, of the design, including desired life, size, shape, and appearance.
2. Determine the environment in which the element will be placed, considering such factors as corrosion potential and temperature.

3. Determine the nature and characteristics of the loads to be carried by the element, such as
   - Static, dead, slowly applied loads.
   - Dynamic, live, varying, repeated loads that may potentially cause fatigue failure.
   - Shock or impact loads.
4. Determine the magnitudes for the loads and the operating conditions, such as
   - Maximum expected load.
   - Minimum expected load.
   - Mean and alternating levels for fluctuating loads.
   - Frequency of load application and repetition.
   - Expected number of cycles of loading.
5. Analyze how loads are to be applied to determine the type of stresses produced, such as direct normal stress, bending stress, direct shear stress, torsional shear stress, or some combination of stresses.
6. Propose the basic geometry for the element, paying particular attention to
   - Its ability to carry the applied loads safely.
   - Its ability to transmit loads to appropriate support points. Consider the *load paths*.
   - The use of efficient shapes according to the nature of the loads and the types of stresses encountered. This applies to the general shape of the element and to each of its cross sections. Achieving efficiency involves optimizing the amount and the type of material involved. Section 20–2 gives some suggestions for efficient design of frames and members in bending and torsion.
   - Providing appropriate attachments to supports and to other elements in the machine or structure.
   - Providing for the positive location of other components that may be installed on the element being designed. This may call for shoulders, grooves, holes, retaining rings, keys and keyseats, pins, or other means of fastening or holding parts.
7. Propose the method of manufacturing the element with particular attention to the precision required for various features and the surface finish that is desired. Will it be cast, machined, ground, or polished, or produced by some other process? These design decisions have important impacts on the performance of the element, its ability to withstand fatigue loading, and the cost to produce it.
8. Specify the material from which the element is to be made, along with its condition. For metals the specific alloy should be specified, and the condition could include such processing factors as hot rolling, cold drawing, and a specific heat treatment. For nonmetals, it is often necessary to consult with vendors to specify the composition and the mechanical and physical properties of the desired material. Consult Chapter 2 and Section 20–2 for additional guidance.
9. Determine the expected properties of the selected material, for example
   - Ultimate tensile strength, s_u.
   - Ultimate compressive strength, s_{uc}, if appropriate.
   - Yield strength, s_y.
   - Ductility as represented by percent elongation.
   - Stiffness as represented by modulus of elasticity, E or G.
10. Specify an appropriate design factor, N, for the stress analysis using the guidelines discussed in Section 5–9.
11. Determine which stress analysis method from those outlined in Sections 5–4 and 5–7 applies to the design being completed.
12. Compute the appropriate design stress for use in the stress analysis. If fatigue loading is involved, the actual expected endurance limit of the material should be computed as outlined in Section 5–6. This requires the consideration of the expected size of the section, the type of material to be used, the nature of the stress, and the desired reliability. Because the size of the section is typically unknown at the start of the design process, an estimate must be made to allow the inclusion of a reasonable size factor, C_s. You should check the estimate at the end of the design process to verify that reasonable values were assumed at this stage of the design.
13. Determine the nature of any stress concentrations that may exist in the design at places where geometry changes occur. Stress analysis should be considered at all such places because of the likelihood of localized high tensile stresses that may produce fatigue failure. If the geometry of the element in these areas is known, determine the appropriate stress concentration factor, K_t. If the geometry is not yet known, it is advisable to estimate the expected magnitude of K_t. The estimate must then be checked at the end of the design process.
14. Complete the required stress analyses at all points where the stress may be high and at changes of cross section to determine the minimum acceptable dimensions for critical areas.
15. Specify suitable, convenient dimensions for all features of the element. Many design decisions are required, such as
   - The use of preferred basic sizes as listed in Table A2–1.
   - The size of any part that will be installed on or attached to the element being analyzed. Examples of this are shown in Chapter 12 on shaft design where gears, chain sprockets, bearings, and other elements are to be installed on the shafts. But many machine elements have similar needs to accommodate mating elements.
   - Elements should not be significantly oversized without good reason in order to achieve an efficient overall design.
   - Sometimes the manufacturing process to be used has an effect on the dimensions. For example, a company may have a preferred set of cutting tools for use in producing the elements. Casting, rolling, or molding processes often have limitations on the dimensions of certain features such as the thickness of ribs, radii produced by machining or bending, variation in cross section within different parts of the element, and convenient handling of the element during manufacture.
   - Consideration should be given to the sizes and shapes that are commercially available in the desired

material. This could result in significant cost reductions both in material and in processing.

- Sizes should be compatible with standard company practices if practical.

16. After completing all necessary stress analyses and proposing the basic sizes for all features, check all assumptions made earlier in the design to ensure that the element is still safe and reasonably efficient. See Steps 7, 12, and 13.
17. Specify suitable tolerances for all dimensions, considering the performance of the element, its fit with mating elements, the capability of the manufacturing process, and cost. Chapter 13 may be consulted. The use of computer–based tolerance-analysis techniques may be appropriate.
18. Check to determine whether some part of the component may deflect excessively. If that is an issue, complete an analysis of the deflection of the element as designed to this point. Sometimes there are known limits for deflection based on the operation of the machine of which the element being designed is a part. In the absence of such limits, the following guidelines may be applied based on the degree of precision desired:

Deflection of a Beam Due to Bending

General machine part:	0.000 5 to 0.003 in/in of beam length
Moderate precision:	0.000 01 to 0.000 5 in/in
High precision:	0.000 001 to 0.000 01 in/in

Deflection (Rotation) Due to Torsion

General machine part:	0.001° to 0.01°/in of length
Moderate precision:	0.000 02° to 0.000 4°/in
High precision:	0.000 001° to 0.000 02°/in

See also Section 20–2 for additional suggestions for efficient design. The results of the deflection analysis may cause you to redesign the component. Typically, when high stiffness and precision are required, deflection, rather than strength, will govern the design.

19. Document the final design with drawings and specifications.
20. Maintain a careful record of the design analyses for future reference. Keep in mind that others may have to consult these records whether or not you are still involved in the project.

5–12 DESIGN EXAMPLES

Example design problems are shown here to give you a feel for the application of the process outlined in Section 5–11. It is not practical to illustrate all possible situations, and you must develop the ability to adapt the design procedure to the specific characteristics of each problem. Also note that there are many possible solutions to any given design problem. The selection of a final solution is the responsibility of you, the designer.

In most design situations, a great deal more information will be available than is given in the problem statements in this book. But, often, you will have to seek out that information. We will make certain assumptions in the examples that allow the design to proceed. In your job, you must ensure that such assumptions are appropriate. The design examples focus on only one or a few of the components of the given systems. In real situations, you must ensure that each design decision is compatible with the totality of the design.

Design Example 5–1 A large electrical transformer is to be suspended from a roof truss of a building. The total weight of the transformer is 32 000 lb. Design the means of support.

Solution

Objective: Design the means of supporting the transformer.

Given: The total load is 32 000 lb. The transformer will be suspended below a roof truss inside a building. The load can be considered to be static. It is assumed that it will be protected from the weather and that temperatures are not expected to be severely cold or hot in the vicinity of the transformer.

Basic Design Decisions: Two straight, cylindrical rods will be used to support the transformer, connecting the top of its casing to the bottom chord of the roof truss. The ends of the rod will be threaded to allow them to be secured by nuts or by threading them into tapped holes. This design example will be concerned only with the rods. It is assumed that appropriate attachment points are available to allow the two rods to share the load equally during service. However, it is possible that only one rod will carry the entire load at some point during installation. Therefore, each rod will be designed to carry the full 32 000 lb.

We will use steel for the rods, and because neither weight nor physical size is critical in this application, a plain, medium-carbon steel will be used. We specify SAE 1040 cold-drawn steel. From Appendix 3, we find that it has a yield strength of 71 ksi and moderately high ductility as represented by its 12% elongation. The rods should be protected from corrosion by appropriate coatings.

The objective of the design analysis that follows is to determine the size of the rod.

Analysis: The rods are to be subjected to static direct normal tensile stress. Assuming that the threads at the ends of the rods are cut or rolled into the nominal diameter of the rods, the critical place for stress analysis is in the threaded portion.

Use the direct tensile stress formula, $\sigma = F/A$. We will first compute the design stress and then compute the required cross-sectional area to maintain the stress in service below that value. Finally, a standard thread will be specified from the data in Appendix Table A2–2(b) for American Standard threads.

Results

In the basic tensile stress equation, $\sigma = F/A$. The stress state of the 3D stress element is $\sigma_1 = \sigma = F/A$, $\sigma_2 = \sigma_3 = 0$.

From Section 5–4, Design for Static Loading, Equation (5–7) applies for MSST, the maximum shear stress theory. Letting $\sigma_1 = F/A = \sigma_d$,

$$\sigma_{\rm d} = \sigma_1 \leq \frac{s_y}{N}$$

From Section 5–9, for the design factor, we can specify $N = 3$, typical for general machine design with some uncertainty about installation procedures. Then,

$$\sigma_d = s_y/N = (71\,000 \text{ psi})/3 = 23\,667 \text{ psi}$$

Now we can solve for the required cross-sectional area of each rod.

$$A = F/\sigma_d = (32\,000 \text{ lb})/(23\,667 \text{ lb/in}^2) = 1.35 \text{ in}^2$$

A standard size thread will now be specified. Appendix Table A2–2(b) lists the tensile stress area for American Standard threads. A $1\frac{1}{2}$–6 UNC thread ($1\frac{1}{2}$-in-diameter rod with 6 threads per in) has a tensile stress area of 1.405 in^2 which should be satisfactory for this application.

Comments

The final design specifies a $1\frac{1}{2}$-in-diameter rod made from SAE 1040 cold-drawn steel with $1\frac{1}{2}$–6 UNC threads machined on each end to allow the attachment of the rods to the transformer and to the truss. Note that for this problem, both the MSST [Equation (5–7)] and the distortion energy theory DET [Equation (5–10)] yield the same result because the loading is uniaxial tension.

Design Example 5–2

A part of a conveyor system for a production operation is shown in Figure 5–17. Design the pin that connects the horizontal bar to the fixture. The empty fixture weighs 85 lb. A cast iron engine block weighing 225 lb is hung on the fixture to carry it from one process to another, where it is then removed. It is expected that the system will experience many thousands of cycles of loading and unloading of the engine blocks.

Solution

Objective

Design the pin for attaching the fixture to the conveyor system.

Given

The general arrangement is shown in Figure 5–17. The fixture places a shearing load that is alternately 85 lb and 310 lb (85 + 225) on the pin many thousands of times in the expected life of the system.

Basic Design Decisions

It is proposed to make the pin from SAE 1020 cold-drawn steel. "Design Properties of Carbon and Alloy Steels" lists $s_y = 51$ ksi and $s_u = 61$ ksi. The steel is ductile with 15% elongation. This material is inexpensive, and it is not necessary to achieve a particularly small size for the pin.

The connection of the fixture to the bar is basically a clevis joint with two tabs at the top of the fixture, one on each side of the bar. There will be a close fit between the tabs and the bar to minimize bending action on the pin. Also, the pin will be a fairly close fit in the holes while still allowing rotation of the fixture relative to the bar.

Analysis

The Goodman criterion, expressed in Equation (5–32) in Section 5–7, applies for completing the design analysis because fluctuating shearing stresses are experienced by the pin. Therefore, we will have to determine relationships for the mean and alternating stresses (τ_m and τ_a) in terms of the applied loads and the cross-sectional area of the bar. Note that the pin is in double shear, so two cross sections resist the applied shearing force. In general, $\tau = F/2A$.

Now we will use the basic forms of Equations (5–1) and (5–2) to compute the values for the mean and alternating forces on the pin:

$$F_m = (F_{\max} + F_{\min})/2 = (310 + 85)/2 = 198 \text{ lb}$$

$$F_a = (F_{\max} - F_{\min})/2 = (310 - 85)/2 = 113 \text{ lb}$$

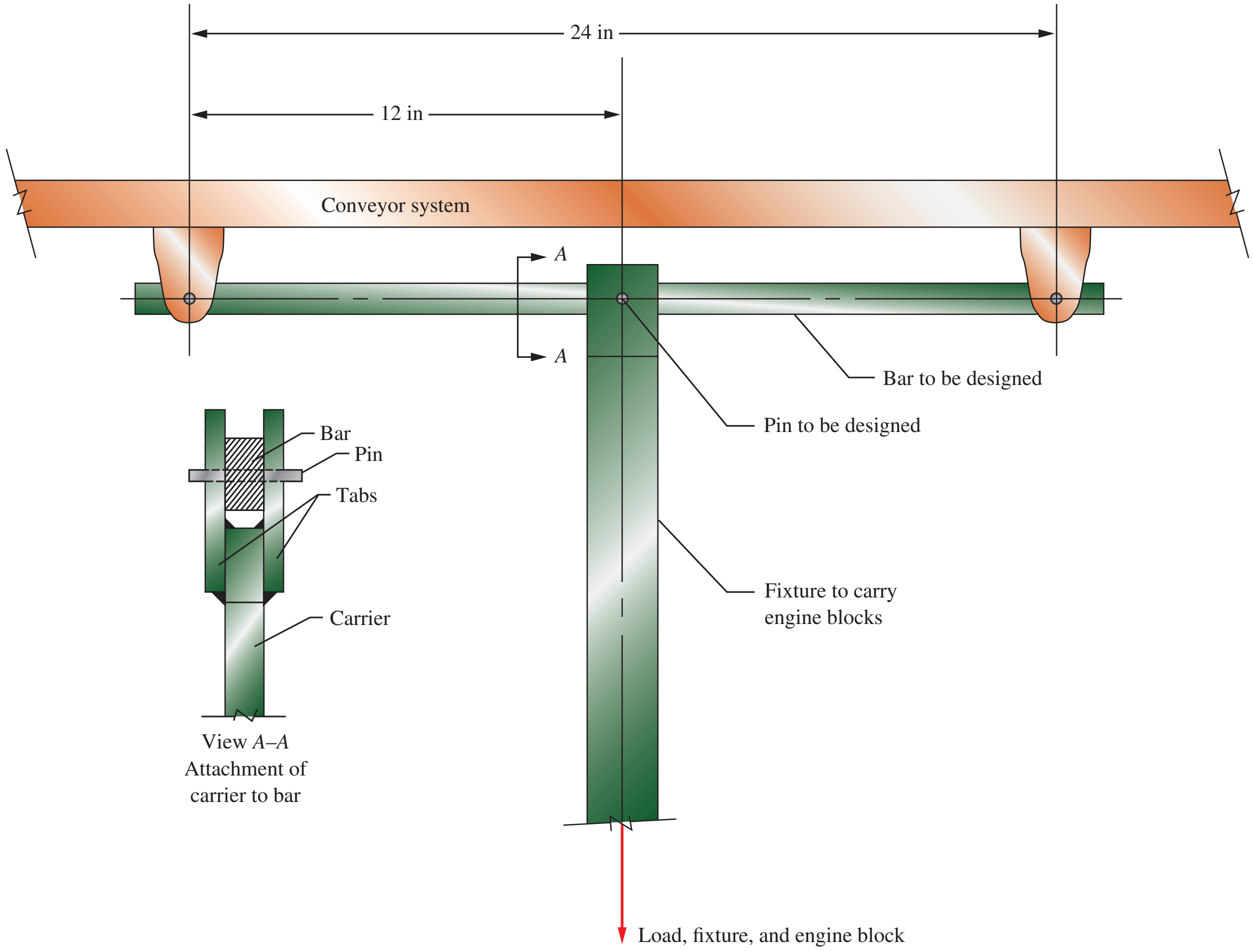


FIGURE 5–17 Conveyor system

The stresses will be found from $\tau_m = \dfrac{F_m}{2A}$ and $\tau_a = \dfrac{F_a}{2A}$.

From stress transformation, the three mean principal stresses are as follows:

$$\sigma_{m1} = \frac{F_m}{2A}, \sigma_{m2} = 0, \sigma_{m3} = -\frac{F_m}{2A}$$

Similarly, the three alternating principal stresses are as follows:

$$\sigma_{a1} = \frac{F_a}{2A}, \sigma_{a2} = 0, \sigma_{a3} = -\frac{F_a}{2A}$$

We can now apply the Goodman criterion [Equation (5–32)]

$$\frac{K_t\sigma'_a}{s_n{}'} + \frac{\sigma'_m}{s_u} = \frac{1}{N}$$

$$\frac{K_t(\sigma_{a1} - \sigma_{a3})}{s_n{}'} + \frac{\sigma_{m1} - \sigma_{m3}}{s_u} = \frac{1}{N}$$

Therefore,

$$\frac{K_tF_a}{As'_n} + \frac{F_m}{As_u} = \frac{1}{N}$$

The material strength values needed are $s_u = 61\ 000$ psi and $s_n{}'$. We must find the value of s'_n using the method from Section 5–6. We find from Figure 5–11 that $s_n = 23$ ksi for the machined pin having a value of $s_u = 61$ ksi. It is expected that the pin will be fairly small, so we will use $C_s = 1.0$. The material is wrought steel rod, so $C_m = 1.0$. Let's use $C_{st} = 1.0$ to be conservative because there is little information about such factors for direct shearing stress. A high reliability is desired for this application, so let's use $C_R = 0.75$ to produce a reliability of 0.999 (see Table 5–3). Then

$$s'_n = C_R(s_n) = (0.75)(23 \text{ ksi}) = 17.25 \text{ ksi} = 17\,250 \text{ psi}$$

Because the pins will be of uniform diameter, $K_t = 1.0$. Let's use $N = 4$ because mild shock is expected. Therefore,

$$\frac{K_t F_a}{A s_n'} + \frac{F_m}{A s_u} = \frac{1}{N}$$

$$\frac{(1.0)(113 \text{ lb})}{A\left(17\,250 \dfrac{\text{lb}}{\text{in}^2}\right)} + \frac{198 \text{ lb}}{A\left(61\,000 \dfrac{\text{lb}}{\text{in}^2}\right)} = \frac{1}{4}$$

Results Based on the above analysis, we can solve for the required area, $A = 0.03919 \text{ in}^2$

Then the required diameter is $D = 0.223$ in.

Final Design Decisions and Comments The computed value for the minimum required diameter for the pin, 0.223 in, is quite small. Other considerations such as bearing stress and wear at the surfaces that contact the tabs of the fixture and the bar indicate that a larger diameter would be preferred. Let's specify $D = 0.50$ in for the pin at this location. The pin will be of uniform diameter within the area of the bar and the tabs. It should extend beyond the tabs, and it could be secured with cotter pins or retaining rings.

This completes the design of the pin. But the next design example deals with the horizontal bar for this same system. There are pins at the conveyor hangers to support the bar. They would also have to be designed. However, note that each of these pins carries only half the load of the pin in the fixture connection. These pins would experience less relative motion as well, so wear should not be so severe. Thus, let's use pins with $D = 3/8 \text{ in} = 0.375$ in at the ends of the horizontal bar.

Design Example 5–3 A part of a conveyor system for a production operation is shown in Figure 5–17. The complete system will include several hundred hanger assemblies like this one. Design the horizontal bar that extends between two adjacent conveyor hangers and that supports a fixture at its midpoint. The empty fixture weighs 85 lb. A cast iron engine block weighing 225 lb is hung on the fixture to carry it from one process to another, where it is then removed. It is expected that the bar will experience several thousand cycles of loading and unloading of the engine blocks. Design Example 5–2 considered this same system with the objective of specifying the diameter of the pins. The pin at the middle of the horizontal bar where the fixture is hung has been specified to have a diameter of 0.50 in. Those at each end where the horizontal bar is connected to the conveyor hangers are each 0.375 in.

Solution Objective Design the horizontal bar for the conveyor system.

Given The general arrangement is shown in Figure 5–17. The bar is simply supported at points 24 in apart. A vertical load that is alternately 85 lb and 310 lb (85 + 225) is applied at the middle of the bar through the pin connecting the fixture to the bar. The load will cycle between these two values many thousands of times in the expected life of the bar. The pin at the middle of the bar has a diameter of 0.50 in, while the pins at each end are 0.375 in.

Basic Design Decisions It is proposed to make the bar from steel in the form of a rectangular bar with the long dimension of its cross section vertical. Cylindrical holes will be machined on the neutral axis of the bar at the support points and at its center to receive cylindrical pins that will attach the bar to the conveyor carriers and to the fixture. Figure 5–18 shows the basic design for the bar.

The thickness of the bar, t, should be fairly large to provide a good bearing surface for the pins and to ensure lateral stability of the bar when subjected to the bending stress. A relatively thin bar would tend to buckle along its top surface where the stress is compressive. As a design decision, we will use a thickness of $t = 0.50$ in. The design analysis will determine the required height of the bar, h, assuming that the primary mode of failure is stress due to bending. Other possible modes of failure are discussed in the comments at the end of this example.

An inexpensive steel is desirable because several hundred bars will be made. We specify SAE 1020 hot-rolled steel having a yield strength of $s_y = 30$ ksi and an ultimate strength of $s_u = 55$ ksi (Appendix 3).

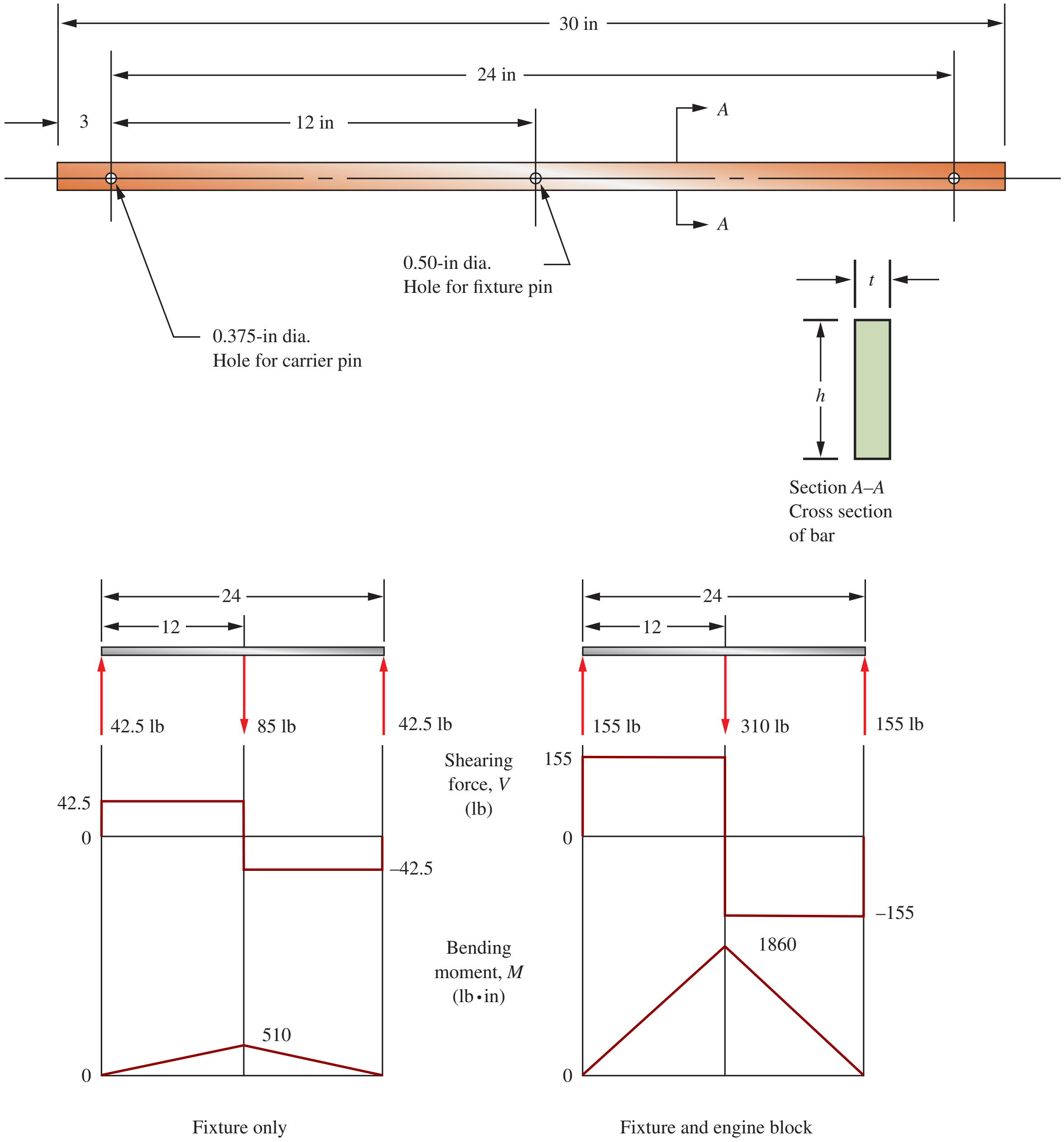


FIGURE 5–18 Basic design of the horizontal bar and the load, shearing force, and bending moment diagrams

Analysis The Goodman criterion applies for completing the design analysis because fluctuating normal stress due to bending is experienced by the bar. Equation (5–32) will be used:

$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t\sigma_a}{s'_n}$$

In general, the bending stress in the bar will be computed from the flexure formula:

$$\sigma = M/S$$

where M = bending moment

S = section modulus of the cross section of the bar = $th^2/6$ (Appendix 1)

Our approach will be to first determine the values for both the mean and the alternating bending moments experienced by the bar at its middle. Then the yield and endurance limit values for the steel will be found. Reference 5 in Chapter 3 indicates that a small hole, with diameter d, in a plate-beam does not weaken the beam if the ratio d/h is less than 0.50. That is, if $d/h < 0.50$, $K_t = 1.0$. We will make that assumption and check it later. Based on the application conditions, let's use $N = 4$ as advised in item 4 in Section 5–9 because the actual use pattern for this conveyor system in a factory environment is somewhat uncertain and shock loading is likely.

Bending Moments Figure 5–18 shows the shearing force and bending moment diagrams for the bar when carrying just the fixture and then both the fixture and the engine block. The maximum bending moment occurs at the middle of the bar where the load is applied. The values are $M_{max} = 1860$ lb·in with the engine block on the fixture and $M_{min} = 510$ lb·in for the fixture alone. Now the values for the mean and alternating bending moments are calculated using modified forms of Equations (5–1) and (5–2):

$$M_m = (M_{max} + M_{min})/2 = (1860 + 510)/2 = 1185 \text{ lb·in}$$

$$M_a = (M_{max} - M_{min})/2 = (1860 - 510)/2 = 675 \text{ lb·in}$$

The stresses will be found from $\sigma_m = \dfrac{M_m}{S}$ and $\sigma_a = \dfrac{M_a}{S}$. Note that the stress element at the location of interest is subjected to uniaxial tension/compression without any shear. For 1D loading.

$$\sigma_{m1} = \sigma_m = \frac{M_m}{S}, \sigma_{m2} = \sigma_{m3} = 0$$

$$\sigma_{a1} = \sigma_a = \frac{M_a}{S}, \sigma_{a2} = \sigma_{a3} = 0$$

As such, we can now apply the Goodman criterion from Equation (5–32) with

$$\sigma'_m = \sigma_m = \frac{M_m}{S}$$

$$\sigma'_a = \sigma_a = \frac{M_a}{S}$$

Material Strength Values The material strength properties required are the ultimate strength s_u and the estimated actual endurance limit s'_n. We know that the ultimate strength $s_u = 55$ ksi. We now find s'_n using the method outlined in Section 5–6.

Size factor, Cs: From Section 5–6, Equation (5–10) defines an equivalent diameter, D_e, for the rectangular section as

$$D_e = 0.808\sqrt{ht}$$

We have specified the thickness of the bar to be $t = 0.50$ in. The height is unknown at this time. As an estimate, let's assume $h \approx 2.0$ in. Then,

$$D_e = 0.808\sqrt{ht} = 0.808\sqrt{(2.0 \text{ in})(0.50 \text{ in})} = 0.808 \text{ in}$$

We can now use Figure 5–12 or the equations in Table 5–4 to find $C_s = 0.90$. This value should be checked later after a specific height dimension is proposed.

Material factor, C_m: Use $C_m = 1.0$ for the wrought, hot-rolled steel.

Stress-type factor, C_{st}: Use $C_{st} = 1.0$ for repeated bending stress.

Reliability factor, C_R: A high reliability is desired. Let's use $C_R = 0.75$ to achieve a reliability of 0.999 as indicated in Table 5–3.

The value of $s_n = 20$ ksi is found from Figure 5–11 for hot-rolled steel having an ultimate strength of 55 ksi.

Now, applying Equation (5–21) from Section 5–6, we have

$$s'_n = (C_m)(C_{st})(C_R)(C_s)s_n = (1.0)(1.0)(0.75)(0.90)(20 \text{ ksi}) = 13.5 \text{ ksi}$$

Solution for the Required Section Modulus At this point, we have specified all factors in Equation (5–32) except the section modulus of the cross section of the bar that is involved in each expression for stress as shown above. We will now solve the equation for the required value of S.

Recall that we showed earlier that $\sigma_m = M_m/S$ and $\sigma_a = M_a/S$. Then

$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t\sigma_a}{s'_n} = \frac{M_m}{Ss_u} + \frac{K_tM_a}{Ss'_n} = \frac{1}{S}\left[\frac{M_m}{s_u} + \frac{K_tM_a}{s'_n}\right]$$

$$S = N\left[\frac{M_m}{s_u} + \frac{K_tM_a}{s'_n}\right] = 4\left[\frac{1185 \text{ lb·in}}{55\,000 \text{ lb/in}^2} + \frac{1.0(675 \text{ lb·in})}{13\,500 \text{ lb/in}^2}\right]$$

$$S = 0.286 \text{ in}^3$$

Results

The required section modulus has been found to be $S = 0.286 \text{ in}^3$. We observed earlier that $S = th^2/6$ for a solid rectangular cross section, and we decided to use this form to find an initial estimate for the required height of the section, h. We have specified $t = 0.50$ in. Then the estimated minimum acceptable value for the height h is

$$h = \sqrt{6S/t} = \sqrt{6(0.286 \text{ in}^3)/(0.50 \text{ in})} = 1.85 \text{ in}$$

The table of preferred basic sizes in the decimal-inch system (Table A2–1) recommends $h = 2.00$ in. We should first check the earlier assumption that the ratio $d/h < 0.50$ at the middle of the bar. The actual ratio is

$$d/h = (0.50 \text{ in})/(2.00 \text{ in}) = 0.25 \text{ (okay)}$$

This indicates that our earlier assumption that $K_t = 1.0$ is correct. Also, our assumed value of $C_s = 0.90$ is correct because the actual height, $h = 2.0$ in, is identical to our assumed value.

We will now compute the actual value for the section modulus of the cross section with the hole in it. See Figure A15–6 in the appendix.

$$S = \frac{t(h^3 - d^3)}{6h} = \frac{(0.50 \text{ in})[(2.00 \text{ in})^3 - (0.50 \text{ in})^3]}{6(2.00 \text{ in})} = 0.328 \text{ in}^3$$

This value is larger than the minimum required value of 0.286 in^3. Therefore, the size of the cross section is satisfactory with regard to stress due to bending.

Final Design Decisions and Comments

In summary, the following are the design decisions for the horizontal bar of the conveyor hanger shown in Figure 5–18.

1. ***Material:*** SAE 1020 hot-rolled steel.
2. ***Size:*** Rectangular cross section. Thickness $t = 0.50$ in; height $h = 2.00$ in.
3. ***Overall design:*** Figure 5–18 shows the basic features of the bar.
4. ***Other considerations:*** Remaining to be specified are the tolerances on the dimensions for the bar and the finishing of its surfaces. The potential for corrosion should be considered and may call for paint, plating, or some other corrosion protection. The size of the cross section can likely be used with the as-received tolerances on thickness and height, but this is somewhat dependent on the design of the fixture that holds the engine block and the conveyor hangers. So the final tolerances will be left open pending later design decisions. The holes in the bar for the pins should be designed to produce a close sliding fit with the pins, and the details of specifying the tolerances on the hole diameters for such a fit are discussed in Chapter 13. See also the discussion on lug joints in Section 3–21.
5. ***Other possible modes of failure:*** The analysis used in this problem assumed that failure would occur due to the bending stresses in the rectangular bar. The dimensions were specified to preclude this from happening. Other possible modes are discussed as follows:

   a. ***Deflection of the bar as an indication of stiffness:*** The type of conveyor system described in this problem should not be expected to have extreme rigidity because moderate deflection of members should not impair its operation. However, if the horizontal bar deflects so much that it appears to be rather flexible, it would be deemed unsuitable. This is a subjective judgment. We can use Case (a) in Table A14–1 to compute the deflection.

   $$y = FL^3/48EI$$

   In this design,

   $$F = 310 \text{ lb} = \text{maximum load on the bar}$$
   $$L = 24.0 \text{ in} = \text{distance between supports}$$
   $$E = 30 \times 10^6 \text{ psi} = \text{modulus of elasticity of steel}$$
   $$I = th^3/12 = \text{moment of inertia of the cross section}$$
   $$I = (0.50 \text{ in})(2.00 \text{ in})^3/12 = 0.333 \text{ in}^4$$

   Then,

   $$y = \frac{(310 \text{ lb})(24.0 \text{ in})^3}{48(30 \times 10^6 \text{ lb/in}^2)(0.333 \text{ in}^4)} = 0.0089 \text{ in}$$

   This value seems satisfactory. In Section 5–11, some guidelines were given for deflection of machine elements. One stated that bending deflections for general machine parts should be limited to the range of 0.0005 to 0.003 in/in of beam length. For the bar in this design, the ratio of y/L can be compared to this range:

$$y/L = (0.0089 \text{ in})/(24.0 \text{ in}) = 0.0004 \text{ in/in of beam length}$$

Therefore, this deflection is well within the recommended range.

b. ***Buckling of the bar:*** When a beam with a tall, thin, rectangular cross section is subjected to bending, it would be possible for the shape to distort due to buckling before the bending stresses would cause failure of the material. This is called *elastic instability*, and a complete discussion is beyond the scope of this book. However, Reference 14 shows a method of computing the critical buckling load for this kind of loading. The pertinent geometrical feature is the ratio of the thickness t of the bar to its height h. It can be shown that the bar as designed will not buckle.

c. ***Bearing stress on the inside surfaces of the holes in the beam:*** Pins transfer loads between the bar and the mating elements in the conveyor system. It is possible that the bearing stress at the pin–hole interface could be large, leading to excessive deformation or wear. Reference 4 in Chapter 3 indicates that the allowable bearing stress for a steel pin in a steel hole is $0.90s_y$.

$$\sigma_{bd} = 0.90s_y = 0.90(30\,000 \text{ psi}) = 27\,000 \text{ psi}$$

The actual bearing stress at the center hole is found using the projected area, $D_p t$.

$$\sigma_b = F/D_p t = (310 \text{ lb})/(0.50 \text{ in})(0.50 \text{ in}) = 1240 \text{ psi}$$

Thus the pin and hole are very safe for bearing.

Design Example 5–4

A bracket is made by welding a rectangular bar to a circular rod, as shown in Figure 5–19. Design the bar and the rod to carry a static load of 250 lb.

Solution

Objective: The design process will be divided into two parts:

1. Design the rectangular bar for the bracket.
2. Design the circular rod for the bracket.

Rectangular Bar

Given: The bracket design is shown in Figure 5–19. The rectangular bar carries a load of 250 lb vertically downward at its end. An effectively fixed support is provided by the weld at its left end where the loads are transferred to the circular rod. The bar acts as a cantilever beam, 12 in long. The design task is to specify the material for the bar and the dimensions of its cross section.

Basic Design Decisions: We will use steel for both parts of the bracket because of its relatively high stiffness, the ease of welding, and the wide range of strengths available. Let's specify SAE 1340 annealed steel having $s_y = 63$ ksi and $s_u = 102$ ksi (Appendix 3). The steel is highly ductile, with a 26% elongation.

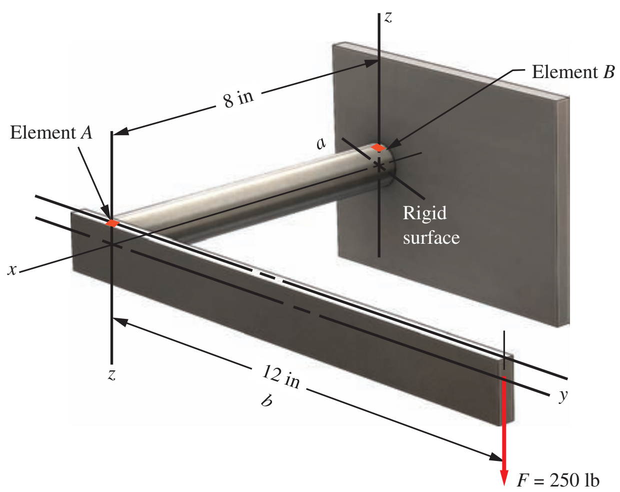


FIGURE 5–19 Bracket design

The objective of the design analysis that follows is to determine the size of the cross section of the rectangular bar. Assuming that the bar acts as a cantilever and the loading and processing conditions are well known, we will use a design factor of $N = 2$ because of the static load.

Analysis and Results

The free-body diagram of the cantilever bar is shown in Figure 5–20, along with the shearing force and bending moment diagrams. This should be a familiar case, leading to the judgment that the maximum tensile stress occurs at the top of the bar near to where it is supported by the circular rod. This point is labeled element A in Figure 5–20. The maximum bending moment there is $M = 3000$ lb·in.

The stresses at A are

$$\sigma_1 = \sigma_A = \frac{M}{S}, \sigma_2 = \sigma_3 = 0$$

where S = section modulus of the cross section of the bar.

We will first compute the minimum allowable value for S and then determine the dimensions for the cross section.

The MSST criterion, Equation (5–7), applies because of the static loading. We will first compute the design stress from

$$\sigma_d = s_y/N$$

$$\sigma_d = s_y/N = (63\,000 \text{ psi})/2 = 31\,500 \text{ psi}$$

Now we must ensure that the expected maximum stress $\sigma_A = M/S$ does not exceed the design stress. We can substitute $\sigma_A = \sigma_d$ and solve for S.

$$S = M/\sigma_d = (3000 \text{ lb}\cdot\text{in})/(31\,500 \text{ lb/in}^2) = 0.095 \text{ in}^3$$

The relationship for S is

$$S = th^2/6$$

As a design decision, let's specify the approximate proportion for the cross-sectional dimensions to be $h = 3t$. Then,

$$S = th^2/6 = t(3t)^2/6 = 9t^3/6 = 1.5t^3$$

The required minimum thickness is then

$$t = \sqrt[3]{S/1.5} = \sqrt[3]{(0.095 \text{ in}^3)/1.5} = 0.399 \text{ in}$$

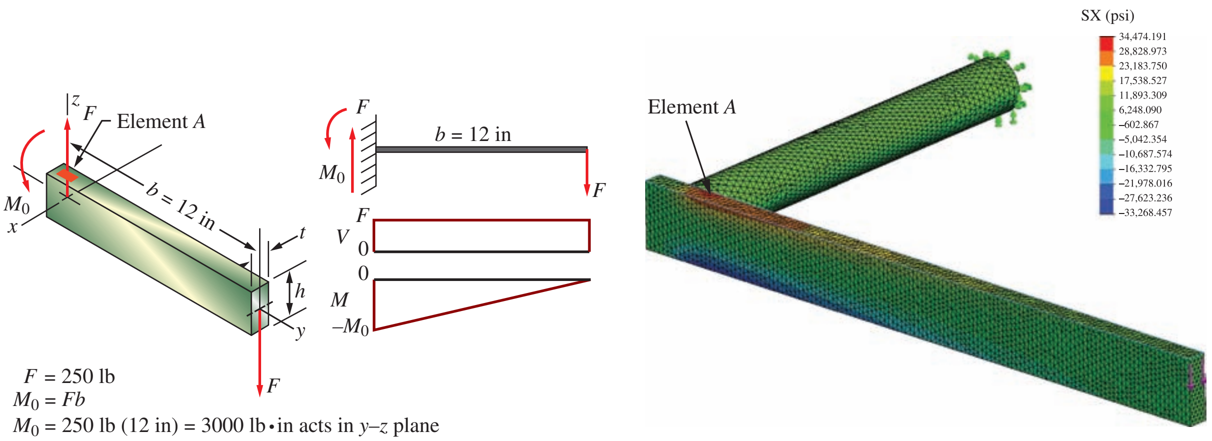
The nominal height of the cross section should be, approximately,

$$h = 3t = 3(0.399 \text{ in}) = 1.20 \text{ in}$$

Final Design Decisions and Comments

In the fractional-inch system, preferred sizes are selected to be $t = 3/8$ in $= 0.375$ in and $h = 1\frac{1}{4}$ in $= 1.25$ in (see Table A2–1). Note that we chose a slightly smaller value for t but a slightly larger value for h. We must check to see that the resulting value for S is satisfactory.

$$S = th^2/6 = (0.375 \text{ in})(1.25 \text{ in})^2/6 = 0.0977 \text{ in}^3$$



(*a*) Free-body diagram of the bar

(*b*) Finite-element model of the bar

FIGURE 5–20 Free-body diagram of bar

This is larger than the required value of 0.095 in^3, so the design is satisfactory.

Part (b) of Figure 5–20 shows a finite-element analysis (FEA) model for the rectangular bar only. The variations in color on the bar represent stress levels with red being the highest and blue the lowest. This model verifies the observation that the highest stress is at Element *A*.

Circular Rod

Given The bracket design is shown in Figure 5–19. The design task is to specify the material for the rod and the diameter of its cross section.

Basic Design Decisions Let's specify SAE 1340 annealed steel, the same as that used for the rectangular bar. Its properties are $s_y = 63$ ksi and $s_u = 102$ ksi.

Analysis and Results Figure 5–21 is the free-body diagram for the rod. The rod is loaded at its left end by the reactions at the end of the rectangular bar, namely, a downward force of 250 lb and a moment of 3000 lb·in. The figure shows that the moment acts as a torque on the circular rod, and the 250-lb force causes bending with a maximum bending moment of 2000 lb·in at the right end. Reactions are provided by the weld at its right end where the loads are transferred to the support. The rod then is subjected to a combined stress due to torsion and bending. Element *B* on the top of the rod is subjected to the maximum combined stress.

The manner of loading on the circular rod is identical to that analyzed earlier. It was shown that when only bending and torsional shear occur, a procedure called the ***equivalent torque method*** can be used to complete the analysis. First we define the equivalent torque, T_e:

$$T_e = \sqrt{M^2 + T^2} = \sqrt{(2000\ \text{lb}\cdot\text{in})^2 + (3000\ \text{lb}\cdot\text{in})^2} = 3606\ \text{lb}\cdot\text{in}$$

Then the shear stress in the bar is

$$\tau = T_e/Z_p$$

where Z_p = polar section modulus

For a solid circular rod,

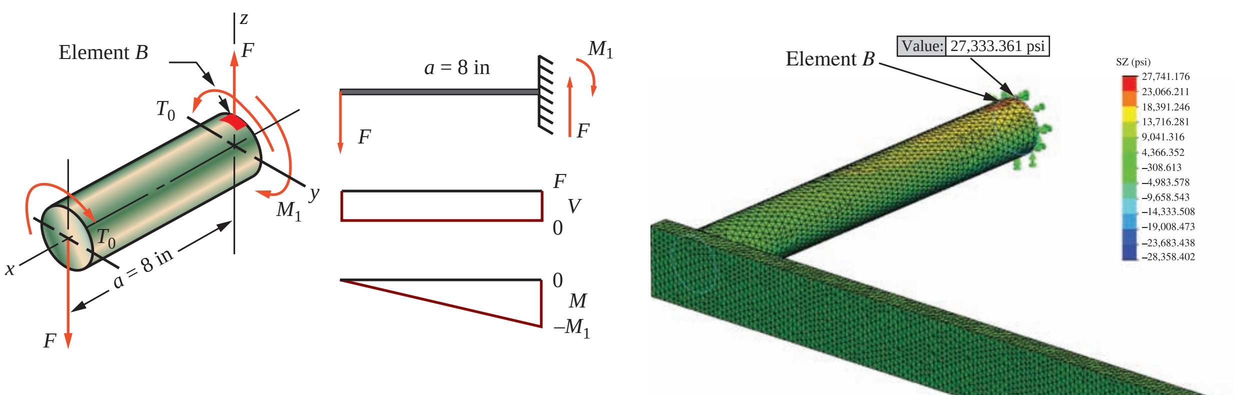
$$Z_p = \pi D^3/16$$

Our approach is to determine the design shear stress and T_e and then solve for Z_p. The maximum shear stress theory of failure can be applied and the design shear stress is

$$\tau_d = 0.50 s_y/N = (0.5)(63\,000\ \text{psi})/2 = 15\,750\ \text{psi}$$

We let $\tau = \tau_d$ and solve for Z_p:

$$Z_p = T_e/\tau_d = (3606\ \text{lb}\cdot\text{in})/(15\,750\ \text{lb/in}^2) = 0.229\ \text{in}^3$$



F = 250 lb

$M_1 = Fa$ = 250 lb(8 in) = 2000 lb·in (acts in y–z plane)

$T_0 = Fb$ = 250 lb(12 in) = 3000 lb·in (see Figure 5–19. $T_0 = M_0$)

(*a*) Free-body diagram of the circular rod

(*b*) Finite-element analysis model for the circular rod

FIGURE 5–21 Free-body diagram of rod

Now that we know Z_p, we can compute the required diameter from

$$D = \sqrt[3]{16Z_p/\pi} = \sqrt[3]{16(0.229\ \text{in}^3)/\pi} = 1.053\ \text{in}$$

This is the minimum acceptable diameter for the rod.

Final Design Decisions and Comments

The circular rod is to be welded to the side of the rectangular bar, and we have specified the height of the bar to be $1\frac{1}{4}$ in. Let's specify the diameter of the circular rod to be machined to 1.10 in. This will allow welding all around its periphery.

Figure 5–21(b) shows a finite-element analysis (FEA) model for the circular bar only. The variations in color on the bar represent stress levels with red being the highest and blue the lowest. This model verifies the observation that the highest stress is at Element *B*.

5–13 STATISTICAL APPROACHES TO DESIGN

The design approaches presented in this chapter are somewhat deterministic in the sense that data are taken to be discrete values, and analyses use the data to determine specific results. The method of accounting for uncertainty with regard to the data themselves lies with the selection of an acceptable value for the design factor represented by the final design decision. Obviously this is a subjective judgment. Often decisions that are made to ensure the safety of a design cause many designs to be quite conservative.

Competitive pressures call for ever more efficient, less conservative designs. Throughout this book, recommendations are made to seek more reliable data for loads, material properties, and environmental factors, providing more confidence in the results of design analyses and allowing lower values for the design factor as discussed in Section 5–9. A more robust and reliable product results from testing samples of the actual material to be used in the product; performing extensive measurements of loads to be experienced; investing in more detailed performance testing, experimental stress analysis, and finite-element analysis; exerting more careful control of manufacturing processes; and life testing of prototype products in realistic conditions where possible. All of these measures typically come with significant additional costs, and difficult decisions must be made about whether or not to implement them.

In combination with the approaches listed previously, a greater use of statistical methods (also called *stochastic methods*) is emerging to account for the inevitable variability of data by determining the mean values of critical parameters from several sets of data and quantifying the variability using the concepts of distributions and standard deviations. References 5 and 11 provide guidance in these methods. Section 2–4 provided a modest discussion of this approach to account for the variability of materials property data.

Industries such as automotive, aerospace, construction equipment, and machine tools devote considerable resources to acquiring useful data for operating conditions that will aid designers in producing more efficient designs.

Samples of Statistical Terminology and Tools

- Statistical methods analyze data to present useful information about the source of the data.
- Stochastic methods apply probability theories to characterize variability in the data.
- Data sets can be analyzed to determine the mean (average), the range of variation, and the standard deviation.
- Inferences can be made about the nature of the distribution of the data, such as normal or lognormal.
- Linear regression and other means of curve fitting can be employed to represent a set of data by mathematical functions.
- The distributions for applied loads and stresses can be compared with the distribution for the strength of the material to determine to what degree they overlap and the probability that a certain number of failures will occur. See Figure 5–22.
- The reliability of a component or a complete product can be quantified.

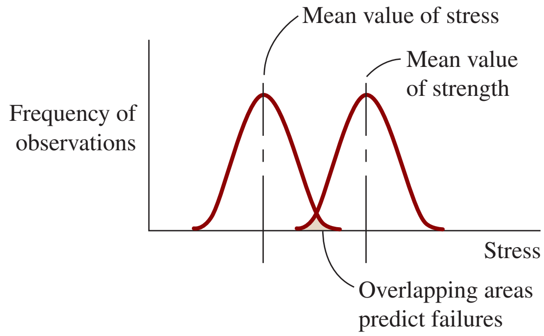


FIGURE 5–22 Illustration of statistical variation in failure potential

- The optimum assignment of tolerances can be made to reasonably ensure satisfactory performance of a product while allowing as broad a range of tolerances as practical.

5–14 FINITE LIFE AND DAMAGE ACCUMULATION METHOD

The fatigue design methods described thus far in this chapter have the goal to design a component for infinite life by using the estimated actual endurance limit as the basis for design. Applied repeated stress levels below this value will provide infinite life. Furthermore, the analyses were based on the assumption that the loading pattern was uniform over the life of the component. That is, the mean and alternating stresses do not vary over time.

There are, however, many common examples for which a finite life is adequate for the application and where the loading pattern does vary with time. Consider the following.

Finite Life Examples

First, let's discuss the concept of finite life. Refer to the endurance strength curves in Figure 5–10 in Section 5–5. Data are plotted on a stress versus number of cycles to failure graph (σ *vs.* N) with both axes using logarithmic scales. For the materials that exhibit an endurance limit, it can be seen that the limit occurs for a life of approximately 10^6 cycles. How long would it take to accumulate 1 million cycles of load applications? Here are some examples.

Bicycle brake lever: Assume that the brake is applied every 5.0 minutes while riding 4.0 hours per day every day of each year. It would take more than 57 years to apply the brake 1 million times.

Lawn mower height-adjustment mechanism: Consider a lawn mower used by a commercial lawn maintenance company. Assume that the cutting height for the mower is adjusted to accommodate varying terrain three times per mowing and that the mower is used 40 times per week for all 12 months per year. It would take 160 years to accumulate 1 million cycles of load on the height adjustment mechanism.

Automotive lift in a service station: Assume that the service technician lifts four automobiles per hour, 10 hours per day, 6 days per week, each week of the year. It would take more than 80 years to accumulate 1 million cycles of load on the lift mechanism.

Each of these examples indicates that it may be appropriate to design the load-carrying members of the example systems for something less than infinite life. But many industrial examples do require design for infinite life. Here is an example.

Parts feeding device: On an automated assembly system, a feeding device inserts 120 parts per minute. If the system operates 16 hours per day, 6 days per week, each week of the year, it would take only 8.7 days to accumulate 1 million cycles of loading. It would see 35.9 million cycles in one year.

When it can be justified to design for a finite life less than the number of cycles corresponding to the endurance limit, you will need data similar to that in Figure 5–10 for the actual material to be used in the component. Testing the material yourself is preferred, but it would be a time-consuming and costly exercise to acquire sufficient data to make statistically valid σ-N curves. References 2–4, 9, 11, and 13 may provide suitable data, or an additional literature search may be required. Once reliable data are identified, use the endurance strength at the specified number of cycles as the starting point for computing the estimated actual endurance strength as described in Section 5–6. Then use that value in subsequent analyses as described in Section 5–10.

Varying Stress Amplitude Examples

Here we are looking for examples where the component experiences cyclical loading for a large number of cycles but for which the amplitude of the stress varies over time.

Bicycle brake lever: Let's reconsider the braking action on a bicycle. Sometimes you need to bring the bike to a stop very quickly from a high speed, requiring a rather high force on the brake lever. Other times you may apply a lighter force to simply slow down a bit to safely negotiate a curve.

Automotive suspension member: Suspension parts such as strut, spring, shock absorber, control arm, or fastener pass loads from the wheel to the frame of a car. The magnitude of the load depends on vehicle speed, the condition of the road, and driver action. Roads may be smoothly paved, potholed, or rough-surfaced gravel. The vehicle may even be driven off-road where violent peaks of stress will be encountered.

Machine tool drive system: Consider the lifetime of a milling machine. Its primary function is to cut metal, and it takes a certain amount of torque to drive the cutter depending on the machinability of the material, the depth of cut, and the feed rate. Surely the torque will vary significantly from job to job. During part of its operating time, there may be no cutting action at all as a new part is positioned or as the operation completes one cut and adjusts before making another. At times the cutter will encounter locally harder material, requiring higher torque for a short period of time.

Cranes, power shovels, bulldozers, and other construction equipment: Here there are obviously varying loads as the equipment is used for numerous tasks, such as hoisting large steel beams or small bracing members, digging through hard clay or soft sandy soil, rapidly grading a hillside or performing the final smoothing of a driveway, or encountering a tree stump or large rock.

How would you determine the loads that these devices would experience over time? One method involves building a prototype and instrumenting critical elements with strain gages, load cells, or accelerometers. Then the system would be "put through its paces" over a wide range of tasks while loads and stresses are recorded as a function of time. Similar vehicles could be monitored to determine the frequency that different kinds of loading would be encountered over its expected life. Combining such data would produce a record from which the total number of cycles of stress at any given level can be estimated. Statistical techniques such as spectrum analysis, Fast Fourier Transform analysis, and time compression analysis produce charts that summarize stress amplitude and frequency data that are useful for fatigue and vibration analysis. See Reference 11 for extended discussion of these techniques.

Damage Accumulation Method

The principle of *damage accumulation* is based on the assumption that any given level of stress applied for one cycle of loading contributes to a certain amount of damage to a component. Refer again to Figure 5–10 and observe the *S–N* curve for the alloy steel, SAE 4340 at a hardness of HB275 and having an ultimate strength of 1048 MPa (152 ksi). Reading a few points from that curve gives the following set of data:

Stress amplitude	Number of cycles to failure
710 MPa (103 ksi)	1.0×10^3
600 MPa (87 ksi)	1.0×10^4
505 MPa (73 ksi)	1.0×10^5
415 MPa (60 ksi)	1.0×10^6 [Considered the endurance limit, s_n]
<415 MPa	∞

There is a theoretically infinite life for stresses below the endurance limit and no damage occurs.

Now, let's say that the component experiences 480 cycles of stress at the 600 MPa (87 ksi) level. It can be said that it suffers a damage of the ratio of $480/(1.0 \times 10^4) = 0.048$. Being subjected to 250 cycles of stress at 710 MPa (103 ksi) produces a damage of $250/(1.0 \times 10^3) = 0.250$. The cumulative damage of both events is the sum, $0.048 + 0.250 = 0.298$. When the sum of these types of calculations is equal to 1.0, it is predicted that the component would fail. Therefore, this brief example predicts that approximately 30% of the component's life has been accumulated.

This kind of logic can be used to predict the total life of a component subjected to a sequence of loading levels. Let n_i represent the number of cycles of a specific stress level experienced by a component. Let N_i be the number of cycles to failure for this stress level as indicated by a σ–N curve such as those shown in Figure 5–10. Then the damage contribution from this loading is

$$D_i = n_i/N_i$$

When several stress levels are experienced for different numbers of cycles, the cumulative damage, D_c, can be represented as,

$$D_c = \sum_{i=1}^{i=k} (n_i/N_i) \tag{5–35}$$

Failure is predicted when $D_c \geq 1.0$. This process is called the *Miner linear cumulative-damage rule* or simply *Miner's rule* in honor of his work in 1945. An example problem will now demonstrate the application of Miner's rule.

The example data in Figure 5–10 and listed above are taken for completely reversed and repeated bending stress on a small [0.30 in (7.62 mm)] diameter circular bar with a mirror polished surface as used in the standard R. R. Moore test device (Figure 5–3). For other conditions (the typical situation), the *S–N* curve can be adjusted by first computing the actual expected endurance limit of the material, s'_n, using Equation 5–21 from Section 5–6 as has been done throughout this chapter. Then a ratio of s'_n/s_n can be applied to stress data from the *S–N* curves before determining the number of cycles to failure. The process is illustrated in Example Problem 5–5.

Example Problem 5–5

Determine the cumulative damage experienced by a ground circular rod, 38 mm diameter, subjected to the combination of the cycles of loading and varying levels of reversed, repeated bending stress shown in Table 5–5.

The bar is made from SAE 4340, HB275 alloy steel with an ultimate strength of 1048 MPa and an endurance limit of 430 MPa. Use Figure 5–10 for the *S–N* curve.

TABLE 5–5 Loading Pattern for Example Problem 5–5

Stress Level		Number of Cycles
MPa	ksi	n_i
650	94.3	2 000
600	87.0	3 000
500	72.5	10 000
350	50.8	25 000
300	43.5	15 000

Solution

Given SAE 4340, HB 275 alloy steel rod. $s_u = 1048$ MPa $D = 38$ mm. Ground surface.
Endurance strength data (*S-N*) shown in Figure 5–10 $s_n = 430$ MPa.
Loading is reversed, repeated bending. Load history shown in Table 5–5.

Analysis First adjust the *S-N* data for actual conditions using methods of Section 5–6. Use Miner's rule to estimate the portion of life used by the loading pattern.

Results For SAE 4340, HB 275, $s_u = 1048$ MPa
From Figure 5–11, basic $s_n = 475$ MPa for ground surface
Material factor, $C_m = 1.00$ for wrought steel
Type-of-stress factor, $C_{st} = 1.0$ for reversed, rotating bending stress
Reliability factor, $C_R = 0.81$ (Table 5–3) for $R = 0.99$ (Design decision)
Size factor, $C_s = 0.84$ (Figure 5–12 and Table 5–4 for $D = 38$ mm)
Estimated actual endurance limit, s'_n – Computed:

$$s'_n = s_n\, C_m\, C_{st}\, C_R\, C_s = (475\text{ MPa})(1.0)(1.0)(0.81)(0.84) = 323\text{ MPa}$$

This is the estimate for the actual endurance limit of the steel. In Figure 5–10, the endurance limit for the standard specimen is 430 MPa. The ratio of the actual to the standard data is 323/430 = 0.75. We can now adjust the entire *S-N* curve by this factor. The result is shown in Figure 5–23.

Now we can read the number of cycles of life, N_i, corresponding to each of the given loading levels from Table 5–5. The combined data for the number of applied load cycles, n_i, and the life cycles, N_i, are now used in Miner's rule, Equation 5–35, to determine the cumulative damage, D_c. Results are shown in Table 5–6.

TABLE 5–6 Cumulative Life Calculations for Example Problem 5–5

Stress Level		Number of Cycles	Life Cycles	
MPa	ksi	n_i	N_i	n_i/N_i
650	94.3	2000	1.1×10^4	0.182
600	87.0	3000	1.8×10^4	0.167
500	72.5	10 000	5.8×10^4	0.172
350	50.8	25 000	5.6×10^5	0.045
300	43.5	15 000	∞	0.000
				Total 0.566

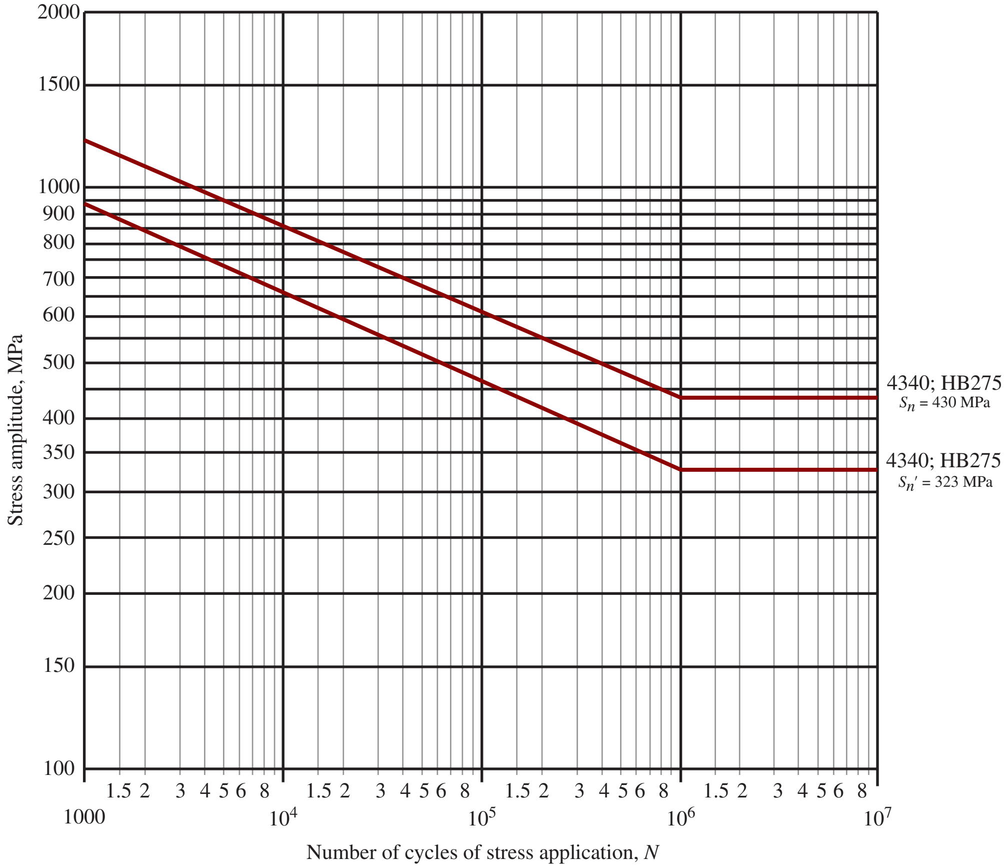


FIGURE 5–23 Endurance strength for Example Problem 5–5

Comment We can conclude from this number that approximately 57% of the life of the component has been accumulated by the given loading. For these data, the greatest damage occurs from the 650 MPa loading for 2000 cycles. An almost equal amount of damage is caused by the 500 MPa loading for 10 000 cycles. Note that the cycles of loading at 300 MPa contributed nothing to the damage because they are below the endurance limit of the steel.

REFERENCES

1. Anderson, Ted L. *Fracture Mechanics: Fundamentals and Applications*. 3rd ed. Boca Raton, FL: CRC Press, 2005.
2. Campbell, F. C. *Fatigue and Fracture: Understanding the Basics*. Materials Park, OH: ASM International, 2012.
3. Balandin, Dimitri V., Nikolaj N. Bolotnik, and Walter D. Pilkey. *Optimal Protection from Impact, Shock and Vibration*. Boca Raton, FL: CRC Press, 2001.
4. Boyer, Howard E. *Atlas of Fatigue Curves*. Materials Park, OH: ASM International, 1986.
5. Budynas, Richard G., and Keith J. Nisbett. *Shigley's Mechanical Engineering Design*. 10th ed. New York: McGraw-Hill, 2015.
6. deSilva, Clarence W. *Vibration and Shock Handbook*. Boca Raton, FL: CRC Press, 2005.
7. Dowling, Norman E. *Mechanical Behavior of Materials*. 4th ed. Upper Saddle River, NJ: Pearson/Prentice-Hall, 2013.
8. Juvinall, Robert C., and Kurt M. Marshek. *Fundamentals of Machine Component Design*. 5th ed. New York: John Wiley & Sons, 2011.
9. Meyers, Marc A., and K. Krishan Chawla. *Mechanical Behavior of Materials*. 2nd ed. Cambridge, UK: Cambridge University Press, 2009.
10. Paez, Thomas, and Allan Piersol. *Harris' Shock and Vibration Handbook*. 6th ed. New York, NY: McGraw-Hill, 2010.
11. Yung-Li Lee, Hong-Tae Kang, and Mark E. Barkey. *Metal Fatigue Analysis Handbook: Practical Problem-Solving Techniques for Computer-Aided Engineering*. New York: Elsevier, 2011.
12. Spotts, Merhyle F., Terry E. Shoup, and Lee E. Hornberger. *Design of Machine Elements*. 8th ed. Upper Saddle River, NJ: Pearson/Prentice-Hall, 2004.
13. McEvily, Arthur J., and Jirapong Kasivitamnuay. *Metal Failures: Mechanisms, Analysis, Prevention*. 2nd ed. New York: Wiley, 2013.

14. Young, Warren C., Richard G. Budynas, and Ali Sadegh. *Roark's Formulas for Stress and Strain*. 8th ed. New York, NY: McGraw-Hill, 2012.
15. Zahavi, Eliaju, and Vladimir Torbilo. *Fatigue Design—Life Expectancy of Machine Parts*. Boca Raton, FL: CRC Press, 1996.
16. Huston, Ronald, and Harold Josephs. *Practical Stress Analysis in Engineering Design*. 3rd ed. Boca Raton, FL: CRC Press, 2009.

INTERNET SITES RELATED TO DESIGN

1. **eFatigue.com** An online source for stress concentration factor information and a fatigue calculator.
2. **Safe Technology** A developer of the fe-safe™ line of software for fatigue analysis that can be linked with finite-element models. Uses several analysis methods such as stress-life, strain-life, von Mises stress, multiaxial fatigue, and damage accumulation.
3. **HBM-nCode** Provider of software and measurement systems for acquiring and processing data for fatigue analysis. Systems also work with finite-element analysis data systems. Uses stress-life, strain-life, multiaxial, weld analysis, and other methods.
4. **ProsigLtd** A producer of the DATS Fatigue Life and Analysis software along with hardware for measurement and testing of vibration and noise of vehicles and industrial equipment. From the home page, select *What we do* and then choose *Analysis Software*.
5. **Comsol Multiphysics** A producer of simulation software covering many types of physical phenomena including structural mechanics, thermal analysis, computational fluid mechanics, and others. An overview of the range of structural mechanics analysis methods is illustrated in the several videos under the menu *Video Gallery*.
6. **AutoFEM Software, LLP** A producer of finite-element analysis software linked with AutoCAD. Includes five modules: static, fatigue, frequency, buckling, and thermal analyses. Offers free AutoFEM Analysis Lite for users of AutoCAD with somewhat restricted capabilities.
7. **ANSYS** Structural analysis finite-element software that includes linear and nonlinear static and dynamic analysis of structures and mechanical devices. Also includes the nCode Design Life fatigue analysis software.
8. **MSC** Software Provider of desktop versions of many types of NASTRAN software for static and dynamic analysis of structures and moving devices, including fatigue analysis.

PROBLEMS

Stress Ratio

For each of Problems 1–9, draw a sketch of the variation of stress versus time, and compute the maximum stress, minimum stress, mean stress, alternating stress, and stress ratio, R. For Problems 6–9, analyze the beam at the place where the maximum stress would occur at any time in the cycle.

1. A link in a mechanism is made from a round bar having a diameter of 10.0 mm. It is subjected to a tensile force that varies from 3500 to 500 N in a cyclical fashion as the mechanism runs.
2. A strut in a space frame has a rectangular cross section of 10.0 mm by 30.0 mm. It sees a load that varies from a tensile force of 20.0 kN to a compressive force of 8.0 kN.
3. A link in a packaging machine mechanism has a square cross section of 0.40 in on a side. It is subjected to a load that varies from a tensile force of 860 lb to a compressive force of 120 lb.
4. A circular rod with a diameter of 3/8 in supports part of a storage shelf in a warehouse. As products are loaded and unloaded, the rod is subjected to a tensile load that varies from 1800 to 150 lb.
5. A part of a latch for a car door is made from a circular rod having a diameter of 3.0 mm. With each actuation, it sees a tensile force that varies from 780 to 360 N.
6. A part of the structure for a factory automation system is a beam that spans 30.0 in as shown in Figure P5–6. Loads are applied at two points, each 8.0 in from a support. The left load $F_1 = 1800$ lb remains constantly applied, while the right load $F_2 = 1800$ lb is applied and removed frequently as the machine cycles. Evaluate the beam at both *B* and *C*.

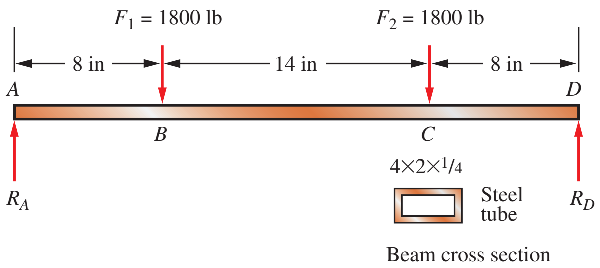


FIGURE P5–6 (Problems 6 and 23)

7. A cantilevered boom is part of an assembly machine and is made from an American Standard steel beam, S4×7.7. A tool with a weight of 500 lb moves continuously from the end of the 60-in beam to a point 10 in from the support.
8. A part of a bracket in the seat assembly of a bus is shown in Figure P5–8. The load varies from 1450 to 140 N as passengers enter and exit the bus.

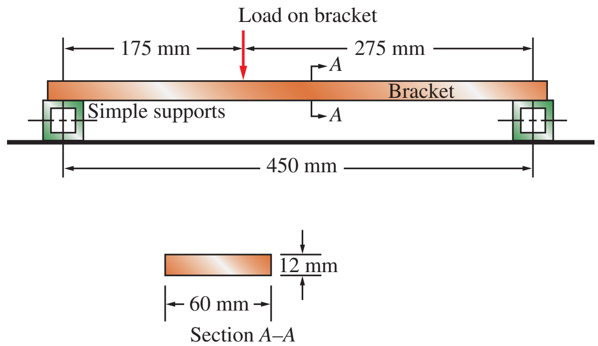


FIGURE P5–8 Seat bracket (Problems 8, 19, and 20)

9. A flat steel strip is used as a spring to maintain a force against part of a cabinet latch in a commercial printer, as shown in Figure P5–9. When the cabinet door is open, the spring is deflected by the latch pin by an amount $y_1 = 0.25$ mm. The pin causes the deflection to increase to 0.40 mm when the door is closed.

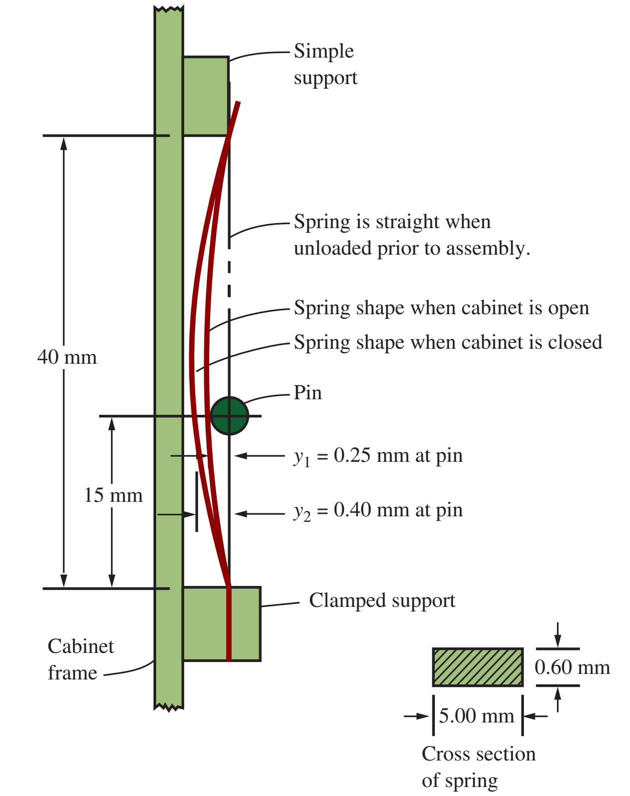


FIGURE P5–9 Cabinet latch spring (Problems 9 and 22)

Endurance Limit

For Problems 10–14, use the method outlined in Section 5–4 to determine the expected actual endurance limit for the material.

10. Compute the estimated endurance limit for a 0.75-in-diameter rod made from SAE 1040 cold-drawn steel. It is to be subjected to repeated and reversed bending stress. A reliability of 99% is desired.
11. Compute the estimated actual endurance limit for SAE 5160 OQT 1300 steel rod with a diameter of 20.0 mm. It is to be machined and subjected to repeated and reversed bending stress. A reliability of 99% is desired.
12. Compute the estimated actual endurance limit for SAE 4130 WQT 1300 steel bar with a rectangular cross section of 20.0 mm by 60 mm. It is to be machined and subjected to repeated and reversed bending stress. A reliability of 99% is desired.
13. Compute the estimated actual endurance limit for SAE 301 stainless steel rod, 1/2 hard, with a diameter of 0.60 in. It is to be machined and subjected to repeated axial tensile stress. A reliability of 99.9% is desired.
14. Compute the estimated actual endurance limit for a machined rectangular steel bar (ASTM A242) 3.5 in high by 0.375 in thick, subjected to repeated and reversed bending stress. A reliability of 99% is desired.

Design and Analysis

15. A link in a mechanism is to be subjected to a tensile force that varies from 3500 to 500 N in a cyclical fashion as the mechanism runs. It has been decided to use SAE 1040 cold-drawn steel. Complete the design of the link, specifying a suitable cross-sectional shape and dimensions.
16. A circular rod is to support part of a storage shelf in a warehouse. As products are loaded and unloaded, the rod is subjected to a tensile load that varies from 1800 to 150 lb. Specify a suitable shape, material, and dimensions for the rod.
17. A strut in a space frame sees a load that varies from a tensile force of 20.0 kN to a compressive force of 8.0 kN. Specify a suitable shape, material, and dimensions for the strut.
18. A part of a latch for a car door is to be made from a straight bar. With each actuation, it sees a tensile force that varies from 780 to 360 N. Small size is important. Complete the design, and specify a suitable shape, material, and dimensions for the rod.
19. A part of a bracket in the seat assembly of a bus is shown in Figure P5–8. The load varies from 1450 to 140 N as passengers enter and exit the bus. The bracket is made from SAE 1020 hot-rolled steel. Determine the resulting design factor.
20. For the bus seat bracket described in Problem 19 and shown in Figure P5–8, propose an alternate design for the bracket, different from that shown in the figure, to achieve a lighter design with a design factor of approximately 4.0.
21. A cantilevered boom is part of an assembly machine. A tool with a weight of 500 lb moves continuously from the end of the 60-in beam to a point 10 in from the support. Specify a suitable design for the boom, giving the material, the cross-sectional shape, and the dimensions.
22. A flat steel strip is used as a spring to maintain a force against part of a cabinet latch in a commercial printer as shown in Figure P5–9. When the cabinet door is open, the spring is deflected by the latch pin by an amount $y_1 = 0.25$ mm. The pin causes the deflection to increase to 0.40 mm when the door is closed. Specify a suitable material for the spring if it is made to the dimensions shown in the figure.
23. A part of the structure for a factory automation system is a beam that spans 30.0 in, as shown in Figure P5–6. Loads are applied at two points, each 8.0 in from a support. The left load $F_1 = 1800$ lb remains constantly applied, while the right load $F_2 = 1800$ lb is applied and removed frequently as the machine cycles. If the rectangular tube is made from ASTM A500 Grade B steel, is the proposed design satisfactory? Improve the design to achieve a lighter beam.
24. Figure P5–24 shows a hydraulic cylinder that pushes a heavy tool during the outward stroke, placing a compressive load of 400 lb in the piston rod. During the return stroke, the rod pulls on the tool with a force of 1500 lb. Compute the resulting design factor for the 0.60-in-diameter

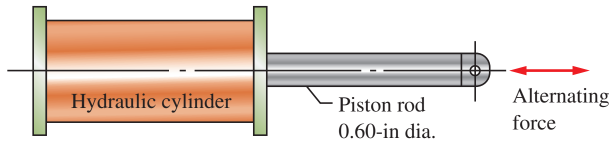


FIGURE P5–24 (Problem 24)

rod when subjected to this pattern of forces for many cycles. The material is SAE 4130 WQT 1300 steel. If the resulting design factor is much different from 4.0, determine the size of the rod that would produce $N = 4.0$.

25. The cast iron cylinder shown in Figure P5–25 carries only an axial compressive load of 75 000 lb. (The torque $T = 0$.) Compute the design factor if it is made from gray cast iron, Grade 40A, having a tensile ultimate strength of 40 ksi and a compressive ultimate strength of 140 ksi.
26. Repeat Problem 25, except using a tensile load with a magnitude of 12 000 lb.
27. Repeat Problem 25, except using a load that is a combination of an axial compressive load of 75 000 lb and a torsion of 20 000 lb·in.

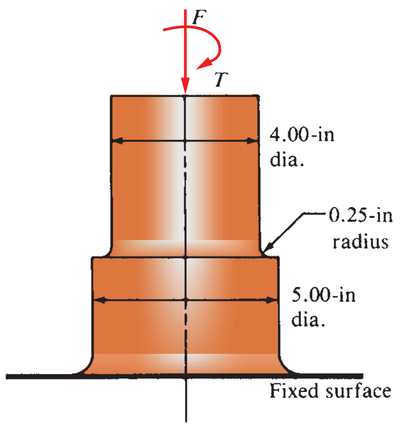


FIGURE P5–25 (Problems 25, 26, and 27)

28. The shaft shown in Figure P5–28 is supported by bearings at each end, which have bores of 20.0 mm. Design the shaft to carry the given load if it is steady and the shaft is stationary. Make the dimension a as large as possible while keeping the stress safe. Determine the required

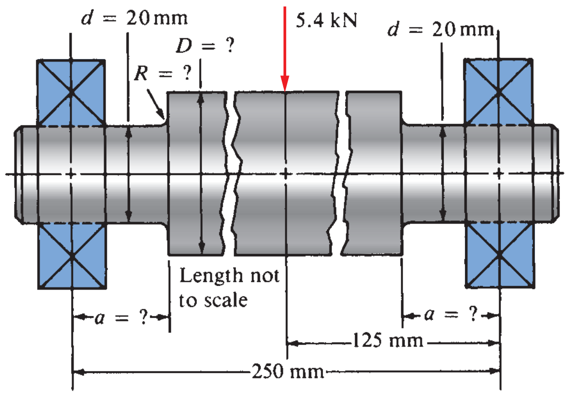


FIGURE P5–28 (Problems 28, 29, and 30)

diameter in the middle portion. The maximum fillet permissible is 2.0 mm. Use SAE 1137 cold-drawn steel. Use a design factor of 3.

29. Repeat Problem 28, except using a rotating shaft.
30. Repeat Problem 28, except using a shaft that is rotating and transmitting a torque of 150 N·m from the left bearing to the middle of the shaft. Also, there is a profile keyseat at the middle under the load.
31. Figure P5–31 shows a proposed design for a seat support. The vertical member is to be a standard hollow circular shape selected from either Appendices 15–17 (steel pipe) or 15–18 (mechanical tubing). The two loads are static and act simultaneously. The material is similar to SAE 1020 hot-rolled steel. Use a design factor of 3.

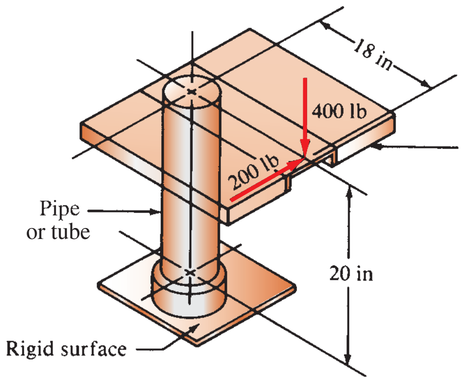


FIGURE P5–31 (Problem 31)

32. A torsion bar is to have a solid circular cross section. It is to carry a fluctuating torque from 30 to 65 N·m. Use SAE 4140 OQT 1000 for the bar, and determine the required diameter for a design factor of 2. Attachments produce a stress concentration of 2.5 near the ends of the bar.
33. Determine the required size for a square bar to be made from SAE 1213 cold-drawn steel. It carries a constant axial tensile load of 1500 lb and a bending load that varies from zero to a maximum of 800 lb at the center of the 48-in length of the bar. Use a design factor of 3.
34. Repeat Problem 33, but add a constant torsional moment of 1200 lb·in to the other loads.

In some of the following problems, you are asked to compute the design factor resulting from the design proposed for the given loading. Unless stated otherwise, assume that the element being analyzed has a machined surface. If the design factor is significantly different from $N = 3$, redesign the component to achieve approximately $N = 3$. (Some problems use figures found in Chapter 3.)

35. A tensile member in a machine structure is subjected to a steady load of 4.50 kN. It has a length of 750 mm and is made from a steel tube, SAE 1040 hot-rolled, having an outside diameter of 18 mm and an inside diameter of 12 mm. Compute the resulting design factor.
36. A steady tensile load of 5.00 kN is applied to a square bar, 12 mm on a side and having a length of 1.65 m. Compute the stress in the bar and the resulting design factor if it is made from (a) SAE 1020 hot-rolled steel; (b) SAE 8650 OQT 1000 steel; (c) ductile iron A536 (60-40-18);

(d) aluminum alloy 6061-T6; (e) titanium alloy Ti-6Al-4V, annealed; (f) rigid PVC plastic; and (g) phenolic plastic.

37. An aluminum rod, made from alloy 6061-T6, is made in the form of a hollow square tube, 2.25 in outside with a wall thickness of 0.125 in. Its length is 16.0 in. It carries an axial compressive force of 12 600 lb. Compute the resulting design factor. Assume that the tube does not buckle.

38. Compute the design factor in the middle portion only of the rod AC in Figure P3–8 if the steady vertical force on the boom is 2500 lb. The rod is rectangular, 1.50 in by 3.50 in, and is made from SAE 1144 cold-drawn steel.

39. Compute the forces in the two angled rods in Figure P3–9 for a steady applied force, $F = 1500$ lb, if the angle θ is 45°. Then design the middle portion of each rod to be circular and made from SAE 1040 hot-rolled steel. Specify a suitable diameter.

40. Repeat Problem 39 if the angle θ is 15°.

41. A circular bar of SAE 4140 OQT 1000 steel steps from 12 mm to 10 mm with a fillet radius of 1.5 mm and carries a repeated/reversed 7500 N axial force. Compute the design factor.

42. Compute the torsional shear stress in a circular shaft having a diameter of 50 mm when subjected to a torque of 800 N·m. If the torque is completely reversed and repeated, compute the resulting design factor. The material is SAE 1040 WQT 1000.

43. If the torque in Problem 42 fluctuates from zero to the maximum of 800 N·m, compute the resulting design factor.

44. Compute the torsional shear stress in a circular shaft 0.40 in in diameter that is due to a steady torque of 88.0 lb·in. Specify a suitable aluminum alloy for the rod.

45. Compute the required diameter for a solid circular shaft if it is transmitting a maximum of 110 hp at a speed of 560 rpm. The torque varies from zero to the maximum. There are no other significant loads on the shaft. Use SAE 4130 WQT 700.

46. Specify a suitable material for a hollow shaft with an outside diameter of 40 mm and an inside diameter of 30 mm when transmitting 28 kilowatts (kW) of steady power at a speed of 45 radians per second (rad/s).

47. Repeat Problem 46 if the power fluctuates from 15 to 28 kW.

48. Figure P5–48 shows part of a support bar for a heavy machine, suspended on springs to soften applied loads. The tensile load on the bar varies from 12 500 lb to a minimum of 7500 lb. Rapid cycling for many million cycles is expected. The bar is made from SAE 6150 OQT 1300 steel. Compute the design factor for the bar in the vicinity of the hole.

49. Figure P3–61 shows a valve stem from an engine subjected to an axial tensile load applied by the valve spring. The force varies from 0.80 to 1.25 kN. Compute the resulting design factor at the fillet under the shoulder. The valve is made from SAE 8650 OQT 1300 steel.

50. A conveyor fixture shown in Figure P3–62 carries three heavy assemblies (1200 lb each). The fixture is machined from SAE 1144 OQT 900 steel. Compute the resulting design factor in the fixture, considering stress concentrations at the fillets and assuming that the load acts axially. The load will vary from zero to the maximum as the conveyor is loaded and unloaded.

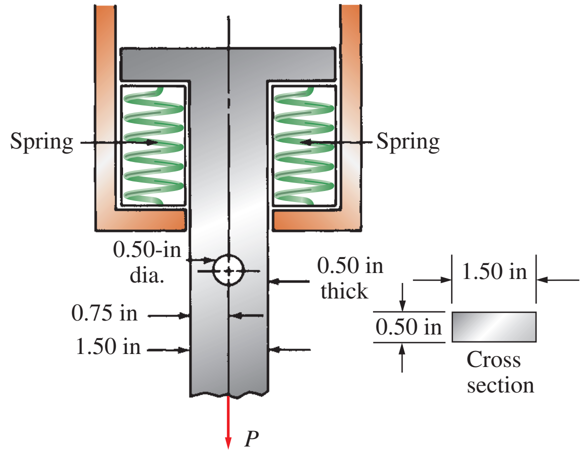


FIGURE P5–48 (Problem 48)

51. For the flat plate in tension in Figure P3–63, compute the minimum resulting design factor, assuming that the holes are sufficiently far apart that their effects do not interact. The plate is machined from stainless steel, UNS S17400 in condition H1150. The load is repeated and varies from 4000 to 6200 lb.

For Problems 52–56, select a suitable material for the member, considering stress concentrations, for the given loading to produce a design factor of $N = 3$.

52. Use Figure P3–64. The load is steady. The material is to be some grade of gray cast iron, ASTM A48.

53. Use Figure P3–65. The load varies from 20.0 to 30.3 kN. The material is to be titanium.

54. Use Figure P3–66. The torque varies from zero to 2200 lb. in. The material is to be steel.

55. Use Figure P3–67. The bending moment is steady. The material is to be ductile iron, ASTM A536.

56. Use Figure P3–68. The bending moment is completely reversed. The material is to be stainless steel.

57. Figure P5–67 shows part of an automatic screwdriver designed to drive several million screws. The maximum torque required to drive a screw is 100 lb·in. Compute the design factor for the proposed design if the part is made from SAE 8740 OQT 1000.

58. The beam in Figure P5–58 carries two steady loads, $P = 750$ lb. Evaluate the design factor that would result if the beam were made from class 40A gray cast iron.

59. A tension link is subjected to a repeated, one-direction load of 3000 lb. Specify a suitable material if the link is to be steel and is to have a diameter of 0.50 in.

60. One member of an automatic transfer device in a factory must withstand a repeated tensile load of 800 lb and must not elongate more than 0.010 in in its 25.0-in length. Specify a suitable steel material and the dimensions for the rod if it has a square cross section.

61. Figure P5–61 shows two designs for a beam to carry a repeated central load of 1200 lb. Which design would have the highest design factor for a given material?

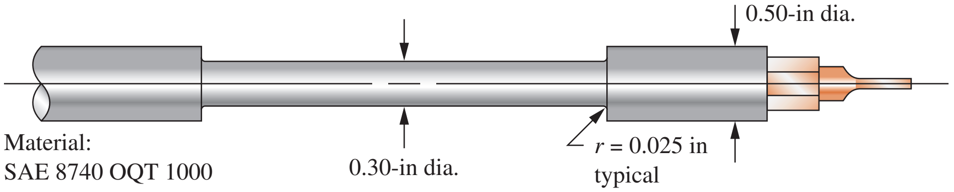


FIGURE P5–57 Screwdriver for Problem 57

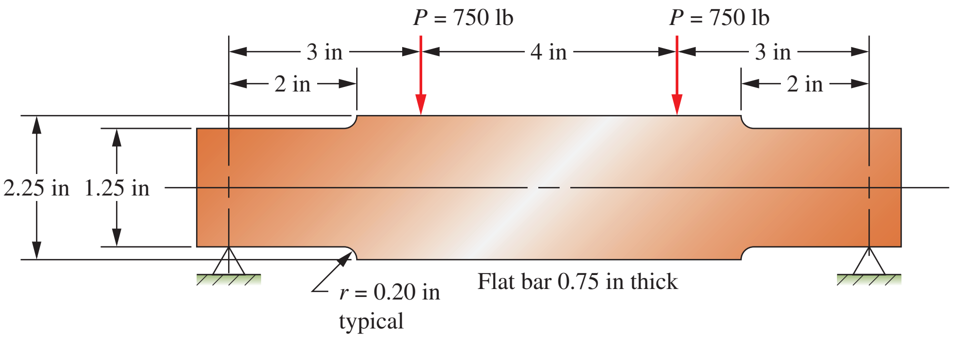


FIGURE P5–58 Beam for Problem 58

62. Refer Figure P5–61. By reducing the 8.0-in dimension, redesign the beam in Part (b) of the figure so that it has a design factor equal to or higher than that for the design in Part (a).

63. Refer Figure P5–61. Redesign the beam in Part (b) of the figure by first increasing the fillet radius to 0.40 in and then by reducing the 8.0-in dimension so that the new design has a design factor equal to or higher than that for the design in Part (a).

64. The part shown in Figure P5–64 is made from SAE 1040 HR steel. It is to be subjected to a repeated, one-direction force of 5000 lb applied through two 0.25-in-diameter pins in the holes at each end. Compute the resulting design factor.

65. For the part described in Problem 64, make at least three improvements in the design that will significantly reduce the stress without increasing the weight. The dimensions marked © are critical and cannot be changed. After the redesign, specify a suitable material to achieve a design factor of at least 3.

66. The link shown in Figure P5–66 is subjected to a tensile force that varies from 3.0 to 24.8 kN. Evaluate the design factor if the link is made from SAE 1040 CD steel.

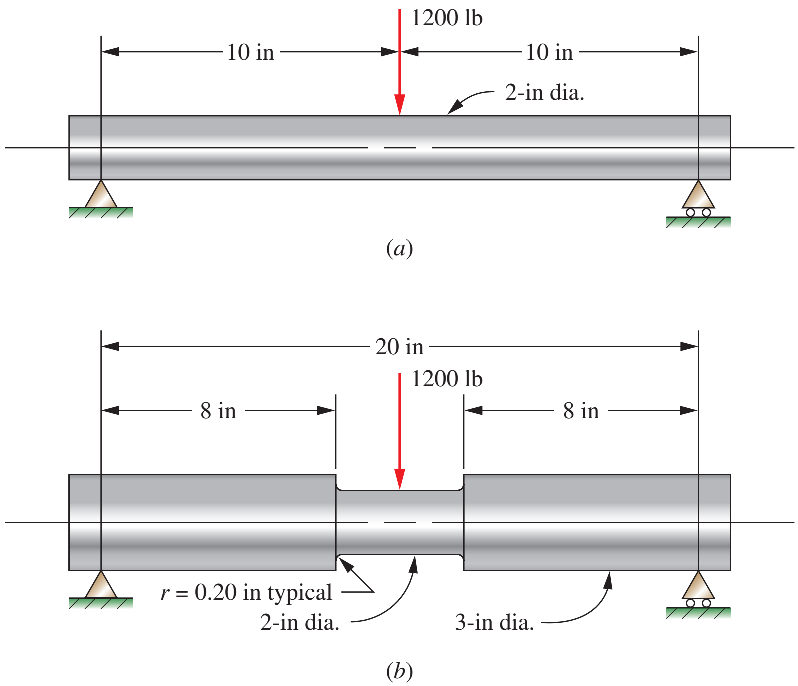


FIGURE P5–61 Beam for Problems 61, 62, and 63

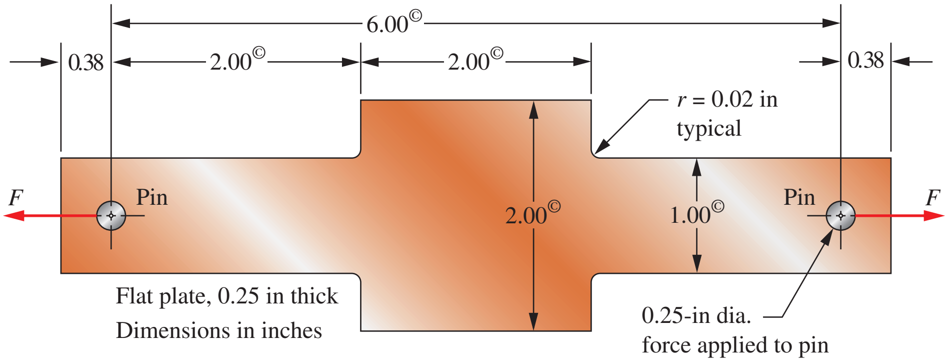


FIGURE P5–64 Beam for Problems 64 and 65

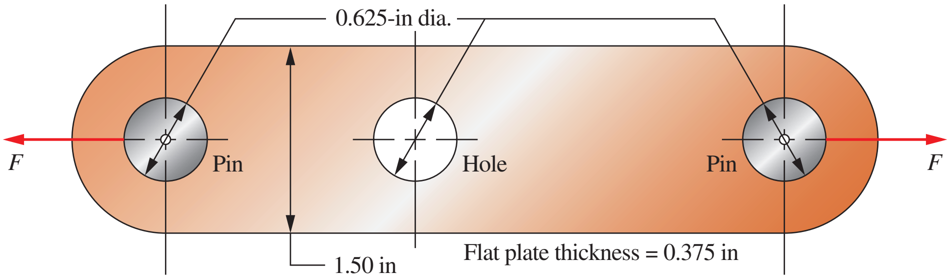


FIGURE P5–66 Link for Problem 66

67. The beam shown in Figure P5–67 carries a repeated, reversed load of 400 N applied at section C. Compute the resulting design factor if the beam is made from SAE 1340 OQT 1300.

68. For the beam described in Problem 67, change the steel's tempering temperature to achieve a design factor of at least 2.5.

69. The cantilever shown in Figure P5–69 carries a downward load that varies from 300 to 700 lb. Compute the resulting design factor if the bar is made from SAE 1050 HR steel.

70. For the cantilever described in Problem 69, increase the size of the fillet radius to improve the design factor to at least 3.0 if possible.

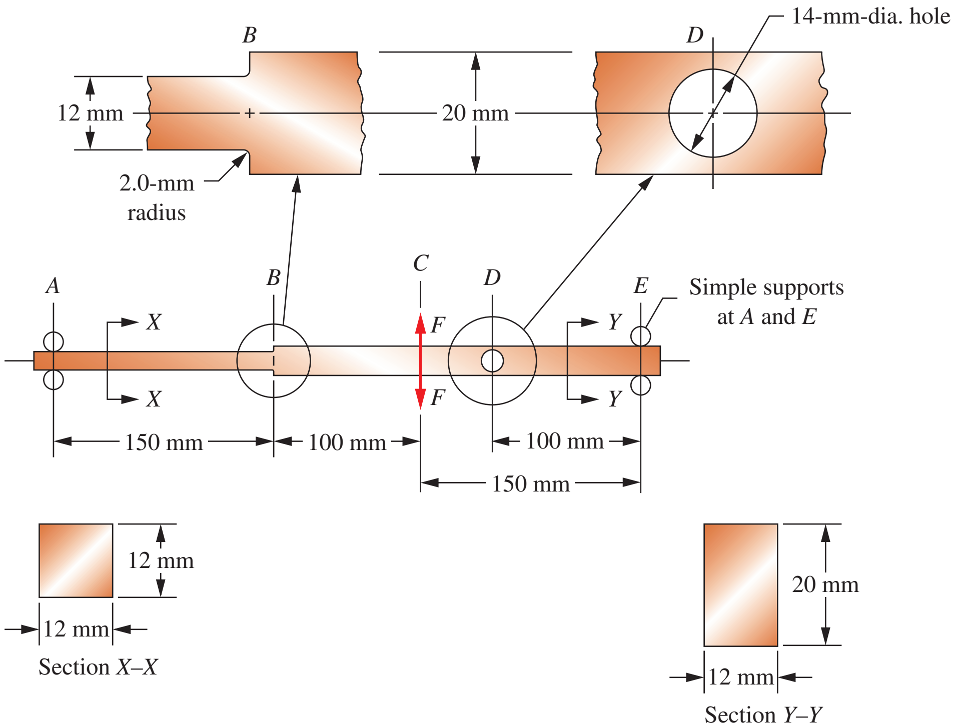


FIGURE P5–67 Beam for Problems 67 and 68

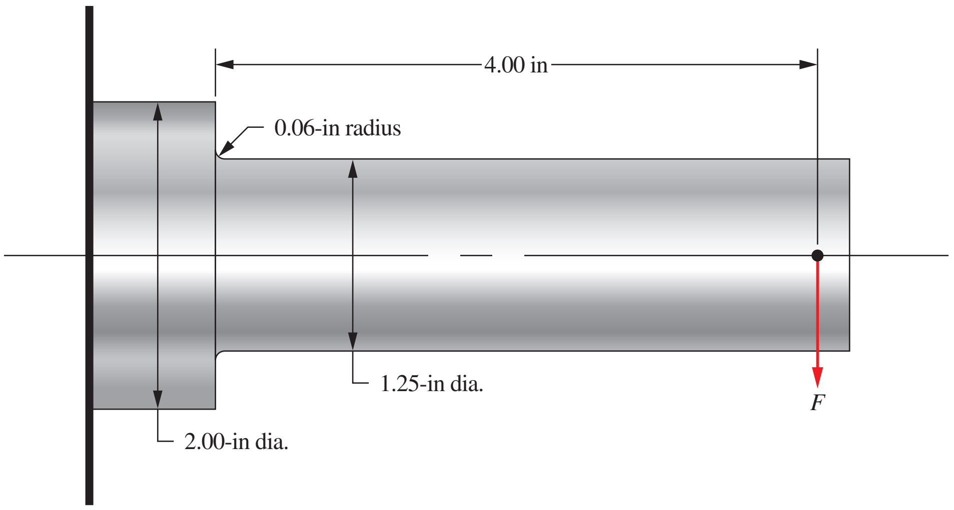


FIGURE P5–69 Cantilever for Problems 69, 70, and 71

71. For the cantilever described in Problem 69, specify a suitable material to achieve a design factor of at least 3.0 without changing the geometry of the beam.

72. Figure P5–72 shows a rotating shaft carrying a steady downward load of 100 lb at *C*. Specify a suitable material.

73. The stepped rod shown in Figure P5–73 is subjected to a direct tensile force that varies from 8500 to 16 000 lb. If the rod is made from SAE 1340 OQT 700 steel, compute the resulting design factor.

74. For the rod described in Problem 73, complete a redesign that will achieve a design factor of at least 3.0.

75. The beam shown in Figure P5–75 carries a repeated, reversed load of 800 lb alternately applied upward and downward. If the beam is made from SAE 1144 OQT 1100, specify the smallest acceptable fillet radius at *A* to ensure a design factor of 3.0.

76. For the beam described in Problem 75, design the section at *B* to achieve a minimum design factor of 3.0. Specify the shape, dimensions, and fillet radius where the smaller part joins the 2.00-by-2.00-in section.

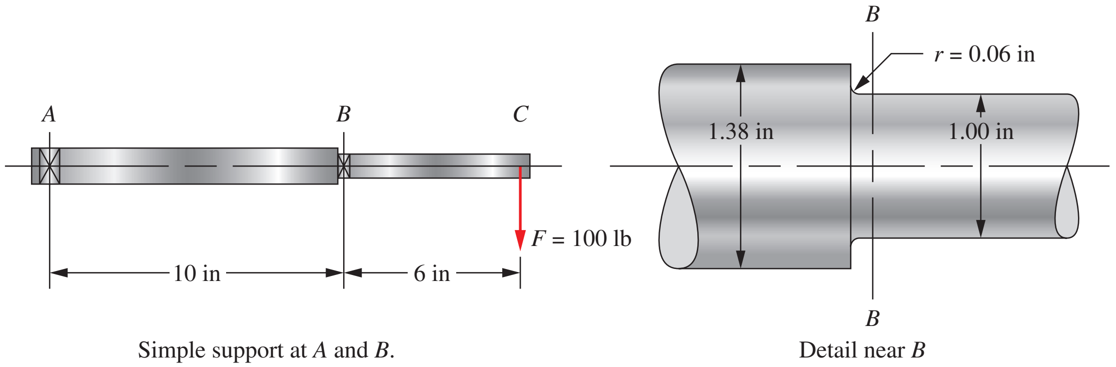


FIGURE P5–72 Shaft for Problem 72

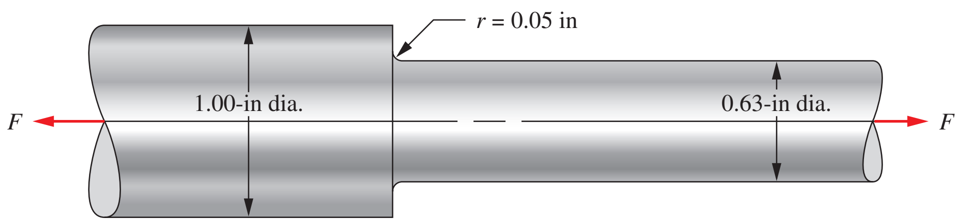


FIGURE P5–73 Rod for Problems 73 and 74

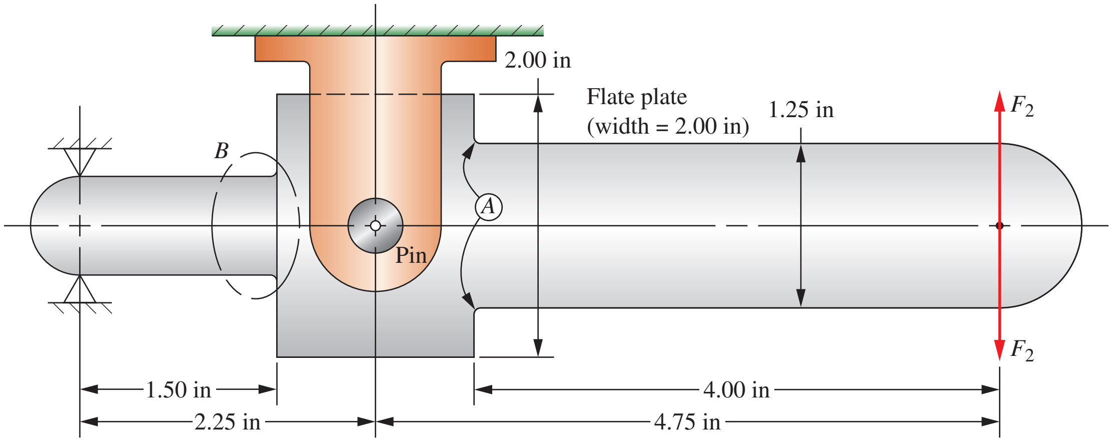


FIGURE P5–75 Beam for Problems 75 and 76

Design Problems

For each of the following problems, complete the requested design to achieve a minimum design factor of 3.0. Specify the shape, the dimensions, and the material for the part to be designed. Work toward an efficient design that will have a low weight.

77. The link shown in Figure P5–77 carries a load of 3000 N that is applied and released many times. The link is machined from a square bar, 12.0 mm on a side, from SAE 1144 OQT 1100 steel. The ends must remain 12.0-mm square to facilitate the connection to mating parts. It is desired to reduce the size of the middle part of the link to reduce weight. Complete the design.

78. Complete the design of the beam shown in Figure P5–78 to carry a large hydraulic motor. The beam is attached to the two side rails of the frame of a truck. Because of the vertical accelerations experienced by the truck, the load on the beam varies from 1200 lb upward to 5000 lb downward. One-half of the load is applied to the beam by each foot of the motor.

79. A tensile member in a truss frame is subjected to a load that varies from 0 to 6500 lb as a traveling crane moves across the frame. Design the tensile member.

80. A hanger for a conveyor system extends outward from two supports, as shown in Figure P5–80. The load at

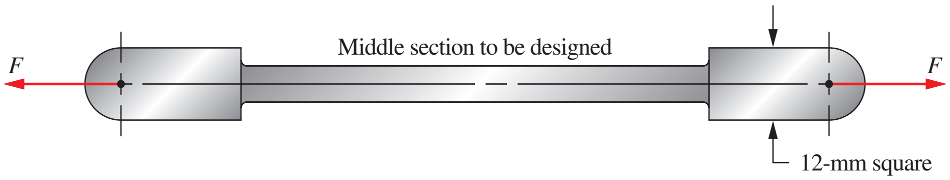


FIGURE P5–77 Link for Problem 77

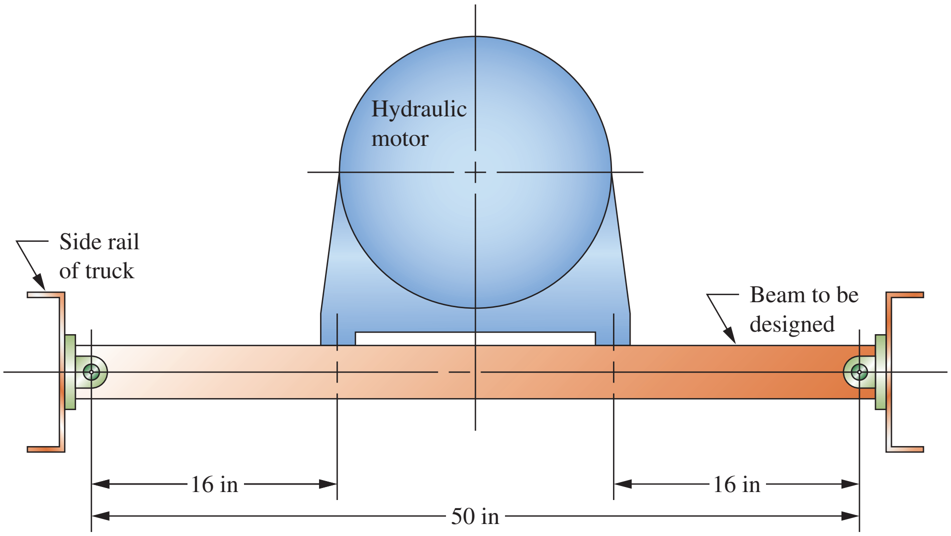


FIGURE P5–78 Beam for Problem 78

the right end varies from 600 to 3800 lb. Design the hanger.

81. Figure P5–81 shows a yoke suspended beneath a crane beam by two rods. Design the yoke if the loads are applied and released many times.

82. For the system shown in Figure P5–81, design the two vertical rods if the loads are applied and released many times.

83. Design the connections between the rods and the yoke and the crane beam shown in Figure P5–81.

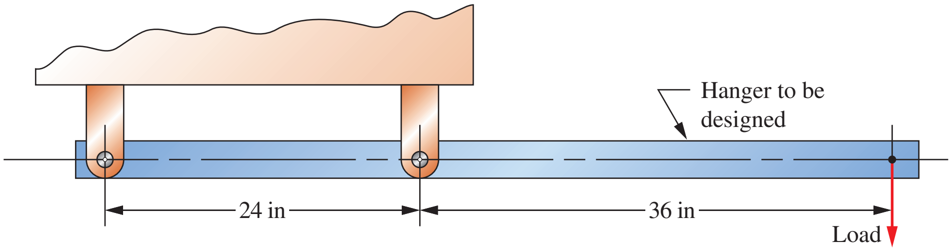


FIGURE P5–80 Hanger for the conveyor system for Problem 80

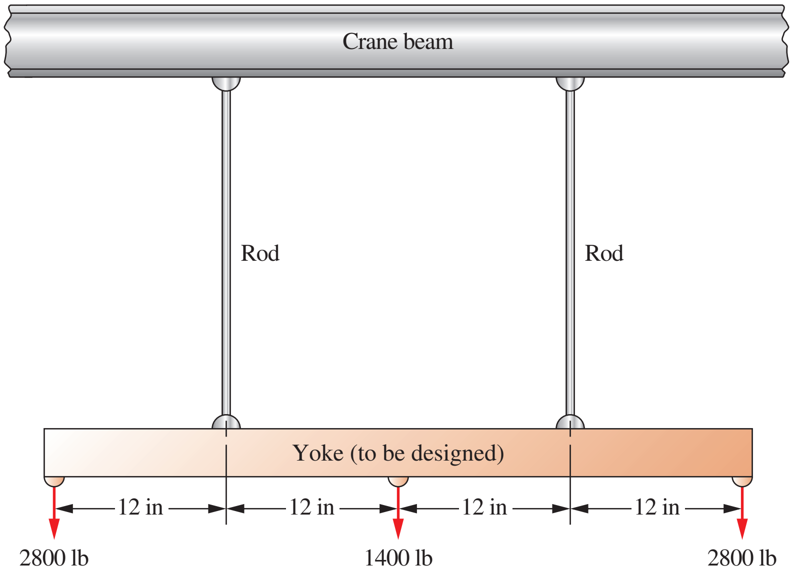


FIGURE P5–81 Yoke and rods for Problems 81, 82, and 83

CHAPTER
SIX

COLUMNS

THE BIG PICTURE Columns

Discussion Map

- A column is a long, slender member that carries an axial compressive load and that fails due to buckling rather than due to failure of the material of the column.

Discover

Find at least 10 examples of columns. Describe them and how they are loaded, and discuss them with your colleagues.

Try to find at least one column that you can load conveniently by hand, and observe the buckling phenomenon.

Discuss with your colleagues the variables that seem to affect how a column fails and how much load it can carry before failing.

This chapter will help you acquire some of the analytical tools necessary to design and analyze columns.

A *column* is a structural member that carries an axial compressive load and that tends to fail by elastic instability, or buckling, rather than by crushing of material. *Elastic instability* is the condition of failure in which the shape of the column is insufficiently rigid to hold it straight under load. At the point of buckling, a radical deflection of the axis of the column occurs suddenly. Then, if the load is not reduced, the column will collapse. Obviously this kind of catastrophic failure must be avoided in structures and machine elements.

Columns are ideally straight and relatively long and slender. If a compression member is so short that it does not tend to buckle, failure analysis must use the methods presented in Chapter 5. This chapter presents several methods of analyzing and designing columns to ensure safety under a variety of loading conditions.

Take a few minutes to visualize examples of column buckling. Find any object that appears to be long and slender—for example, a meter stick, a plastic ruler, a long wooden dowel with a small diameter, a drinking straw, or a thin metal or plastic rod. Carefully apply a downward load on your column while resting the bottom on a desk or the floor. Try to make sure that it does not slide. Gradually increase the load, and observe the behavior of the column until it begins to bend noticeably in the middle. This large deformation is called *buckling*. Then hold that level of load. Don't increase it much beyond that level, or the column will likely break!

Now release the load; the column should return to its original shape. The material should not have broken or yielded. But wouldn't you have considered the column to have failed at the point of buckling? Wouldn't it be important to keep the applied load well below the load that caused the buckling to be initiated?

Now look around you. Think of things you are familiar with, or take time to go out and find other examples of columns. Remember, look for relatively long, slender, load-carrying members subjected to compressive loads. Consider parts of furniture, buildings, cars, trucks, toys, play structures, industrial machinery, and construction machinery. Try to find at least 10 examples. Describe their appearance: the material from which they are made, the way they are supported, and the way they are loaded. Do this activity in the classroom or with colleagues; bring the descriptions to class next session for discussion.

Notice that you were asked to find *relatively long*, *slender*, load-carrying members. How will you know when a member is long and slender? At this point, you should just use your judgment. If the column is available and you are strong enough to load it to buckling, go ahead and try it. Later in this chapter, we will quantify what the terms *long* and *slender* mean.

If the columns you have seen did not actually collapse, what property of the material is highly related to the phenomenon of buckling failure? Remember that the failure was described as *elastic instability*. Then it should seem that the *modulus of elasticity* of the material is a key property, and it is. Review the definition of this property from Chapter 1, and look up representative values in the tables of material properties in Appendices 3–13.

Also note that we specified that the columns are to be initially straight and that loads are to be applied axially. What if these conditions are not met? What if the column is a little crooked before loading? Do you think that it would carry as much compressive loading as one that was straight? Why or why not? What if the column is loaded *eccentrically*, that is, the load is directed off-center, away from the centroidal axis of the column? How will that affect the load-carrying ability? How does the manner of supporting the ends of the column affect its load-carrying ability? What standards exist that guide designers when dealing with columns?

These and other questions will be addressed in this chapter. Any time you are involved in a design in which a compressive load is applied, you should think about analyzing it as a column. The following **You Are the Designer** situation is a good example of such a machine design problem.

6–1 OBJECTIVES OF THIS CHAPTER

After completing this chapter, you will be able to:

1. Recognize that any relatively long, slender compression member must be analyzed as a column to prevent buckling.
2. Specify efficient shapes for the cross section of columns.
3. Compute the *radius of gyration* of a column cross section.
4. Specify a suitable value for the *end-fixity factor*, *K*, and determine the *effective length* of a column.
5. Compute the *slenderness ratio* for columns.
6. Select the proper method of analysis or design for a column based on the manner of loading, the type of support, and the magnitude of the slenderness ratio.
7. Determine whether a column is *long* or *short* based on the value of the slenderness ratio in comparison with the *column constant*.
8. Use the *Euler formula* for the analysis and design of long columns.
9. Use the *J. B. Johnson formula* for the analysis and design of short columns.
10. Analyze crooked columns to determine the allowable load.
11. Analyze columns in which the load is applied with a modest amount of eccentricity to determine the maximum predicted stress and the maximum amount of deflection of the centerline of such columns under load.

YOU ARE THE DESIGNER

You are a member of a team that is designing a commercial compactor to reduce the volume of cardboard and paper waste so that the waste can be transported easily to a processing plant. Figure 6–1 is a sketch of the compaction ram that is driven by a hydraulic cylinder under several thousand pounds of force. The connecting rod between the hydraulic cylinder and the ram must be designed as a column because it is a relatively long, slender compression member. What shape should the cross section of the connecting rod be? From what material should it be made? How is it to be connected to the ram and to the hydraulic cylinder? What are the final dimensions of the rod to be? You, the designer, must specify all of these factors. ■

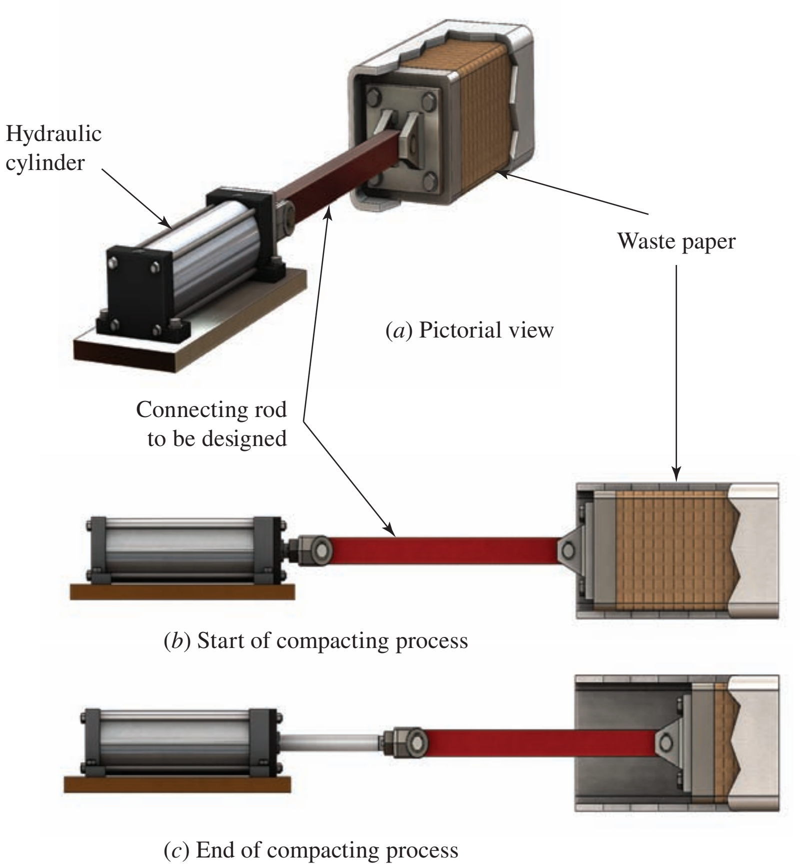


FIGURE 6–1 Waste paper compactor

6–2 PROPERTIES OF THE CROSS SECTION OF A COLUMN

The tendency for a column to buckle is dependent on the shape and the dimensions of its cross section, along with its length and the manner of attachment to adjacent members or supports. Cross-sectional properties that are important are as follows:

1. The cross-sectional area, A.
2. The area moment of inertia of the cross section, I, with respect to the axis about which the value of I is minimum.
3. The least value of the radius of gyration of the cross section, r.

The radius of gyration is computed from

➩ Radius of Gyration

$$r = \sqrt{I/A} \qquad (6\text{–}1)$$

A column tends to buckle about the axis for which the radius of gyration and the area moment of inertia are minimum. Figure 6–2 shows a sketch of a column that has a rectangular cross section. The expected buckling axis is Y-Y because both I and r are much smaller for that axis than for the X-X axis. You can demonstrate this phenomenon by loading a common ruler or meter stick with an axial load of sufficient magnitude to cause buckling. See Appendix 1 for formulas for I and r for common shapes. See Appendix 15 for structural shapes.

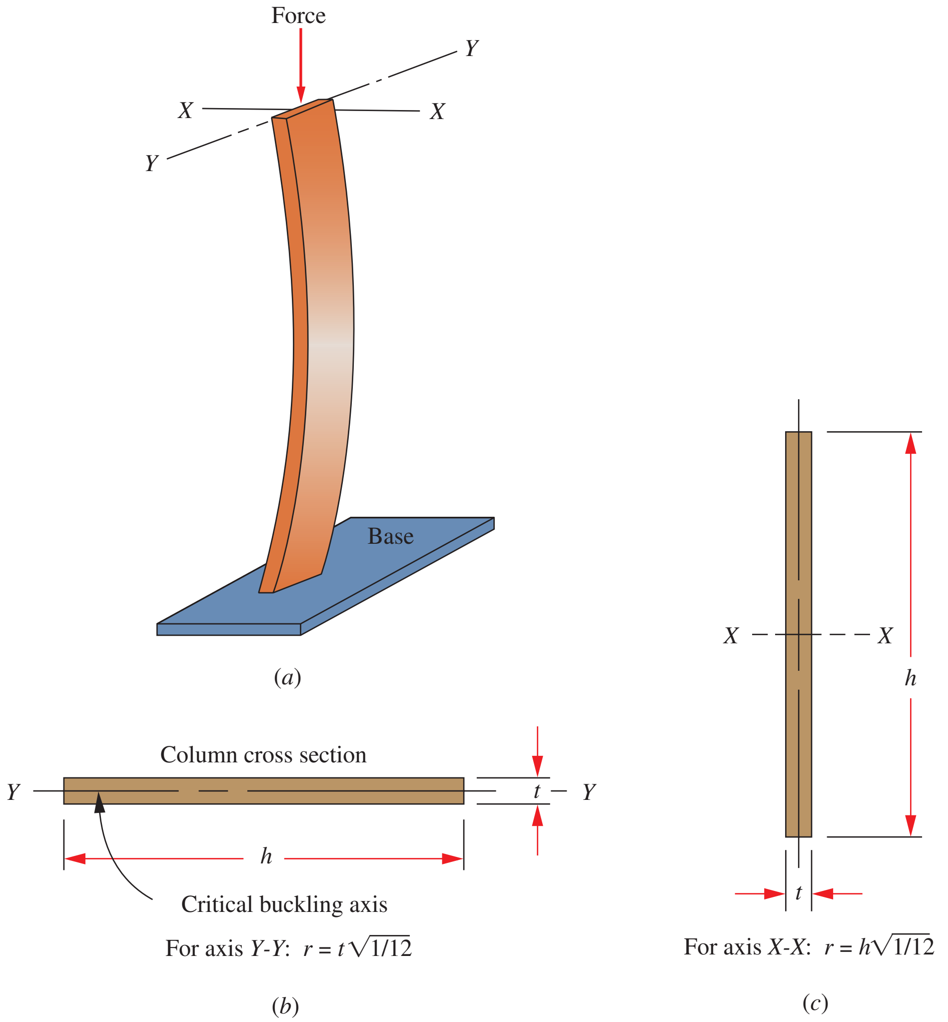


FIGURE 6–2 Buckling of a thin, rectangular column. (a) General appearance of the buckled column. (b) Radius of gyration for Y-Y axis. (c) Radius of gyration for X-X axis

6–3 END FIXITY AND EFFECTIVE LENGTH

The term *end fixity* refers to the manner in which the ends of a column are supported. The most important variable is the amount of restraint offered at the ends of the column to the tendency for rotation. Three forms of end restraint are *pinned*, *fixed*, and *free*.

A *pinned-end* column is guided so that the end cannot sway from side to side, but it offers no resistance to rotation of the end. The best approximation of the pinned end would be a frictionless ball-and-socket joint. A cylindrical pin joint offers little resistance about one axis, but it may restrain the axis perpendicular to the pin axis.

A *fixed end* is one that is held against rotation at the support. An example is a cylindrical column inserted into a tight-fitting sleeve that itself is rigidly supported. The sleeve prohibits any tendency for the fixed end of the column to rotate. A column end securely welded to a rigid base plate is also a good approximation of a fixed-end column.

The *free end* can be illustrated by the example of a flagpole. The top end of a flagpole is unrestrained and unguided, the worst case for column loading.

The manner of support of both ends of the column affects the *effective length* of the column, defined as

➪ Effective Length

$$L_e = KL \tag{6–2}$$

where L = actual length of the column between supports

K = constant dependent on the end fixity, as illustrated in Figure 6–3

The K values are related to the shape of the buckled column. With the pinned-pinned end condition as the base case, it can be observed from the deflected shape in Figure 6–3 that the fixed-fixed, fixed-free, and fixed-pinned columns have effective lengths of $0.5L$, $2.0L$, and $0.7L$, respectively.

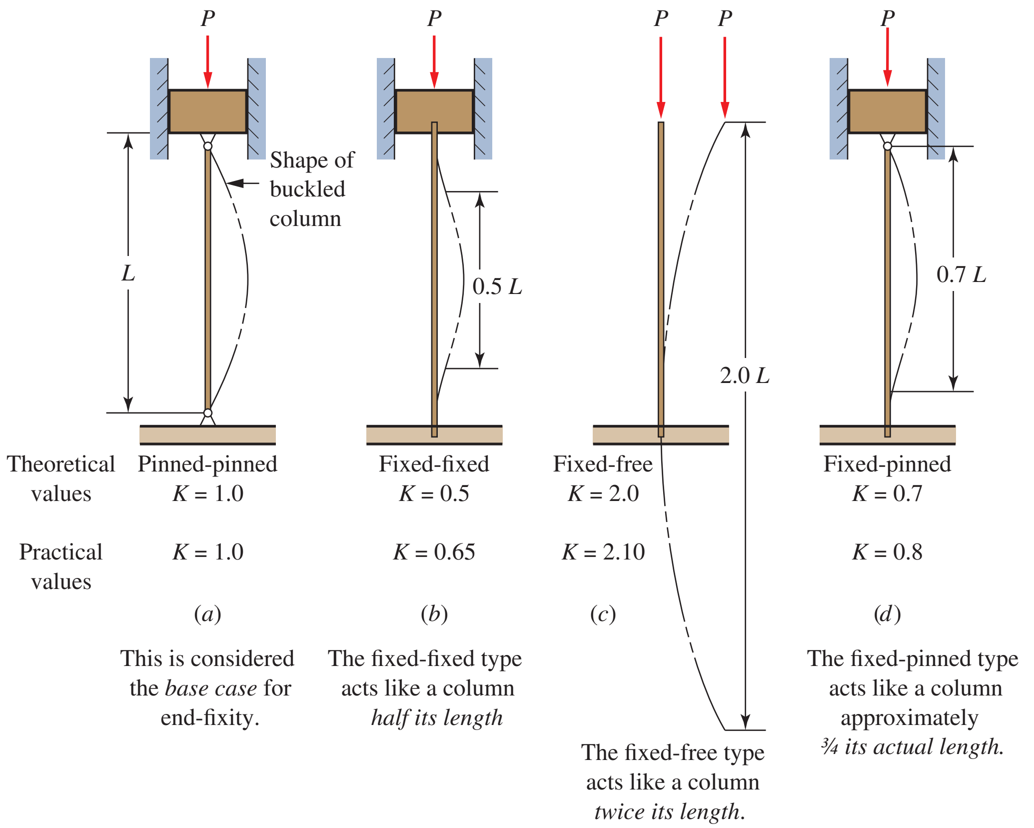


FIGURE 6–3 Values of K for effective length, $L_e = KL$, for different end connections

The first values given for K are theoretical values based on the shape of the deflected column. The second values take into account the expected fixity of the column ends in real, practical structures. It is particularly difficult to achieve a true fixed-end column because of the lack of complete rigidity of the support or the means of attachment. Therefore, the higher value of K is recommended.

6–4 SLENDERNESS RATIO

The *slenderness ratio* is the ratio of the effective length of the column to its least radius of gyration. That is,

➪ Slenderness Ratio, S_r

$$S_r = L_e/r_{\min} = KL/r_{\min} \tag{6–3}$$

We will use the slenderness ratio to aid in the selection of the method of performing the analysis of straight, centrally loaded columns.

6–5 LONG COLUMN ANALYSIS: THE EULER FORMULA

Analysis of a column employs the Euler formula (see Reference 4):

➪ Euler Formula for Long Columns

$$P_{cr} = \frac{\pi^2 EI}{L_e^2} \tag{6–4}$$

The equation gives the critical load, P_{cr}, at which the column would begin to buckle. Notice that the buckling load is dependent only on the geometry (length and cross section) of the column and the stiffness of the material represented by the modulus of elasticity. The strength of the material is not involved at all. For these reasons, it is often of no benefit to specify a high-strength material in a long column application. A lower-strength material having the same stiffness, E, would perform as well.

In a design problem, the objective is to specify a size and a shape of a column cross section to carry a certain load. From Equation (6–4), the area moment of inertia for the required cross section can be easily determined.

An alternative form of the Euler formula is often desirable. Note that, from Equation (6–1),

$$r = \sqrt{\frac{I}{A}} \Rightarrow I = r^2 A \tag{6–5}$$

Substituting (6–5) into (6–4):

$$P_{cr} = \frac{\pi^2 EA}{(L_e/r)^2} = \frac{\pi^2 EA}{S_r^2} \tag{6–6}$$

Design Factor and Allowable Load

Because failure is predicted to occur at a limiting load, rather than a stress, the concept of a design factor is applied differently than it is for most other load-carrying members. Rather than applying the design factor to the yield strength or the ultimate strength of the material, we apply it to the critical load, from Equation (6–4) or (6–6).

For typical machine design applications, a design factor of 3 is used. For stationary columns with well-known loads and end fixity, a lower factor can be used, such as 2.0.

A factor of 1.92 is used in some construction applications.

Conversely, for very long columns, where there is some uncertainty about the loads or the end fixity, or where special dangers are presented, larger factors are advised. (See References 1 and 2.)

In summary, the objective of column analysis and design is to ensure that the load applied to a column is safe, well below the critical buckling load. The following definitions of terms must be understood:

P_{cr} = critical buckling load
P_a = allowable load
P = actual applied load
N = design factor

Then

➪ Allowable Load

$$P_a = P_{cr}/N$$

The actual applied load, P, must be less than P_a.

6–6 TRANSITION SLENDERNESS RATIO

The Euler formula for critical load prediction, Equation (6–6), can be expressed as:

$$\frac{P_{cr}}{A} = \frac{\pi^2 E}{S_r^{\,2}} \tag{6–7}$$

Plotting stress verses slenderness ratio, P_{cr}/A vs. S_r, the Euler formula is an inverse square function as shown in Figure 6–4. It is observed that when a column is "short" (a low slenderness value), Euler formula gives inaccurate predictions with unrealistic high stresses. As such, the J. B. Johnson formula, a parabolic function also shown in Figure 6–4, is used to determine the critical load for short columns. See Section 6–7 for the definition of the J. B. Johnson formula.

The choice of Euler or Johnson formula depends on the value of the slenderness ratio of the column being analyzed in relation to the transition slenderness ratio, S_{rT}, or column constant, C_c. For a given column with the slenderness ratio greater than the column constant, the column is considered "long" and Euler formula should be used. If the slenderness ratio is less than column constant, the column is consider "short" and Johnson formula should be used to predict the critical load.

The Euler–Johnson transition is set at the stress equal to half of the yield strength, s_y, of the material:

$$\frac{s_y}{2} = \frac{\pi^2 E}{S_{rT}^{\,2}} \Rightarrow C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} \tag{6–8}$$

The value of the column constant, or transition slenderness ratio, is dependent on the material properties of modulus of elasticity and yield strength. For any

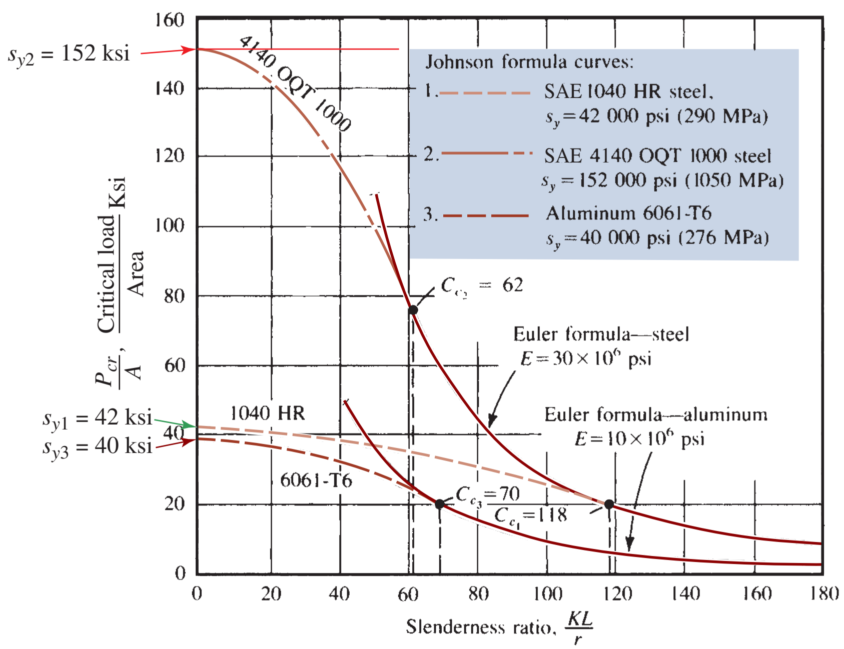


FIGURE 6–4 Euler–Johnson buckling curves

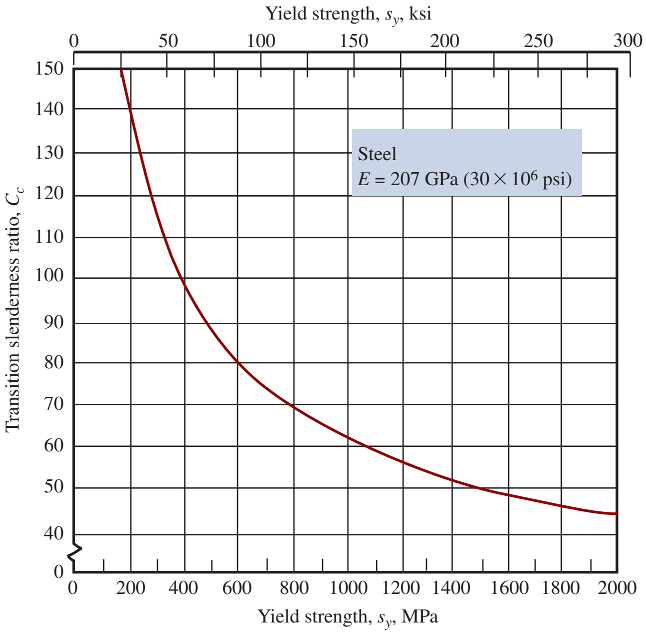


FIGURE 6–5 Transition slenderness ratio C_c vs. yield strength for steel

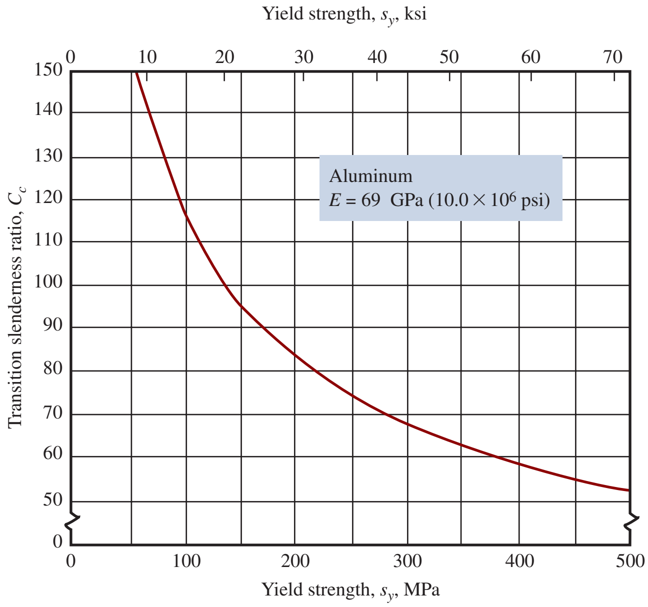


FIGURE 6–6 Transition slenderness ratio C_c vs. yield strength for aluminum

given class of material, for example, steel, the modulus of elasticity is nearly constant. Thus, the value of C_c varies inversely as the square root of the yield strength. Figures 6–5 and 6–6 show the resulting values for steel and aluminum, respectively, for the range of yield strengths expected for each material. The Figures show that the value of C_c decreases as the yield strength increases.

6–7 SHORT COLUMN ANALYSIS: THE J. B. JOHNSON FORMULA

As stated, when the actual slenderness ratio, S_r, for a column is less than the transition value, C_c, then the column is short, and the J. B. Johnson formula should be used. Use of the Euler formula in this range would predict a critical load greater than it really is.

The J. B. Johnson formula is written as follows:

⇨ J. B. Johnson Formula for Short Columns

$$P_{cr} = As_y\left[1 - \frac{s_y S_r{}^2}{4\pi^2 E}\right] \qquad (6\text{–}9)$$

Notice that in Figure 6–4 the Johnson formula curves becomes tangent to the result of the Euler formula at the transition slenderness ratio, the limit of its application. Also, at very low values for the slenderness ratio, the second term of the equation approaches zero, and the critical load approaches the yield load. Curves for three different materials are included in the figure to illustrate the effect of E and s_y on the critical load and the transition slenderness ratio.

The critical load for a short column is affected by the strength of the material in addition to its stiffness, E. As shown in the preceding section, strength is not a factor for a long column when the Euler formula is used. Note that if a column is to be used as part of a building structure, the methods described in References 1 or 2 must be used.

In summary, a procedure for analyzing straight, centrally loaded columns is illustrated below, and Figure 6–7 is a logical flowchart for this procedure.

PROCEDURE FOR ANALYZING STRAIGHT, CENTRALLY LOADED COLUMNS ▼

1. For the given column, compute its actual slenderness ratio.
2. Compute the value of C_c.
3. Compare C_c with KL/r. Because C_c represents the value of the slenderness ratio that separates a long column from a short one, the result of the comparison indicates which type of analysis should be used.
4. If the actual KL/r is greater than C_c, the column is *long*. Use Euler's equation, as described in Section 6–5.
5. If KL/r is less than C_c, the column is *short*. Use the J. B. Johnson formula, described in Section 6–7.